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**CAVITATION INCEPTION
IN THE FREQUENCY RANGE OF
RELEVANCE TO DIAGNOSTIC ULTRASOUND**

**A Dissertation
Presented to the Faculty of the Graduate School
of
Yale University
in Candidacy for the Degree of
Doctor of Philosophy**

**By
Qihong Xu
December 1994**

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ABSTRACT

CAVITATION INCEPTION IN THE FREQUENCY RANGE OF RELEVANCE TO DIAGNOSTIC ULTRASOUND

Qihong Xu

Yale University

1994

Cavitation is generated in distilled water by 2.46 MHz and 4.3 MHz acoustic waves. Several different sizes of polystyrene particle are added as potential nuclei. An improved Active Cavitation Detector (ACD) is used to detect the cavitation by transmitting a tone burst and receiving a backscattered signal from the cavitation. The Mechanical Indices(MIs) corresponding to the cavitation thresholds are calculated. Measured cavitation thresholds and the corresponding MIs are in the range of commercial diagnostic ultrasound equipment. A theory for the transient Doppler frequency shift from a tone burst scattering from a dynamic bubble is developed to lay the ground for extracting cavitation information from a noisy environment. Both digital methods and an analog method are applied to analyze the frequency information from the backscattered signal. The digital methods give quantitative information, whereas the analog method is superior for real time applications.

Ultrasound at 2.46 MHz is used to cause cavitation in blood samples of different volume fractions of red blood cell(RBC). The cavitation thresholds, which are measured by the ACD, increase linearly with the concentration of RBC. The measured cavitation

thresholds and the corresponding MI are also in the upper range of commercial diagnostic ultrasound systems.

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I DEDICATE THIS DISSERTATION TO MY PARENTS.
I AM GRATEFUL THAT THEY LIVED.

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Chapter 1 Introduction

1.1 General Background

Cavitation is defined as a bubble or cavity in a medium which has been generated or set into motion by extraneous factors, such as an acoustic wave. Cavitation caused by an acoustic wave is called acoustic cavitation. The discussion on cavitation in this dissertation will be limited to acoustic cavitation if there is no specific indication.

Cavitation can often be classified as stable and transient cavitation. Stable cavitation is generated by a continuous acoustic wave (CW), and bubble oscillations last for many acoustic cycles. A bubble, which can either preexist or be generated by an acoustic wave, is driven into oscillations about an equilibrium radius. Such oscillations can be nonlinear. Transient cavitation is not a continuous phenomenon, existing only for a short time[Flynn, 1975.]. Transient cavities collapse violently, and the collapse can generate high impulsive pressures. Tomita and Shima have observed the impulsive pressure can be as high as 20 MPa with a bubble of 3.5 mm maximum radius at about 1 mm distance from the bubble center [Tomita and Shima, 1990] . Flint *et al.* have observed that bubble collapse can generate temperatures of thousands of degrees Kelvin [Flint *et al.*, 1991]. These high collapse pressures and temperatures have caused some concern for applications of medical diagnostic ultrasound. There are two main reasons for such concerns: (1) wide spread use of medical ultrasound. (2) high acoustic pressure output used in diagnostic ultrasound. Some ultrasound scanners can generate peak-negative acoustic pressure of 3.9 MPa[Duck, *et al.*, 1985, 1987].

Apfel [Apfel, 1981] and Flynn [Flynn, 1975] have independently determined the theoretical conditions for the generation of transient microcavitation in aqueous media. The generation of cavitation can be described by the cavitation threshold which is the acoustic

pressure amplitude necessary to cause cavitation inception. The cavitation threshold depends on the initial size of the gas bubble, the ambient temperature, and the mechanical properties of host fluid, such as gas concentration, viscosity, surface tension and density. Flynn has calculated that microsecond length pulses of ultrasound can generate transient cavitation for acoustic spatial peak temporal peak(SPT_p) intensities in the range of 10 to 30 W/cm^2 [Flynn, 1982]. Some diagnostic ultrasound operates at acoustic SPT_{pt} intensities over 100 W/cm^2 . There is reason to believe that cavitation can be generated with the application of medical diagnostic ultrasound in aqueous media. Holland and Apfel [Holland and Apfel, 1989] have successfully simplified the theory for threshold prediction of cavitation and given an analytic estimate of transient cavitation thresholds. One of the necessary requirements for the generation of transient cavitation is a stabilized pocket of gas present in aqueous media as a nucleus when the media is insonified. Such nuclei may exist in mammalian tissue. Ter Haar *et al.* have shown the evidence of cavitation *in vivo* caused by 0.75 MHz continuous acoustic waves(CW) used for therapeutic applications [ter Haar, *et al.*, 1986]. Based on theoretical analysis and experimental evidence, the American Institute of Ultrasound in Medicine(AIUM) issued the following statements in 1993:

- 1)Acoustic cavitation may occur with short pulses and has the potential for producing deleterious biological effects.
- 2)Currently available information indicates that pulses with peak pressures greater than 10 MPa ($3300 W/cm^2$) can induce cavitation in mammals.
- 3)With the limited data available, it is not possible to specify threshold pressure amplitudes above which acoustic cavitation will occur in mammals with diagnostically relevant pulse lengths and repetition rates.

1.2 Mechanical Index

Apfel and Holland have theoretically determined that there is a optimal initial nucleus size which yields a minimum peak negative cavitation threshold at a given frequency[Apfel and Holland, 1991]. The cavitation threshold has a simple, close to quadratic relationship with acoustic frequency. Based on this work, they proposed an index, I , for assessing the likelihood for cavitation to occur,

$$I = \frac{(P^*)^a}{f^*},$$

where $a \approx 2.0$, P^* is the normalized derated pressure given by pressure/(1 MPa), and f^* the normalized frequency given by frequency /(1 MHz). Their work has led AIUM and NEMA (National Electrical Equipment Manufactures Association) to recommend the mechanical index be used as an output display standard for industry. This index, MI , which is approximately the square root of I , is defined as:

$$MI = \frac{p}{\sqrt{f_c}}, \text{ where}$$

p is the peak rarefactional pressure (in MPa) derated by 0.3 dB/cm-MHz, and f_c the normalized frequency given by frequency/(1 MHz). Several diagnostic ultrasound scanner manufactures have adopted MI as a display parameter to help the ultrasound operator assess risk.

MI is dimensionless. For a given MI , higher frequency corresponds to higher pressure. Although AIUM has not recommended an upper limit for MI , one manufacture is attempting to increase their outputs to 1.9. If the operating frequency is 4.0 MHz, $MI =$

2.0 gives a peak pressure 4.0 MPa. This value is about the same as the cavitation threshold measured in distilled gaseous water, which will be discussed in chapter 6.

1.3 Work at Yale Physical Acoustics Laboratory

In the last seven years, the Yale Physical Acoustics Laboratory has been studying transient microcavitation in water. Atchley *et al.* used a passive method to measure the thresholds for cavitation produced in water by pulsed ultrasound [Atchley, *et al.*, 1988]. The passive method relied on the scattering of an acoustic wave which generates cavitation, the scattering being caused by transient cavities. Holland automated the passive detection apparatus to determine thresholds for cavitation produced in a fluid by short tone bursts of ultrasound at 0.76, 0.99 and 2.3 MHz [Holland, 1989]. The measurements were done in water, in fluid of higher viscosity, and in diluted whole blood. A main disadvantage of the passive technique is the low sensitivity to the acoustic wave scattered by cavitation. Roy has designed an active cavitation detector (ACD) [Roy, *et al.*, 1990] to measure thresholds for the cavitation in fluids. The ACD, which is based on a particle scattering apparatus originally developed by Roos and Apfel [Roos, *et al.*, 1983, 1988], shares the same principle as active SONAR does by transmitting a probing wave and receiving the backscattered wave from the cavitation bubble. Two transducers are used and confocally positioned. One transducer operates below 5 MHz frequency for causing cavitation and another one operates at 30 MHz frequency for probing the cavitation. According to Rayleigh scattering theory, the scattering strength is proportional to the frequency square. Therefore, compared with the passive technique, the ACD technique can increase the sensitivity by a factor of $\left(\frac{30}{f}\right)^2$, where f is the center frequency of the acoustical wave causing cavitation. For example, if a 2.5 MHz acoustical wave is used to cause the cavitation, the ACD at 30 MHz can theoretically increase the sensitivity by 144 times. In

practice, the sensitivity will not increase that much because of attenuation in the host fluid, which will increase with frequency, and deviations from the Rayleigh scattering approximation. Using the ACD technique and the passive method, Madenshetty studied thresholds for cavitation produced by pulsed tone bursts with a 0.75 MHz and 1% duty cycle. He also studied how the measured thresholds for cavitation vary with the size, kind and number density of particles present in the host fluid. Thresholds measured with the ACD are lower than those detected with passive method, in similar environments. Holland *et al.* investigated the possibility of transient cavitation from short-pulse, diagnostic ultrasound under conditions comparable to clinical ultrasound examinations [Holland, *et al.*, 1992]. A commercial diagnostic ultrasound scanner equipped with both imaging and Doppler modes was used to induce transient cavitation, and the cavitation was detected using the ACD technique. This result further illuminates the possibility of cavitation *in vivo* caused by diagnostic ultrasound.

1.4 Cavitation Detection Based on Frequency Shift Information

The ACD technique has been proven to be effective in detecting cavitation in water or other fluids where the signal to noise ratio(SNR) is significantly over 1. However, because the ACD technique only utilizes amplitude information of the backscattered signal, this technique obviously fails in detecting cavitation if SNR is not over 1, or if the scattered signal from cavitation is imbedded in the scattered signal from a reflector or scatterer. The goal of this project ultimately requires *in vivo* tests, where the “clutter” may cause much stronger scattering than the cavitation does. From the basic principle of the Doppler effect, we know that a moving object can cause a frequency shift in the scattered signal. Because transient cavitation bubbles grow and collapse, the bubble wall velocity can be large enough to produce a detectable frequency shift, which is only associated with cavitation, not particle scattering.

The Doppler frequency shift of the scattered wave from cavitation is different from the ordinary Doppler frequency shift where the velocity of the moving object is a constant. Censor analyzed acoustic scattering from a vibrating object with the assumptions that the vibration amplitude is small and that acoustic scattering is in the range of linear acoustics [Censor, 1972]. However, the results fail to converge to the ordinary Doppler frequency when the surface velocity of vibrating object is a constant. In this dissertation, a quasi-static approximation has been assumed to address this cavitation scattering problem. Both static and acoustic boundaries are combined to solve this moving boundary problem, which yields a result that reduces to the ordinary Doppler frequency when the surface velocity of the vibrating object is a constant.

1.5 Work in the Dissertation

The objective of this work is to study the potential for cavitation in the body in the application of diagnostic ultrasound by performing a fundamental set of experiments involving the observation of submicron cavitation in water seeded with polystyrene particles. In Chapter 2, we briefly review the theories on bubble dynamics. A theory on transient Doppler shift is presented. The apparatus, including ACD, are described in Chapter 3. Because the accuracy of transducer calibration directly affects the absolute value of cavitation thresholds, we present the methods and results of transducer calibration in Chapter 4. Detailed procedures to perform the experiments are described in Chapter 5. We present and discuss experiments and their results in Chapter 6. These experiments are designed in preparation for eventual tests to check the likelihood of cavitation *in vivo*. Two approaches are formulated to reach this objective. First, the thresholds of cavitation in water and in blood are measured by using a modified ACD technique. The cavitation is induced with two transducers, one of 2.46 MHz center frequency, because it is low cost

and effective in causing cavitation, and another one of 4.3 MHz center frequency, because it is more comparable to the frequency of the Doppler mode in clinical ultrasound systems. Polystyrene particles of $0.1\mu m$ to $1.0\mu m$ are added to the host fluid to study the effects of initial nuclei size on thresholds. An experiment in human blood and its dilutions is also conducted. The mechanical indexes associated with the cavitation thresholds are calculated, and are in the range of the application of clinical diagnostic ultrasound. The second part of chapter 6 details the extraction of frequency information of the scattered signal from cavitation. Both digital processing and analog processing methods are used to optimize results. Quantitative frequency information is obtained by using the digital method. On other hand, unlike the digital method which needs post-processing, the analog method is real-time and may have advantages in applications to *in vivo* experiments.

Chapter 2 Some Relevant Scattering and Cavitation Theories

2.1 Review of Relevant Theories on Cavitation and Acoustic Scattering

In this section, we will review some of the theoretical work on acoustic scattering and acoustic cavitation, because these basics allows us to interpret measurements described in Chapter 6. Since there are many publications on acoustic scattering and cavitation, a comprehensive review will not be attempted in this chapter. We will only review the work relevant to our present work and prepare the groundwork for our theoretical work, which will be discussed later in this chapter.

2.1.1 Acoustic Scattering from Fluid Sphere

When a sound wave traveling through a host fluid encounters an inhomogeneity, the sound wave will be scattered. The characteristics of the scattered sound wave are determined by the mechanical properties of both the host fluid and the scatterer. The total sound fields will be the superposition of the incident sound and the scattered sound.

Rayleigh [Rayleigh, 1945] addressed the problem of sound scattering by a fluid object and solved the case when the incident wavelength was much larger than the scatterer size. In Rayleigh's solution, the scattered sound wave is a superposition of a monopole and dipole contributions. Rayleigh's solution is only valid for the far field.

Anderson [Anderson, 1950] obtained the exact analytical solution to the problem of a plane wave scattered by a fluid sphere. Both the incident sound wave and the scattered sound wave can be expressed in terms of eigenfunctions of the wave equation. The scattered wave can be obtained by applying the boundary conditions that both sound pressures and particle velocities are continuous at the boundary of the space. Hay [Hay, 1982] discussed

both thermal and viscous effects on the scattered sound wave. For the above work, the evaluation of the solution usually requires numerical computation.

As shown in Fig. 2.1, a plane wave, $p = p_0 e^{i(\omega t - k r \cos \vartheta)}$, propagates along the x-axis in the medium of density ρ and compressibility κ , where p_0 is the amplitude of acoustic pressure, ω is the angular frequency, and k is the wave number in the host medium. The plane wave is incident on a fluid sphere of radius a , density ρ_{in} and compressibility κ_{in} . The sphere is located at $(0, 0, 0)$ of the Cartesian coordinate system. Neglecting nonlinear wave propagation, we write down the linear wave equations which govern the acoustic pressures both inside the sphere and outside the sphere:

$$\nabla^2 p_{in} - \frac{1}{c_{in}^2} \frac{\partial^2 p_{in}}{\partial t^2} = 0, \quad 2.1$$

$$\nabla^2 (p_o + p) - \frac{1}{c^2} \frac{\partial^2 (p_o + p)}{\partial t^2} = 0, \quad 2.2$$

where p_o stands for the scattered wave outside the sphere and p_{in} inside the sphere, c is the sound velocity in the medium outside the sphere, and c_{in} the velocity inside the sphere.

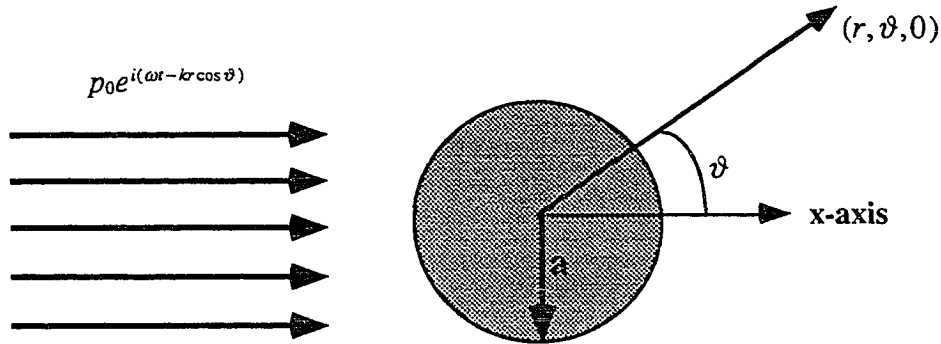


Figure 2.1 Scattering model of a fluid sphere.

The acoustic wave outside the sphere is composed of two parts: the incident wave and the scattered wave. Because Eqs. 2.1 and 2.2 are linear, we can apply the superposition principle,

$$p_{out} = p + p_o, \quad 2.3$$

where p is the incident wave and can be expressed as

$$p = p_0 \sum_{m=0}^{\infty} [(-i)^m (2m+1) P_m(\cos \vartheta) j_m(kr)], \quad 2.4$$

where $P_m(\cos \vartheta)$ is the Legendre function of m th order, $j_m(kr)$ the spherical Bessel function of m th order. The general solution for Eq. 2.1 has the following form,

$$p = \sum_{m=0}^{\infty} (-i)^m (2m+1) A_m P_m(\cos \vartheta) [j_m(kr) \pm n_m(kr)], \quad 2.5$$

where $n_m(kr)$ is the spherical Neuman function of m th order. For $kr \gg 1$, or far field, the scattered wave outside the sphere, p_o , has the form of an outward traveling wave. Therefore, the general solution for p_o is,

$$p_o = \sum_{m=0}^{\infty} (-i)^m (2m+1) B_m P_m(\cos \vartheta) h_m^{(2)}(kr), \quad 2.6$$

where B_m are the coefficients to be determined, $h_m^{(2)}(kr) = j_m(kr) - in_m(kr)$ is the spherical Hankel function of second kind, and $h_m^{(2)}(kr) \approx \frac{1}{kr} e^{-i(kr - \frac{m+1}{2}\pi)}$ as $kr \rightarrow \infty$. To guarantee its convergence, the solution for p_{in} has the following form,

$$p_{in} = \sum_{m=0}^{\infty} [(-i)^m (2m+1) C_m P_m(\cos \vartheta) j_m(kr)], \quad 2.7$$

where C_m are the coefficients to be determined.

The boundary conditions are that the pressures and particle velocities are continuous on the boundary, or

$$(p + p_o)|_{r=a} = p_{in}|_{r=a}, \quad 2.8$$

and

$$\left. \frac{\partial(p + p_o)}{\partial r} \right|_{r=a} = \frac{\rho_{in} c_{in}}{\rho c} \left. \frac{\partial p_{in}}{\partial r} \right|_{r=a}, \quad 2.9$$

where the parameters with subscript *in* are associated with the scatterer.

Solving the system of equations of Eqs 2.8 and 2.7 for B_m , we have

$$B_m = p_0 \frac{j_m'(ka) - \frac{\rho c}{\rho_{in} c_{in}} j_m'(k_{in}a) \frac{j_m(ka)}{j_m(k_{in}a)}}{\frac{\rho c}{\rho_{in} c_{in}} j_m'(k_{in}a) \frac{h_m(ka)}{j_m(k_{in}a)} - h_m'(ka)}. \quad 2.10$$

After algebraic manipulation, B_m can be expressed in the form as in [Anderson, 1950].

$$B_m = -\frac{1}{1 + iD_m}, \quad 2.11$$

$$\text{where } D_m = \frac{\frac{\rho_{in} c_{in}}{\rho c} \frac{n_m'(ka)}{j_m'(ka)} - \frac{n_m(ka)}{j_m(k_{in}a)} \frac{j_m'(k_{in}a)}{j_m'(ka)}}{\frac{j_m(ka)}{j_m(k_{in}a)} \frac{j_m'(k_{in}a)}{j_m'(ka)} - \frac{\rho_{in} c_{in}}{\rho c}}.$$

For $kr \gg 1$, the following approximation is valid,

$$h_m^{(2)}(kr) \approx \frac{1}{kr} e^{-i(kr - \frac{m+1}{2}\pi)}. \quad 2.12$$

The acoustic pressure of the scattered wave outside the scatterer is given by

$$p_o = \frac{p_0}{kr} e^{i(\omega t - kr)} \sum_{m=0}^{\infty} \frac{1}{1 + iD_m} P_m(\cos \vartheta). \quad 2.13$$

The particle velocity on the scatterer surface is

$$u_{in}|_{r=a} = \frac{i}{\rho_{in} c_{in}} p_{in}' = \frac{p_0}{\rho_{in} c_{in}} \sum_{m=0}^{\infty} (-i)^{m+1} (2m+1) \left(\frac{j_m(ka)}{j_m(k_{in}a)} + B_m \frac{h_m^2(ka)}{j_m(k_{in}a)} \right) j_m'(k_{in}a). \quad 2.14$$

For $\vartheta = \pi$, Eq. 2.13 yields the backscattered wave,

$$p_o = \frac{1}{kr} e^{i(\omega t - kr)} \sum_{m=0}^{\infty} \frac{(-1)^m}{1 + iD_m}. \quad 2.15$$

In the case of long wavelength scattering, $ka \ll 1$, which corresponds to our experimental conditions described in Chapter 6, we only need to keep the terms with $m=0$ and $m=1$. Consequently, the amplitude of the scattered wave in Eq. 2.13 can be simplified to

$$|p_o| = \frac{1}{kr} \frac{(ka)^3}{3} \left| \left(1 - \frac{\kappa}{\kappa_{in}} \right) + 3 \frac{\rho - \rho_{in}}{\rho + 2\rho_{in}} \cos \vartheta \right|, \quad 2.16$$

where $\kappa = \frac{1}{\rho c^2}$ is the compressibility. If the frequency of the incident wave and the scatterer radius are fixed, the amplitude of the scattered wave is determined by two factors: the contrast of compressibility, $\frac{\kappa_{in} - \kappa}{\kappa_{in}}$, and the contrast of density, $\frac{\rho - \rho_{in}}{\rho + 2\rho_{in}}$.

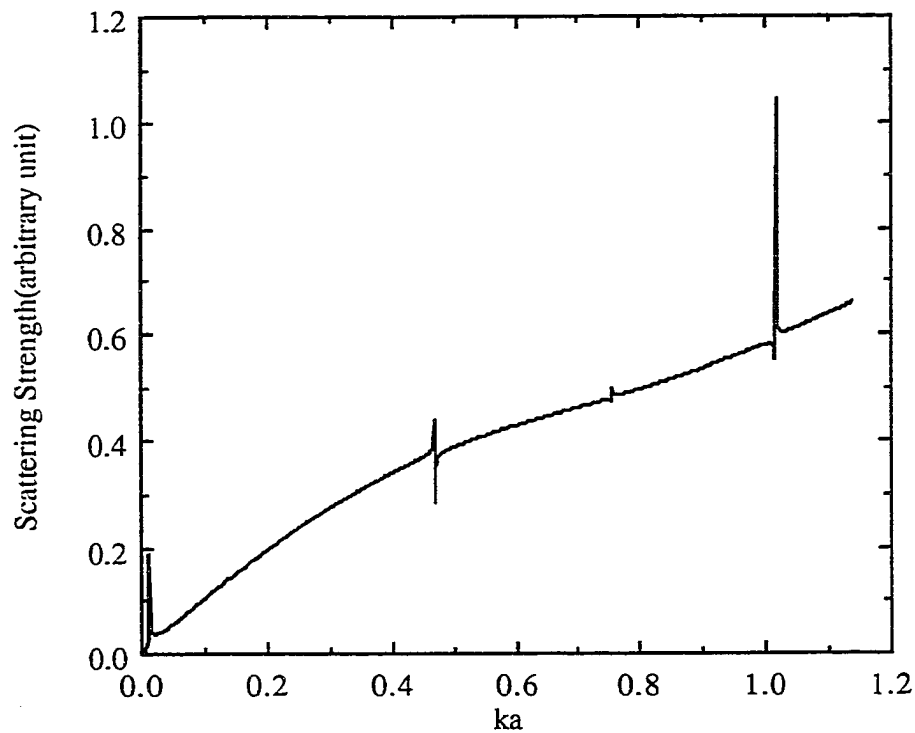


Fig 2.2. Backscattering strength vs ka for air bubble in water. The resonance amplitude does not indicate a true amplitude because of the rounding errors of the numerical computation.

2.1.2 Acoustic Cavitation

Cavitation often refers to the generation of gas- or vapor-filled bubble or cavity in a liquid. Acoustic cavitation is the generation of cavitation by an acoustic field, and will be the focus of our discussion on cavitation.

2.1.2.1 Blake Threshold

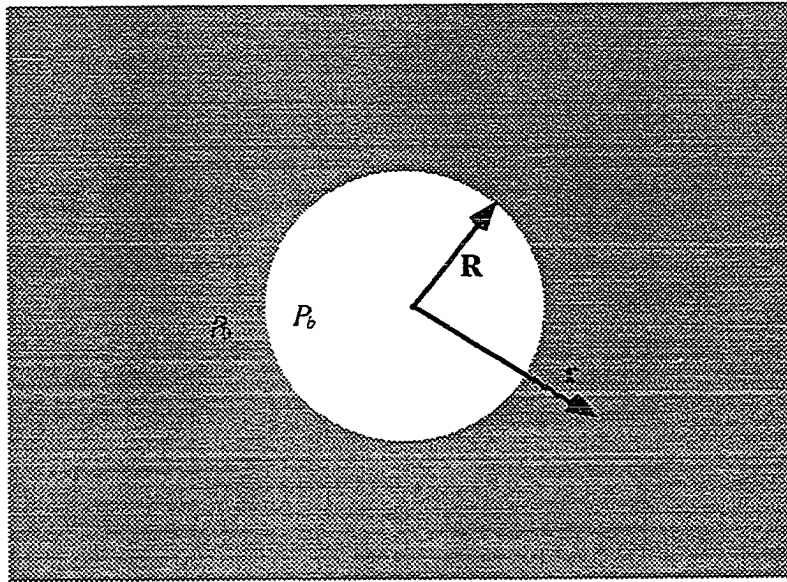


Figure 2.3. Cavitation Model.

The theoretical tensile strength of water is about 100 MPa, but experimental measurements of the tensile strength have always been lower than this value because of impurities, or pre-existing bubbles in the water. Shown in Fig. 2.3 is a gas bubble surrounded by water. If we neglect the vapor pressure of the liquid, the equilibrium equation is,

$$P_b = P_0 + \frac{2\sigma}{R_0}, \quad 2.17$$

in which σ is the surface tension, P_b is the gas pressure inside the bubble and P_0 the ambient pressure in water. If an acoustic pressure of amplitude P_b is applied to the bubble, the bubble radius will change from its initial radius R_b until it reaches a new equilibrium state of radius R_x . Then the equilibrium condition becomes,

$$(P_0 + \frac{2\sigma}{R_B})\left(\frac{R_B}{R_x}\right)^3 = P_0 - P_B + \frac{2\sigma}{R_x}. \quad 2.18$$

To find the minimum external pressure for the bubble to grow, we set

$$\frac{\partial(P_0 - P_B)}{\partial R_x} = 0, \quad 2.19$$

and obtain

$$P_B = P_0 + \frac{8\sigma}{9} \left[\frac{3\sigma}{2\left[P_0 + \frac{2\sigma}{R_B}\right]R_B^3} \right]^{1/2}, \quad 2.20$$

where P_B is the Blake threshold, which is the minimum pressure needed to cause the bubble to grow. Eq. 2.18 can also be approximated by

$$P_B = P_0 + \frac{8\sigma}{9} \left[\frac{3\sigma}{2P_0 R_B^3} \right]^{1/2} \text{ when } \frac{2\sigma}{P_0} \ll P. \quad 2.20$$

2.1.2.2 The Equations Governing Bubble Dynamics

We consider a spherical bubble in an incompressible liquid, and the bubble radius and bubble wall velocity are R and $\dot{R}$, respectively. At a distance, r , the radial velocity in the liquid will be $\dot{r}$, and is given by

$$\dot{r} = \frac{R^2 \dot{R}}{r^2}. \quad 2.21$$

Since the motion is irrotational, the velocity potential is

$$\phi = -\int_r^\infty \dot{r} dr = -\frac{R^2 \dot{R}}{r}. \quad 2.22$$

The motion is governed by the Bernoulli Equation[Young, 1989] as

$$-\frac{\partial \phi}{\partial t} + \frac{u^2}{2} + \frac{p}{\rho} = \frac{p_\infty}{\rho}, \quad 2.23$$

where $p=p(r,t)$ is the local pressure in the liquid, $u=u(r,t)$ is the radial velocity of the liquid at distance r , and $p_\infty = p_\infty(t)$ the pressure is at $r = \infty$.

Substituting Eqs. 2.21 and 2.22 into Eq. 2.23 yields

$$\frac{p - p_\infty}{\rho} = \frac{2R\dot{R}^2 + R^2\ddot{R}}{r} - \frac{R^4\dot{R}^2}{2r^4}. \quad 2.24$$

Letting $r=R$, we get the equation governing the motion of the bubble wall,

$$\frac{p - p_\infty}{\rho} = R\ddot{R} + \frac{3}{2}\dot{R}^2. \quad 2.25$$

This is the fundamental equation by Rayleigh for bubble dynamics in an incompressible, inviscid liquid.

If the bubble is filled with gas, the pressure in the liquid is

$$p = (p_0 + \frac{2\sigma}{R_0}) \left(\frac{2R_0}{R} \right)^{3\gamma}, \quad 2.26$$

where γ is the ratio of specific heats. Eq 25 becomes

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho} \left[\left(p_0 + \frac{2\sigma}{R_0} \right) \left(\frac{2R_0}{R} \right)^{3\gamma} - \frac{2\sigma}{R} - p_\infty \right]. \quad 2.27$$

This equation was first derived by Noltingk and Neppiras[Noltingk *et al.*, 1950]. Poritsky later included the effect of viscosity and augmented Eq. 2.27 to yielded[Poritsky, 1952].

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho} \left[\left(p_0 + \frac{2\sigma}{R_0} \right) \left(\frac{2R_0}{R} \right)^{3\gamma} - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R} - p_\infty \right], \quad 2.28$$

where μ is the shear viscosity of the liquid. Interestingly, Apfel later gave a very simple derivation of Eq. 28, using the principle of energy conservation[Apfel, “Acoustic cavitation,” in *Methods of Experimental Physics*, 1981].

If an acoustic wave is applied to the bubble motion, Eq. 2.28 becomes

$$R\ddot{R} + \frac{3}{2}\dot{R}^2 = \frac{1}{\rho} \left[\left(p_0 + \frac{2\sigma}{R_0} \right) \left(\frac{2R_0}{R} \right)^{3\gamma} - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R} - (P_0 + P_A) \right], \quad 2.29$$

where P_A is the acoustic pressure, which is either a continuous wave or pulsed.

2.1.3 Transient Microcavitation

Transient cavitation can be caused by a single acoustic cycle, and is typified by growth of the bubble to many times its initial size, followed by a rapid and violent collapse. Apfel has been able to approximate the maximum radius, $R_{\max}$, in terms of the initial bubble size, R_0 , static pressure, P_0 , acoustic pressure amplitude, P_a , liquid density, ρ , viscosity, μ , and surface tension, σ . This theory was extended by Holland and Apfel [Holland *et*

al., 1989] to include an estimate of the time delay associated with the Laplace pressure, $\frac{2\sigma}{R_0}$, which led to the following expressions:

$$f = \frac{\frac{1}{3\pi R_0} \sqrt{\frac{P_0 \xi}{\rho}} \left(\sqrt{\frac{p-1}{p}} + \sqrt{\frac{p-p_b}{p}} \right)}{\left(\frac{T'}{T_0(\gamma-1)(\frac{\xi}{3}+1)} \right)^{1/3} - 0.46 + \frac{4\mu}{R_0} \sqrt{\frac{2}{\rho P_0 \xi}} + \frac{2\sigma}{P_0 R_0} \left(\frac{1}{p-p_0} \right)^{1.5} \sqrt{\frac{\xi}{3}}}, \quad 2.30$$

where p_b is the normalized Blake threshold pressure given by

$$p_b \equiv \frac{P_B}{P_0} = 1 + \frac{4}{9} X_B \sqrt{\frac{3X_B}{4(1+X_B)}}, \quad 2.31$$

where $X_B = \frac{2\sigma}{R_0 P_0}$, $p = \frac{P_A}{P_0}$, $\xi = p + p_b - 2 + \sqrt{(p-1)(p-p_b)}$, T_0 is the initial ambient temperature, T' the maximum cavitation collapse temperature, γ the ratio of the specific heats.

2.1.2.4 Rayleigh Formula for the Collapse of A Void

If the gas-filled bubble in Fig. 2.3 is replaced by a void, $R(t)$ will be the void radius and $\dot{R}(t)$ the velocity of the void wall. The liquid velocity at a distance, r , is

$$\dot{r}(t) = \frac{R(t)^2 \dot{R}(t)}{r(t)^2}. \quad 2.32$$

Therefore, the whole kinetic energy of the liquid is

$$\int \frac{1}{2} \dot{r}^2 dm = 2\pi\rho \int_R^\infty \dot{r} r^2 dr = 2\pi\rho \dot{R}^2 R^3, \quad 2.33$$

where $dm = d(\frac{4}{3}\pi r^3 \rho) = 4\pi r^2 \rho dr$ is the infinitesimal element of liquid mass.

When the void shrinks from its initial radius R_0 to R , the work done by the hydrostatic pressure is

$$\Delta E = P\Delta V = P_0 \frac{4}{3}\pi(R_0^3 - R^3), \quad 2.34$$

which is assumed an isothermal process. From an energy point of view, Eq. 2.33 is equal to Eq.2.34, and this yields

$$2\pi\rho\dot{R}^2 R^3 = P_0 \frac{4}{3}\pi(R_0^3 - R^3). \quad 2.35$$

Then,

$$dt = \sqrt{\frac{3\rho}{2P_0}} \sqrt{\frac{R^3}{R_0^3 - R^3}} dR. \quad 2.36$$

The collapse time, t_c , will be (after Rayleigh)

$$t_c = \int_0^t dt = \sqrt{\frac{3\rho}{2P_0}} \int_{R_0}^0 \sqrt{\frac{R^3}{R_0^3 - R^3}} dR \approx 0.91 R_0 \sqrt{\frac{\rho}{P_0}}. \quad 2.37$$

We propose to apply Eq. 2.37 as an estimate for the collapse phase of transient cavitation, thus neglecting the effects of gas or vapor within the bubble. For example, the collapse time for a bubble of $1 \mu m$ radius is $0.09 \mu s$.

2.1.2.5 The Relationship between Bubble Resonance Frequency and Radius

Houghton [Houghton, 1963] showed the relationship between bubble resonance radius, R_r , and bubble resonance frequency, f ,

$$f = \frac{1}{2\pi R_r} \left\{ \frac{3\gamma}{\rho} \left[P_0 + \frac{2\sigma}{R_r} \left(1 - \frac{1}{3\gamma} \right) \right] - \frac{4\mu^2}{\rho^2 R_r^2} \right\}^{1/2}, \quad 2.38$$

where γ is the ratio of specific heats, σ is the surface tension, and μ is the viscosity of the surrounding liquid. The motion of a bubble larger than this size at frequency f will tend to be controlled by the inertia of the surrounding liquid and therefore is unlikely to undergo inertial cavitation.

2.2 Mechanical Index

Eq 2.30 predicts that a transient cavitation will be generated once the amplitude of the acoustic wave exceeds the cavitation threshold at a given frequency. Due to the high outputs of current diagnostic ultrasound scanners [Duke *et al.*, 1987], concern exists for the potential for *in vivo* cavitation production.

To help the users of diagnostic ultrasound assess properly the potential cavitation, Apfel and Holland [Apfel, *et al.*, 1991] proposed an index, I , as a mechanical energy indicator based on Eq 2.30,

$$I = \frac{P^2}{f}, \quad 2.39$$

where P is the peak amplitude of the derated peak negative pressure (in MPa) and f is the center frequency of the output signal (in MHz). Based on Eq. 2.39, a joint committee

composed of members from the American Institute of Ultrasound in Medicine, the Food and Drug Administration, the National Electrical Equipment Manufacturers Association, and other physician representatives proposed that a mechanical index be used as an output display standard, and chose the square root of index proposed by Apfel and Holland:

$$MI = \frac{P_{r,3}(z_{sp})}{\sqrt{f}}; \quad 2.40$$

here $P_{r,3}(z_{sp})$ is the peak rarefactional pressure(in MPa) derated by 0.3 dB/cm-MHz to the location on the beam axis, z_{sp} , where the pulse intensity integral is a maximum, and f is the center frequency in MHz.

2.3 Transient Doppler Frequency Shift

Acoustic scattering from microcavitation is not fully understood because mathematically there is no analytical solution available for describing the interaction between an incident wave and microbubble oscillations because of the complexities of the moving boundary and, sometimes, nonlinear acoustic effects.

It is not our intention to solve the general scattering problem for transient microcavitation. Instead, we simplify the mathematical model by treating the transient microcavitation as a quasi-static event to obtain backscattered sound fields. This simplification is possible because the growth and collapse of a bubble occur in several cycles of the incident 30 MHz probing sound field, and the amplitude of incident probe wave is much smaller than the pressure causing the cavitation, and therefore does not substantially change the growth and collapse of a bubble.

Based on results of a computer simulation [Flynn, *et al.*, 1988], it is reasonable to assume that the maximum boundary velocity, $v_{\max}$, of a bubble is much less than the sound velocity in the surrounding liquid, c_0 . Thus, a linear wave equation

$$\nabla^2 p - c_0^2 \frac{\partial^2 p}{\partial^2 t} = 0, \quad 2.41$$

may be used for describing this scattering problem. We further assume the bubble is spherical. The boundary condition associated with pressures for an inviscid surrounding fluid is then

$$p_{in} - p_{out} = \frac{2\sigma}{R(t)}, \quad 2.42$$

where $R(t)$ is the bubble radius at any instant in time t . By superposing the effects of the incident wave and backscattered wave, we can express $R(t)$ as [see appendix at the end of this chapter.]

$$R(t) = a_0(t) \left[1 - \sum_{m=0}^{\infty} (\zeta_m(t) e^{i\omega_0 t} + \xi_m(t) e^{i\omega_1 t}) P_m(\cos \vartheta) \right], \quad 2.43$$

where $a_0(t)$ is the bubble radius without the perturbation from incident wave, $\zeta_m(t)$ and $\xi_m(t)$ are time-changing functions and satisfy the following conditions

$$\sum_{m=0}^{\infty} |\zeta_m(t) P_m(\cos \vartheta)| < 1 \text{ and } \sum_{m=0}^{\infty} |\xi_m(t) P_m(\cos \vartheta)| < 1. \quad 2.44$$

Ignoring higher order terms, which is justified by assuming that the presence of the probe field adds a small perturbation to the bubble radius, we substitute Eq. 2.43 into Eq. 2.42 to yield,

$$p_{in} - p_{out} = \frac{2\sigma}{a_0(t)} \left[1 + \sum_{m=0}^{\infty} (\zeta_m(t) e^{i\omega_0 t} + \xi_m(t) e^{i\omega_1 t}) P_m(\cos \vartheta) \right]. \quad 2.45$$

The inside pressure, p_{in} , consists of two parts; one is the acoustic field related to incident wave, and the other is the quasi-static pressure, Q_{in} , which drives the bubble to grow and shrink. Therefore, p_{in} has the following form

$$p_{in} = \sum_{m=0}^{\infty} A_m j_m(kr) P_m(\cos \vartheta) e^{i\omega_0 t} + Q_{in}, \quad 2.46$$

where ω_0 is the angular frequency of the incident wave. Because of the bubble growth and collapse, we expect that the frequency of the backscattered wave will be different from that of the incident wave. Thus, the backscattered wave can be expressed as

$$\sum_{m=0}^{\infty} \int E_m(\nu) h_m\left(\frac{\nu}{c_0} r\right) P_m(\cos \vartheta) e^{i\nu t} d\nu, \quad 2.47$$

where $E_m(\nu)$ are the coefficients to be determined. Assuming the incident wave to be plane, the pressure in the surrounding fluid, p_{out} , will be

$$p_{out} = \sum_{m=0}^{\infty} \left[\int E_m(\nu) h_m\left(\frac{\nu}{c_0} r\right) P_m(\cos \vartheta) e^{i\nu t} d\nu \right] + p_0 \sum_{m=0}^{\infty} [(-i)^m j_m(kr) P_m(\cos \vartheta)] e^{i\omega_0 t} + Q_{out}. \quad 2.48$$

Substituting Eq. 2.48 and Eq. 2.46 into Eq. 2.45 yields the following system of equations,

$$Q_{in} - Q_{out} = \frac{2\sigma}{a_0(t)}, \quad 2.49$$

$$\sum_{m=0}^{\infty} [A_m j_m(ka(t)) e^{i\omega_0 t} - \int E_m(\nu) h_m\left(\frac{\nu}{c_0} a(t)\right) e^{i\nu t} d\nu - p_0 (-i)^m j_m(ka(t)) e^{i\omega_0 t}] P_m(\cos \vartheta) =$$

2.50

and

$$a(t) = a_0(t) \left[1 - \sum_{m=0}^{\infty} (\zeta_m(t) e^{i\omega_0 t} + \xi_m(t) e^{i\omega_1 t}) P_m(\cos \vartheta) \right], \quad 2.51$$

where p_0 is the amplitude of the incident wave in the surrounding liquid and Q_{out} is the quasi-static pressure. Eq 2.48 is a quasi-static equation describing the growth and collapse of cavitation bubble [Young, 1989]. Eq 2.50 determines the radius perturbation due to the presence of the incident wave, and has the following approximate solution for $E_m(\nu)$,

$$E_m(\nu) = \frac{2\sigma \xi_m(t)}{a_0(t) h_m\left(\frac{\nu}{c_0} a_0(t)\right)} \delta(\nu - \omega_1), \quad 2.52$$

with $m=0, 1, \dots$

If the bubble is larger than the wavelength, a conventional Doppler frequency shift should be observed,

$$\Delta\omega = 2 \frac{\omega_0}{c_0} \frac{da_0(t)}{dt}. \quad 2.53$$

Using an asymptotic expression for $h_m(z)$,

$$h_m(z) \approx \frac{1}{z} e^{-i(z - \frac{m+1}{2}\pi)}, \text{ for } z \gg 1,$$

we obtain, as expected,

$$\omega_1 = \omega_0 \left(1 + \frac{\dot{a}_0(t)}{c_0} \right). \quad 2.54$$

The higher order terms ($m \geq 2$) in Eq. 2.50 can be dropped when $ka_0(t) \ll 1$. We also notice $\beta_0(t) \propto \frac{\kappa_{in} - \kappa}{\kappa}$ and $\beta_1(t) \propto \frac{3(\rho_{in} - \rho)}{2\rho_{in} + \rho}$, where the subscript, *in*, refers to bubble, and κ and ρ refer to compressibility and density. Therefore, we only need to keep the term $m=0$. Thus, the Doppler frequency shift becomes,

$$\Delta\omega = 2 \frac{\omega_0}{c_0} \frac{da_0(t)}{dt}. \quad 2.55$$

2.4 Detection of Transient Doppler Frequency Shift

The traditional Doppler frequency shift is usually associated with an object moving at a constant velocity or a constant Doppler frequency shift. But in our case, the velocity of bubble wall is not a constant, so the resulting frequency shift is not a constant. Hence, we call this type of Doppler frequency shift “Transient Doppler Shift”. Since the duration of the transient microcavitation is so short, the commercial pulse Doppler technique is not applicable here.

2.4.1 Digital Method

The sinusoidal signal of the backscattering wave usually has about 20 cycles. During this period, the frequency of the sinusoidal signal will experience up and down shifts, which corresponds to bubble growth and collapse. We have used both analog and digital methods to estimate the instantaneous frequency of the backscattered signal. Using the analog method, we can obtain real-time frequency information. Using the digital method, we have more flexibility in analyzing the data and obtaining the desired frequency shift information. Borrowing the idea of Short Time Fourier Transform, we split the

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2.4.1.1 Formulas for Spectrum Estimation Using Eigenanalysis and Prony's Method

We just give the formulas for frequency estimation by using both eigenanalysis and Prony's method. The detailed derivation of these formula were given by [Kay, 1987]. For using eigenanalysis to estimate the frequency of a sinusoidal signal imbedded in white noise, we need to find the frequency which maximizes the value of the following function

$$P(f) = \frac{1}{\sum_{k=M+1}^{k=p+1} |\bar{s}^H \bar{v}_k|^2}, \quad 2.56$$

where $\bar{s}$ is

$$\bar{s} = \begin{bmatrix} 1 \\ \exp(j2\pi fT) \\ \vdots \\ \exp(j2\pi fpT) \end{bmatrix},$$

$\bar{v}_k, k=1, \dots, M$, are the eigenvectors of the p th-order Toeplitz autocorrelation matrix of dimension $(p+1) \times (p+1)$,

$$R_p = \begin{bmatrix} r_{xx}[0] & \dots & r_{xx}^*[p] \\ \vdots & \ddots & \vdots \\ r_{xx}[p] & \dots & r_{xx}[0] \end{bmatrix}, \text{ where } r_{xx}[n] = E\{x[m+n]x^*[m]\}, n = 0, 1, \dots, p, p = 63 \text{ for}$$

the following analysis if it is not specified, and E stands for mathematical expectation.

For using Prony's method to estimate the frequencies of sinusoidal signals imbedded in noise, we have the frequency estimation,

$$f_i = \frac{1}{2\pi T} \tan^{-1} \left\{ \frac{\text{Im}[z_i]}{\text{Re}[z_i]} \right\}, \quad 2.57$$

where z_i , $i = 1, \dots, 2p$, are the roots of the following equation,

$$\sum_{k=1}^{2p} a[k] z^{2p-k} = 0, \quad 2.58$$

where the coefficients $a[k]$, $k = 1, \dots, 2p$ can be determined by the following linear prediction equation of the combination of backward and forward prediction,

$$a[p]x[n-p] + \sum_{k=1}^p \{a[p-k]x[n-p+k] + a[p+k]x[n-p-k]\} = 0, \quad 2.59$$

where $|a[2p]| = 1$.

2.4.1.2 Data Segmentation

In this work, the data are composed of narrow band sinusoidal signals and noise. Two main sources contribute to the noise; one is the electrical circuitry and another is the acoustic noise which is received by the transducer, amplified and fed to a digital oscilloscope (LeCroy 7200). The electrical noise is essentially white, while the acoustic noise is not strictly white due to the interference with the probing wave. This will reduce the sharpness of the spectrum peaks.

The sinusoidal signals are those tone burst sinusoidal waves scattered by the transient microcavitation. It is not expected that the frequencies are constant during the scattering period since the boundary of the scatterer has a certain velocity. The data records are relatively short, and the sinusoidal signals do not undergo many cycles. Therefore, the

Wigner Distribution and its modified versions will not give a clear frequency-time structure.

For those data corresponding to the beginning parts of bubble growth and collapse, it has been expected that the frequency of the backscattered probing wave may have substantial changes during a single period. Since the probing wave is a single frequency signal, the scattered wave should still have a sinusoidal form with time-varying frequency,

$$x[i] = A_0 \sin(2\pi f_i i T) + w[i], \quad i = 0, \dots, M.$$

During a small fraction of the probing wave period, it is not expected that the frequency f_i changes much and can be treated as a constant. Therefore, to track the frequency, the recorded data $x[i]$ can be split into many segments $x_k[n]$, $n = 0, \dots, p$. Each segment length can be only a fraction of the probing wave period and can be decomposed into sinusoidal signals with constant frequencies in an additive white noise,

$$x_k[n] = \sum_{k=1}^M P_k \sin(2\pi f_k n T) + w_k, \quad n = 0, \dots, p$$

in which w_k has a variance independent of k ,

$$E\{w_k[n+m]w_k[n]\} = \begin{cases} \rho_w, & m = 0 \\ 0, & m \neq 0 \end{cases}.$$

The following diagram shows how the data are segmented,

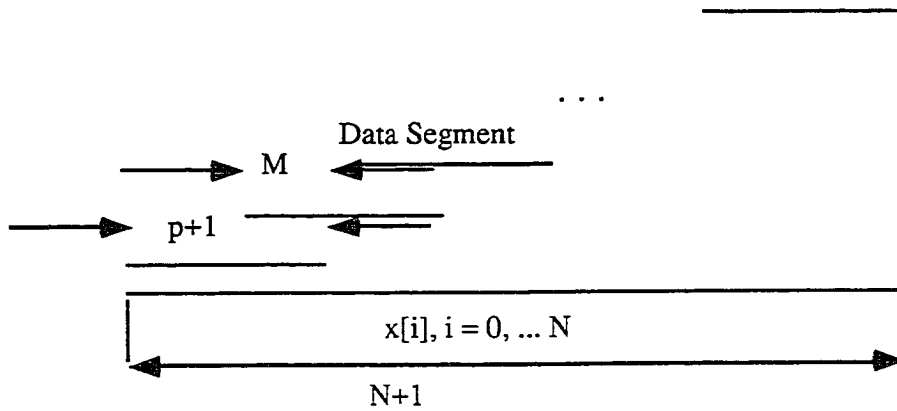


Fig. 2.5. The schematic of data segmentation.

The segments can overlap each other, where the overlapping length M is the same for all segments. As can be easily seen, M can be set from 0 to p . For this work, $M=p$.

2.4.2 Analog Method

Because of large amount of required numerical computations, the digital method will not allow us to track the frequency shift of the backscattered signal in real time. Analog processing technique can track the frequency shift in real time, and is superior in speed to digital techniques. We have developed a simple and very effective analog processing technique and have applied it to extracting the Doppler frequency shift from the backscattered signal. The following example illustrates the principle.

The backscattered signal, $s_{scat}(t)$, can be expressed as

$$s_{scat}(t) = A \cos\{2\pi f(t)t + \theta_0\}, \quad 2.60$$

where θ_0 is the initial phase, $f(t)$ is the non-constant frequency. A reference signal, $s_{ref}(t)$, is

$$s_{ref}(t) = A \cos\{2\pi f_0 t + \theta\}, \quad 2.61$$

where f_0 is the frequency of probing signal of 30 MHz, and θ is the initial phase. Using the two signals as inputs for a multiplier, we have the output of the multiplier

$$s(t) = s_{ref}(t)s_{scat}(t) = \frac{A}{2} \{ \cos(2\pi(f(t) - f_0)t + \vartheta_1) + \cos(2\pi(f(t) + f_0)t + \vartheta_2) \}. \quad 2.62$$

We only need to discuss the first term because the second term can be filtered out by using a low-pass filter. If there is no frequency shift in the backscattered signal, the phase difference will be a constant, and the output of the multiplier is a constant. If the frequency of the backscattered signal is not a constant, the output of the multiplier is a function of time. Because transient cavitation exists for a very short time, the change of $f(t)$ in the backscattered signal will result in a dramatical change of the output of multiplier.

Shown in Fig 2.6 is a schematic of analog processing technique. The mixer functions as a multiplier (AD-FHB, Mini-Circuit Lab., Brooklyn, NY). The backscattered signal from the bubble is fed to the R-port of the mixer, and the reference signal to the L-port of the mixer. The filter input is connected with the I-port of the mixer, and the processed signal from filter output is displayed on the screen of a digital oscilloscope LeCroy 9400. This analog approach was used in the tests reported in Chapter 6.

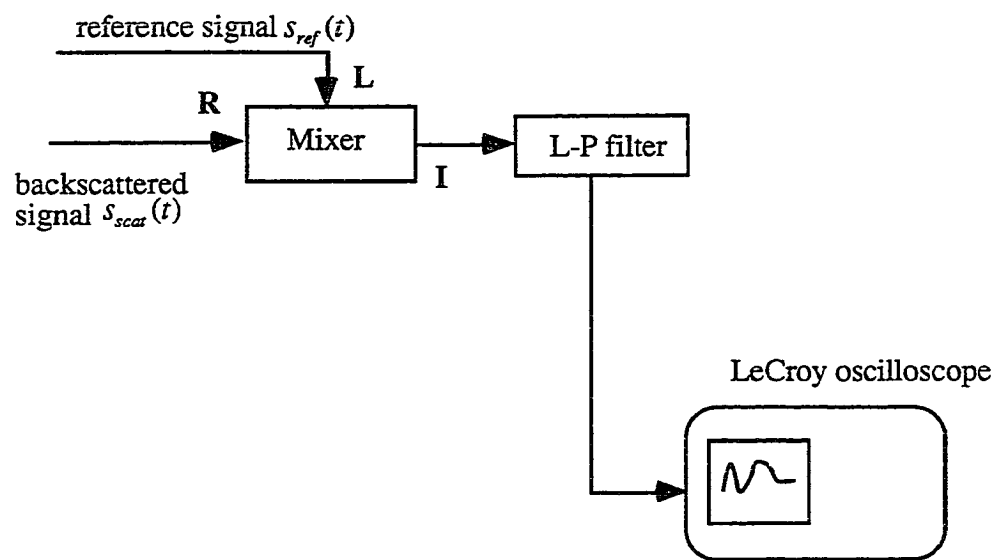


Fig. 2.6. Schematic of analog processing technique.

Appendix

The Calculation of Sphere Radius with a Plane Incident Wave

As discussed in section 2.1.1 , the particle velocity on the boundary is given by

$$u = \frac{i}{\rho_{in} c_{in}} \sum_{m=0}^{\infty} B_m \alpha_m(ka_0) P_m(\cos \vartheta) e^{i\omega t},$$

where $\alpha_m(x) = mj_{m-1}(x) - (m+1)j_{m+1}(x)$ and

$$B_m = p_0(-i)^m(2m+1) \left[\frac{1}{1+iD_m} \frac{h_m(ka_0)}{j_m(k_{in}a_0)} + \frac{j_m(ka_0)}{j_m(k_{in}a_0)} \right].$$

The particle displacements around $r = a_0$ can be obtained by integrating u' with respect to t ,

$$\begin{aligned} \Delta r = \int u(t) dt &= \frac{1}{\rho_{in} c_{in} \omega} \sum_{m=0}^{\infty} B_m \alpha_m(k_{in}a_0) P_m(\cos \vartheta) e^{i\omega t} = \\ &= \frac{p_0(-i)^m}{\rho_{in} c_{in} \omega} e^{i\omega t} \sum_{m=0}^{\infty} \left[\frac{p_0(-i)^m}{\rho_{in} c_{in} \omega} e^{i\omega t} \sum_{m=0}^{\infty} \left[\frac{1}{1+iD_m} \frac{h_m(ka_0)}{j_m(k' a_0)} + j_m(ka) \right] \right]. \end{aligned}$$

Δr can be further written compactly by

$$\Delta r = \sum_{m=0}^{\infty} \xi_m P_m(\cos \vartheta) e^{i\omega t}$$

where

$$\xi_m = \frac{p_0(-i)^m}{\rho_{in} c_{in} \omega} \left[\frac{1}{1+iD_m} \frac{h_m(ka_0)}{j_m(k_{in}a_0)} + \frac{j_m(ka_0)}{j_m(k_{in}a_0)} \right].$$

Chapter 3 Experimental Apparatus

We choose two parameters among our priority list in designing the apparatus- **sensitivity** and **flexibility**. **Sensitivity**-To probe cavitation in a size of microns, the detecting system has to be very sensitive to the mechanical inhomogeneity in the liquid hosting the cavitation. Hence, the sensitivity has the highest priority. As we discussed in chapter 2, the scattering strength for small object is approximately proportional to the frequency of incident wave. The higher frequency the system uses, the smaller the object that the system is capable of detecting. On the other hand, the attenuation of acoustic wave in the liquid increases dramatically as the frequency increases. A proper working frequency for the detecting system can optimize the attenuation and keep the sensitivity. For that reason, Roos and Apfel chose the frequency of 30 MHz as the working frequency to study the acoustic scattering from fluid and solid micro particles [Roos, *et al.*, 1988]. Roy and Madenshetty also used a 30 MHz acoustic signal to measure cavitation thresholds[Roy, *et al.*, 1990]. Some of our electronic design is based on their work.

Flexibility-The system needs to have a certain flexibility so that we can easily change host liquid and cavitating transducer, which will be described later. Several transducers of different frequency were used to function as cavitating transducers. Each of them has a different focal length. The mechanical part of the system must be flexible so that both cavitating transducer and probing transducer can work properly.

Figure 3.1 is a block diagram of the system which is described in detail in Section 3.1.

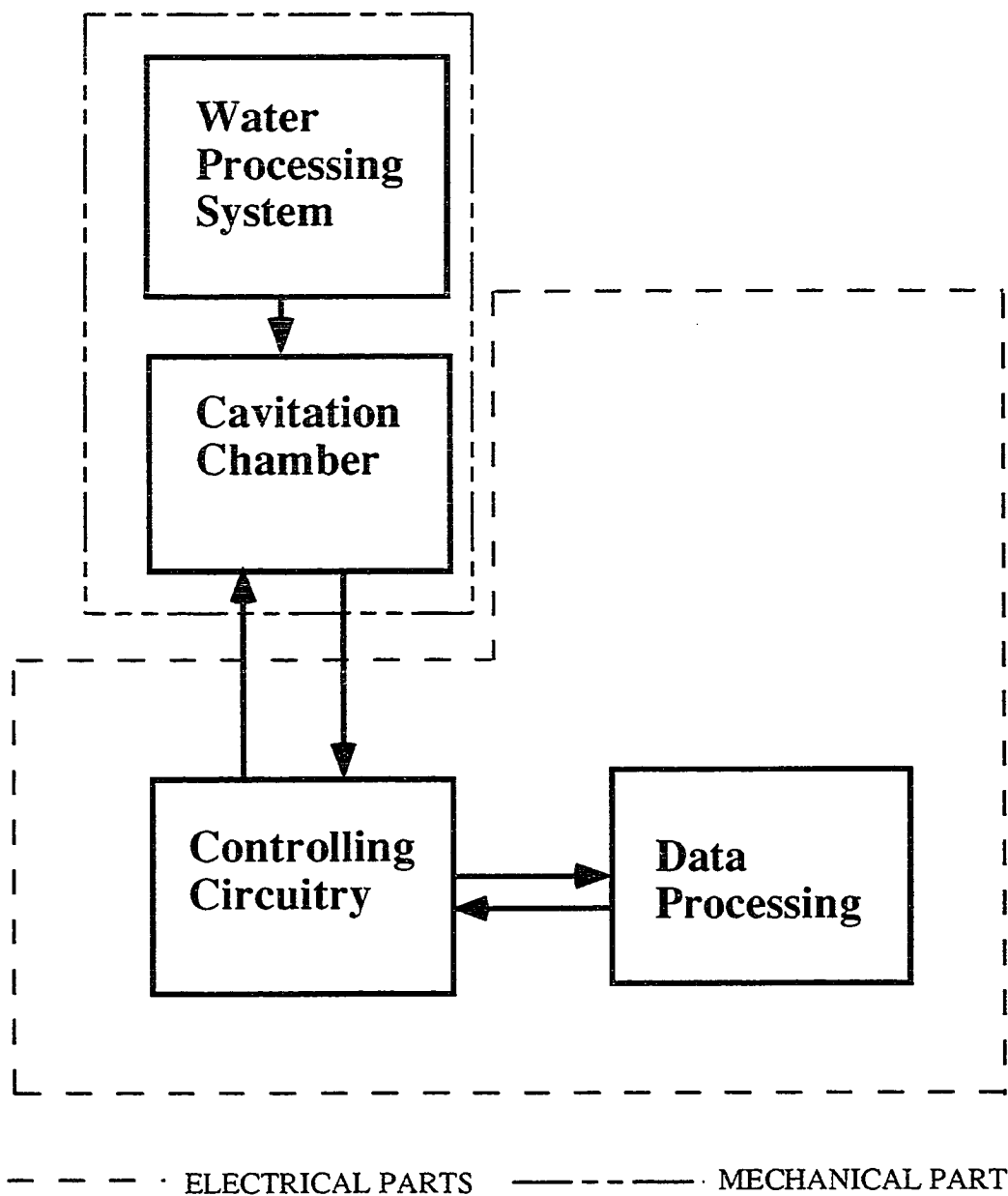


Fig. 3.1 Mechanical part consists of water processing system and cavitation chamber. Each of them will be detailed in the following.

3.1 Mechanical Part

3.1.1 Water Processing System

The water processing system supplies the water in which the transducers are immersed. For cavitation threshold measurement, the "cleanliness" of the water is extremely important.

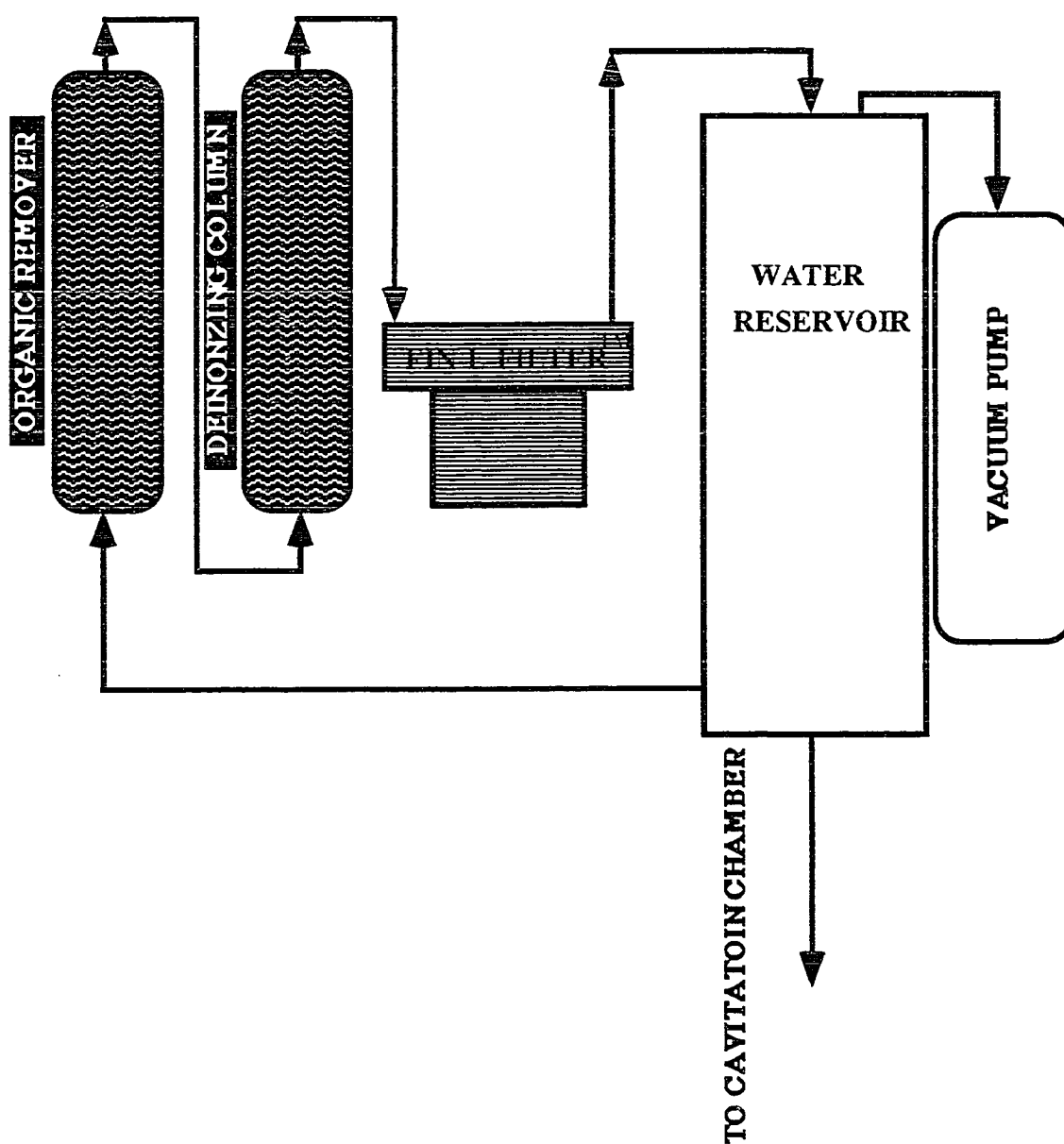


Figure 3.2. Diagram of water processing system

In Fig 3.2, the water reservoir is made of Plexiglas. The reservoir has the volume of 9.5 cm x 30 cm x 45 cm, and 95% of it is filled with distilled water. A centrifugal pump is used to circulate the water. The water coming from the pump passes through a organic

remover(Barnstead D8904), a deionizing column(Barnstead D8902, Barnstead Thermolyne Co., Dubuque, IA) and a filter Fin-L-Filter™ with a mean pore of $0.2 \mu\text{m}$, then circulates to the reservoir. Those components are connected by Nalgene 8000 tubing (inner diameter 0.25" and wall thickness 0.125"). A vacuum pump (Model 1405, Welch Scientific Co., Skokie, IL) is used to degas the water whenever necessary.

Before pumping the water into the water reservoir, we obtain the distilled water through a Corning™ water still (MEGA-PURE™ one liter water still, Corning Glass Works). Then we use Barnstead D2791 NANOpure unit to process the water for organic removal and demineralize it to a resistance of $16\text{M}\Omega/\text{cm}$.

3.1.2 Cavitation Chamber and Transducers

Fig 3.3 is the schematic of cavitation chamber. The water tank is made of Plexiglas of 0.5" thickness and measures 42 cm x 16 cm x 23 cm. It is secured by 1/4"x20 stainless steel screws and sealed with vacuum grease. The cavitating transducer is positioned with an angle of 90° toward the probing transducer. The water tank is partitioned into two units. One with volume of 26.5 cm x 16 cm x 23 cm contains clean water, where both the cavitating transducer and probing transducer are immersed. Another unit with the volume of 15.5 cm x 16 cm x 23 cm contains tap water which absorbs the acoustic energy radiated by the cavitating transducer. The two units are connected through an acoustically transparent membrane made from very thin stainless foil. The reflection coefficient of the foil is about 5% under normal working condition for the 2.46 MHz transducer.

Fig 3.4 is the top view of the cavitation chamber. Both transducers are mounted on 3-D micropositioners (Edmund Scientific Co., Barrington, NJ), which have a resolution of 0.01 mm in x, y and z directions. The probing transducer(Harrisonic Lab., Stamford, CT) is narrow band and focused. It has a center frequency of 30 MHz, 3 dB bandwidth of 2.5 MHz, and focal length of 3 cm. The cavitating transducer is also a focused transducer. To

align the two transducers, we use a brass bead target of 1.3 mm diameter adhered to a tungsten string of 50 μm diameter stretched across a Plexiglas fork. The target is also mounted on one of the 3-D micropositioners. The 30 MHz wavelength is 50 μm which is the same size as the tungsten diameter. The wavelength of 2.5 MHz acoustic wave is 600 μm , and the 4.3 MHz is 380 μm 600, all in water. So, the scattering from the tungsten wire is Rayleigh scattering for the cavitating signals.

Fig 3.6 is the configuration of the cavitating transducer. The transducer consists of a PZT spherical shell segment of 1.5" diameter (Stanvely Sensors, East Hartford, CT) and a home-made holder. One side of the PZT disk has the negative electrode and touches the aluminum frame through which the positive electrode of the PZT disk can form an electrical loop. An O-ring is inserted between the PZT transducer and the Plexiglas, and is gently pressed through the Plexiglas frame to ensure no water leakage from the front surface of the transducer. The Plexiglas frame is connected to the aluminum frame and secured by stainless screws. A rubber pad is inserted between the aluminum frame and the cover to make sure that the transducer is air backed. The positive electrode is connected to a female BNC connector. The PZT transducer is easily removed and replaced. Three different transducers of resonance frequencies of 0.513 MHz, 2.46 MHz and 4.3 MHz have been used. Acoustic characteristics of these transducers will be detailed later in this thesis.

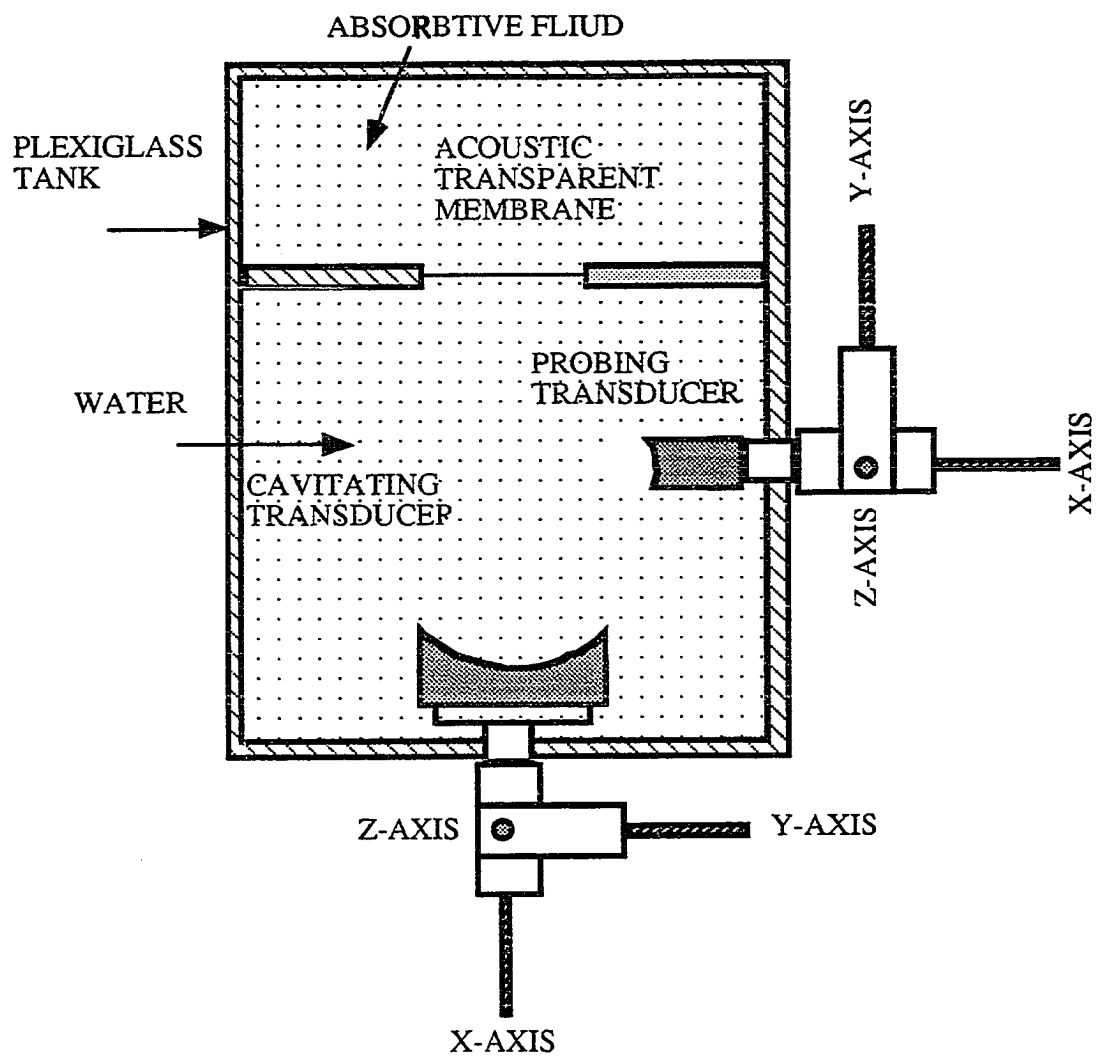


Fig. 3.3 Top view of the cavitation chamber

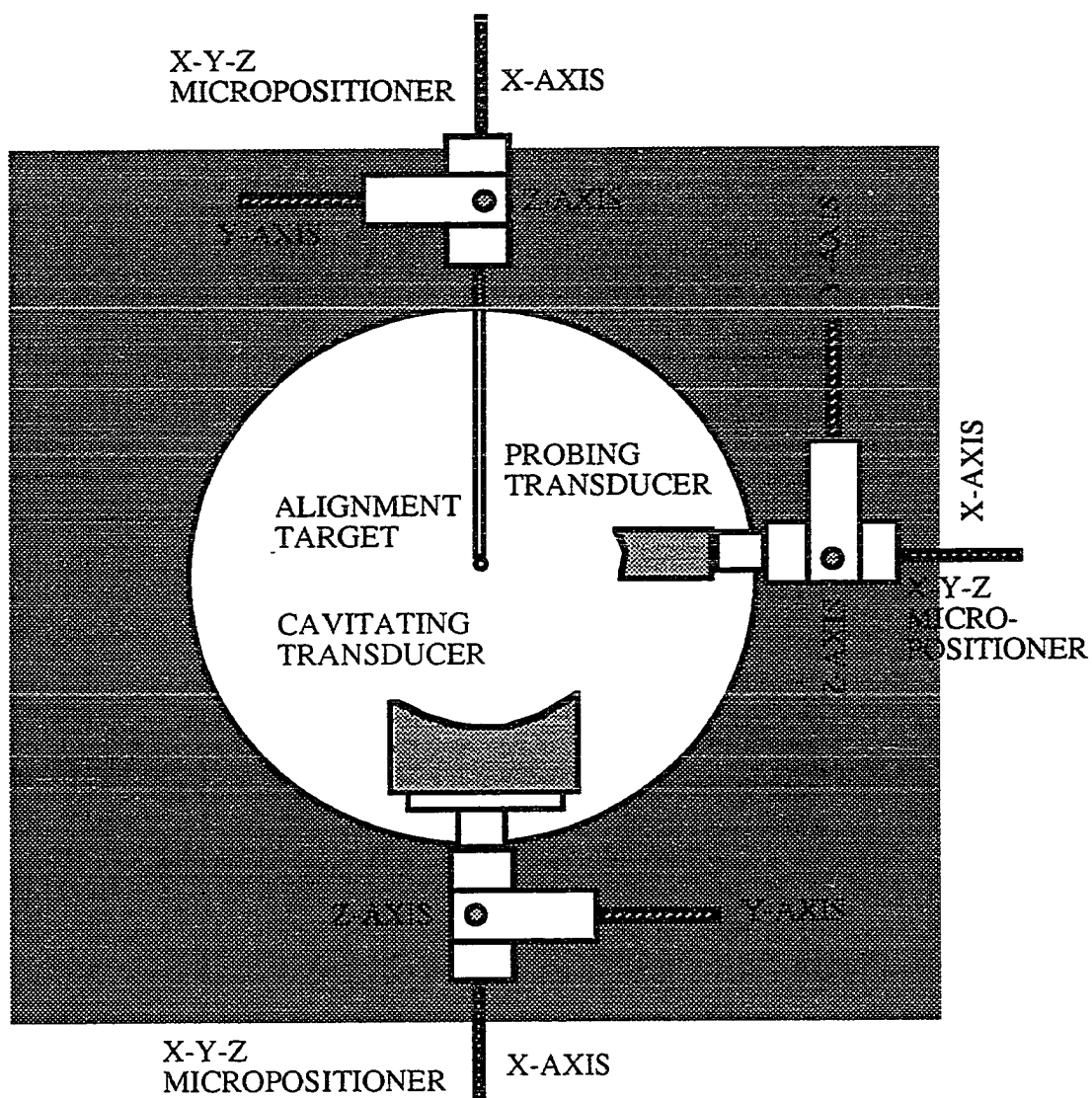


Fig. 3.4 Top view of cavitation chamber with the supporting table.

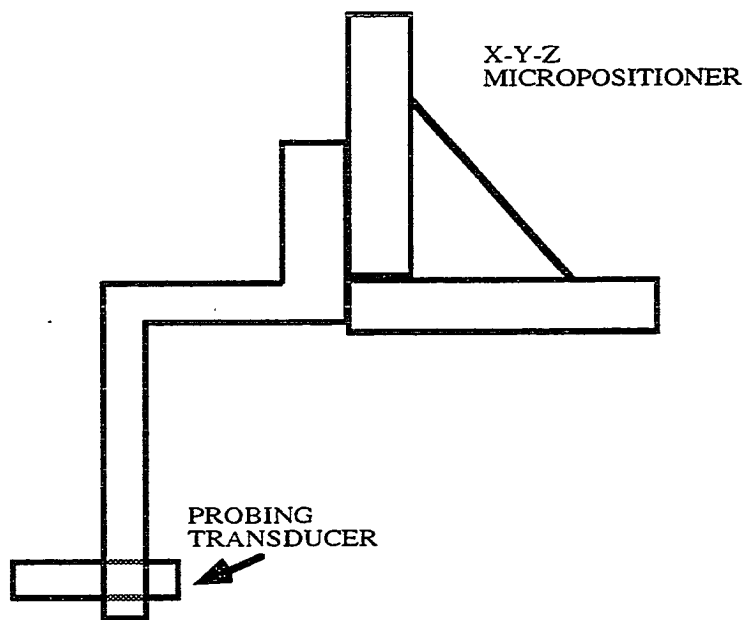


Fig. 3.5a. Side view of the probing transducer and its supporting X-Y-Z micropositioner.

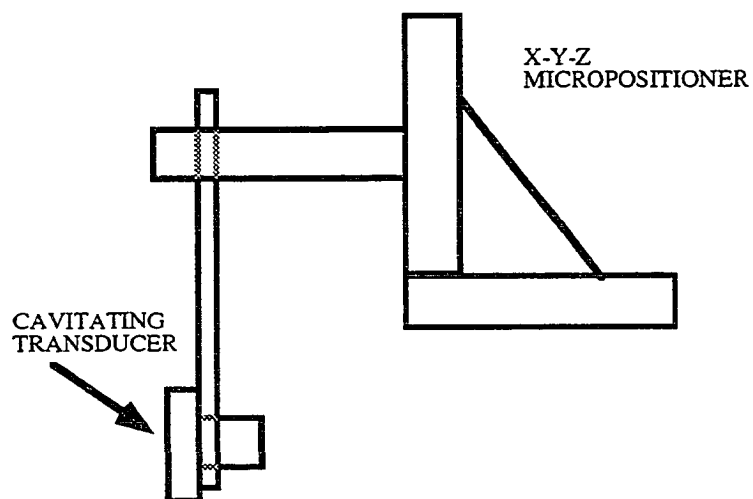


Fig. 3.5b. Side view of the cavitating transducer and its supporting X-Y-Z micropositioner.

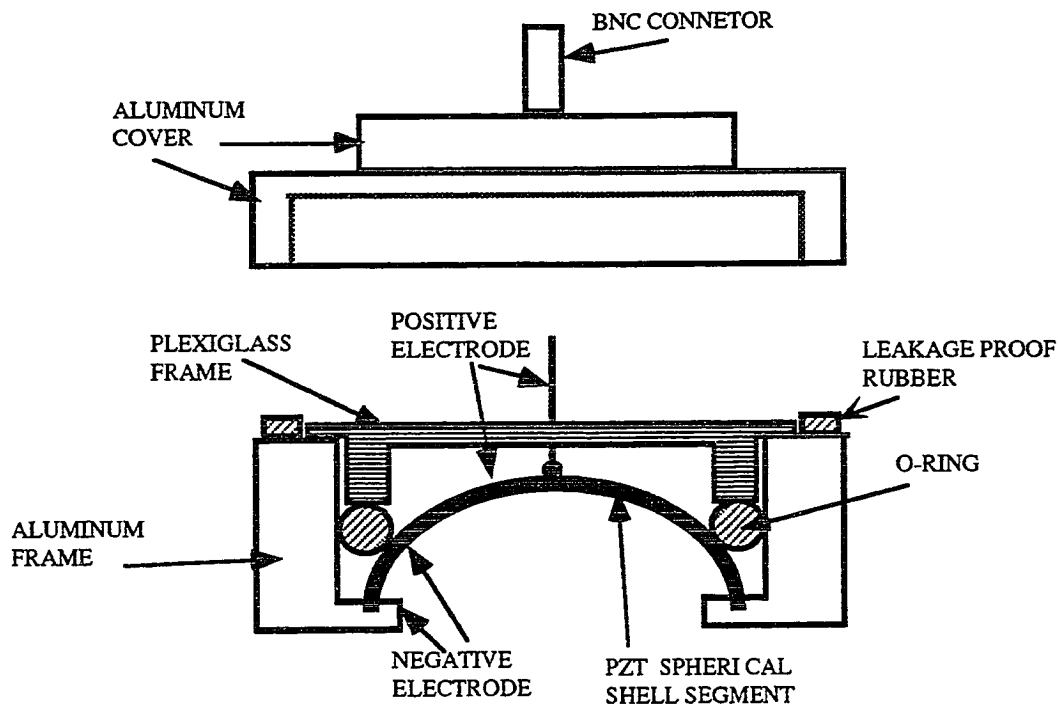


Fig. 3.6. The configuration of the cavitating transducer.

3.2 Electrical Parts

In this section, we describe the electrical system for the generation and detection of microcavitation. Fig. 3.7 is a functional block diagram of the system. As mentioned earlier, two PZT transducers are deployed here and are confocally aligned. One is used to cause cavitation in the confocal region and the other one is used to detect the cavitation. Through the following, we will refer the former as the cavitating transducer and the latter as the probing transducer, and the circuit serving the driving transducer as the driving module and the one serving the probing transducer as the receiving module circuit. The triggering TTL Source (with some modifications based on the unit built by Dr. Ron Roy.) outputs 4 TTL pulses, CAVITATION PULSE, TRANSMITTING PULSE, PROBING PULSE

AND RECEIVING PULSE. All of these pulses have the same pulse repetition frequency (PRF) of 1 kHz. These TTL pulses trigger the driving, probing and receiving module. The TTL pulses are synchronized. There are delays among the TTLs, which are adjustable. The tone burst used for the probing module is longer than the tone burst for the driving module so that the probing signal will not miss the cavitation caused by the driving signal in the confocal region by properly adjusting the delays. The adjustments are determined by the sound speed in the host fluid and the time for the signals from both transducers to travel from the transducer surfaces to the confocal region.

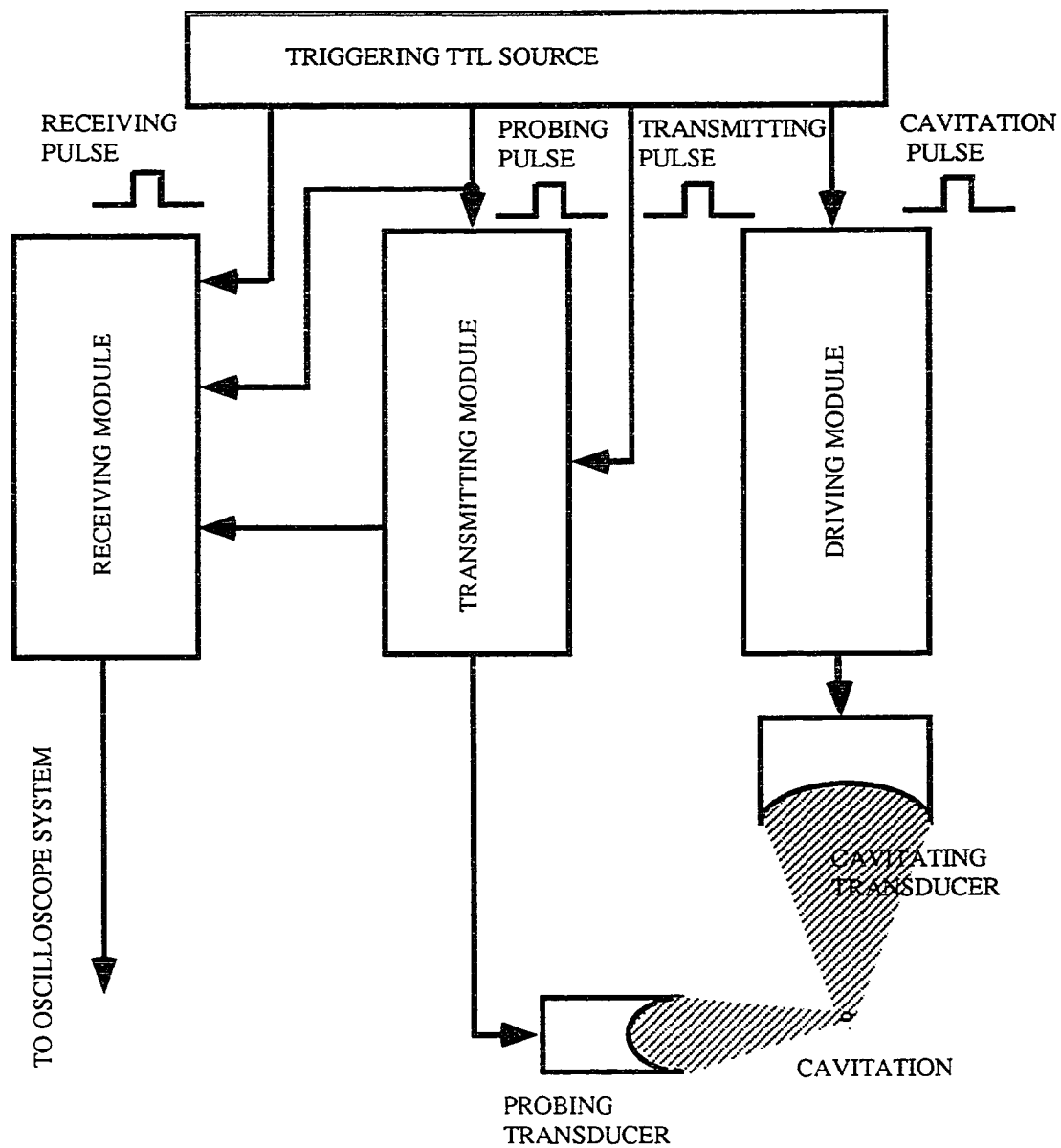
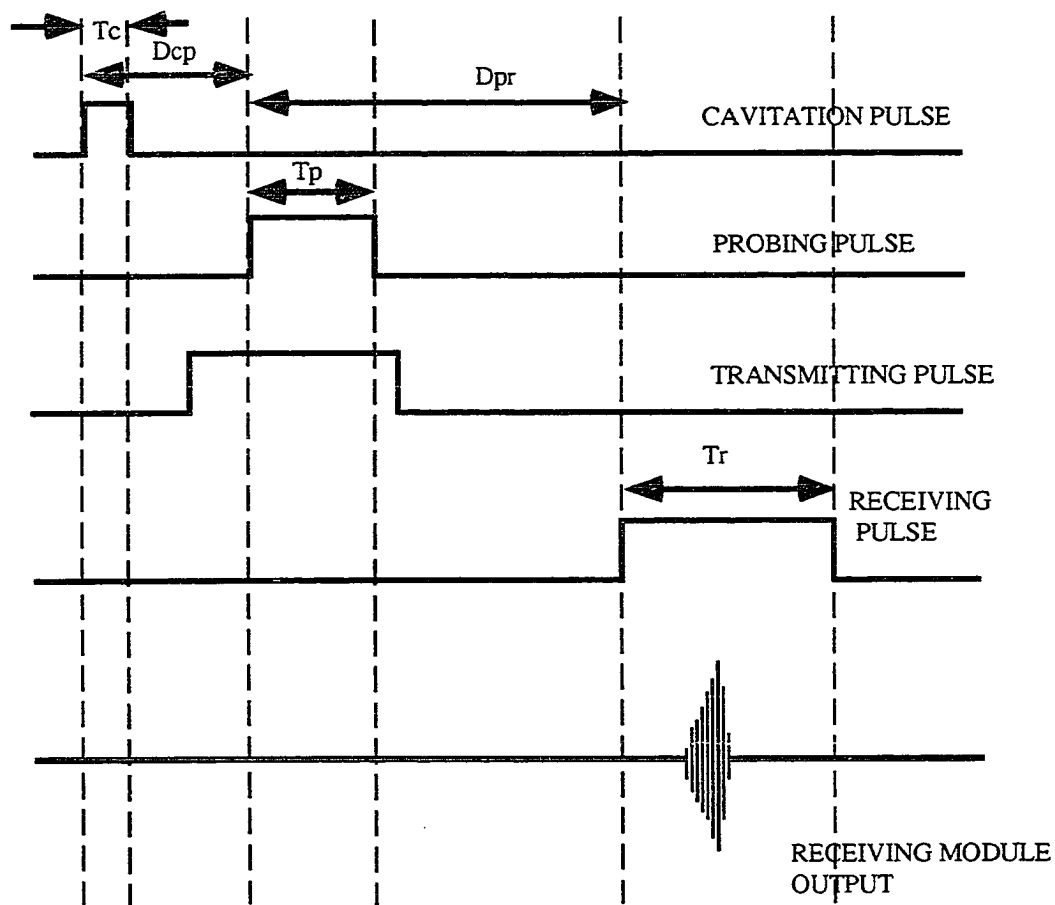


Figure. 3.7 Block diagram of the electrical circuitry



T_c = the width of the CAVITATION PULSE, T_p = the width of the PROBING PULSE, T_r = the width of the RECEIVING PULSE, D_{cp} = the delay between the CAVITATION PULSE and the PROBING PULSE, D_{pr} = the delay between the PROBING PULSE and the RECEIVING PULSE.

Figure 3.8 Timing diagram of the TTLs from the TRIGGERING TTL SOURCE.

As shown in Fig 3.7, a probing pulse is fed to the probing module after a delay D_{cp} , which is relative to the cavitation pulse. Consequently, both the cavitating signal and probing signal can reach the confocal region at the same time. D_{cp} is determined by

$$D_{cp} = \frac{f_p}{c} - \frac{f_d}{c}, \quad 3.1$$

where f_p is the focal length of the probing transducer, f_d the focal length of the cavitating transducer and c the sound speed of the host fluid. After a delay of D_{pr} , the receiving pulse triggers the receiving module, the SPST switches are in the open state, and the backscattered signal from the cavitation will be amplified. The delay D_{pr} is determined by

$$D_{pr} = 2 \frac{f_p}{c} - \frac{f_d}{c}, \quad 3.2$$

the factor 2 is because of the round trip of the probing wave from the transducer surface to the confocal region.

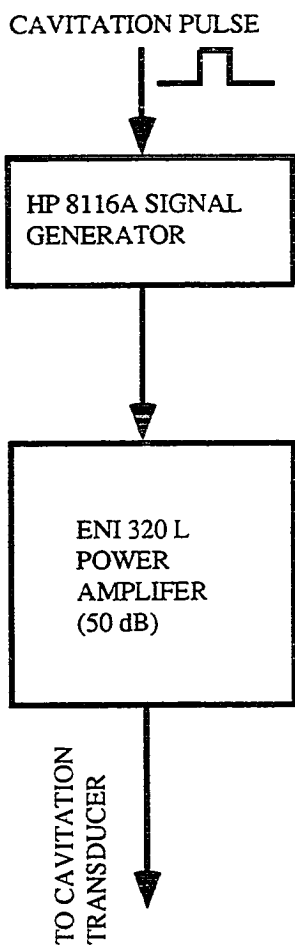


Figure 3.9. The driving module

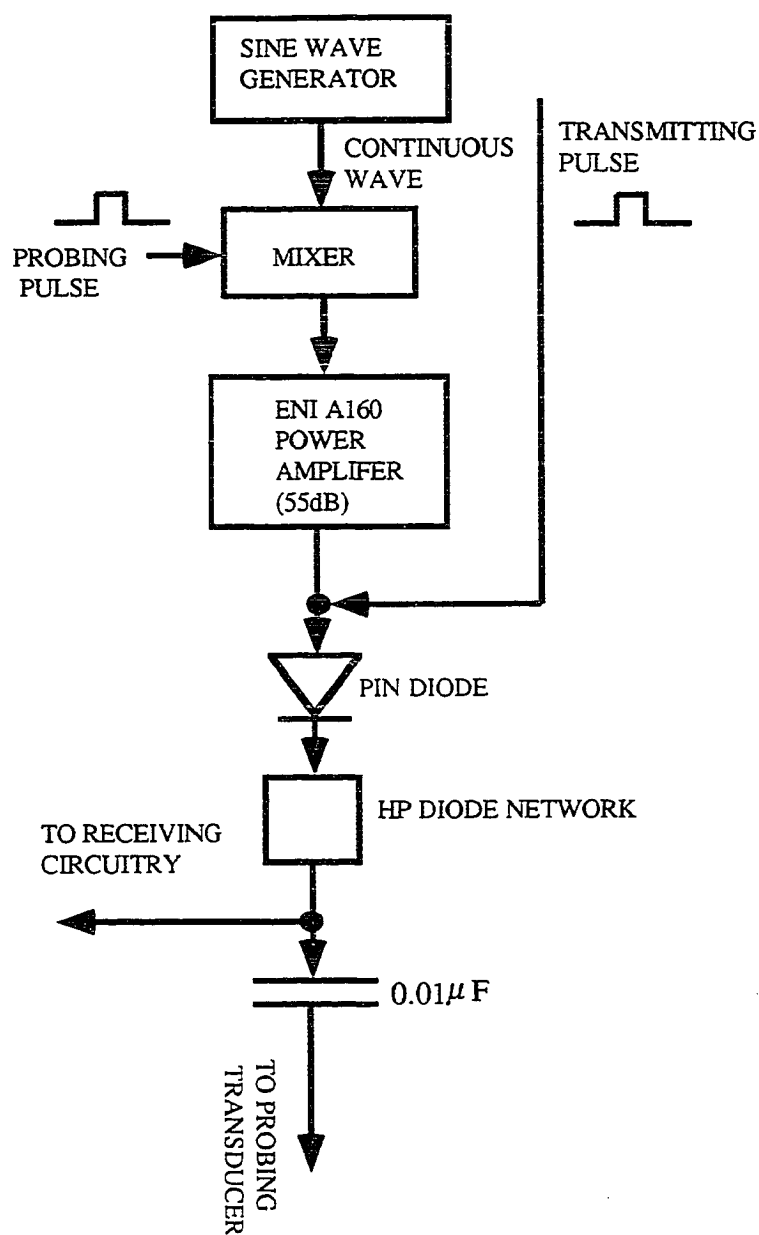


Figure 3.10. The transmitting module

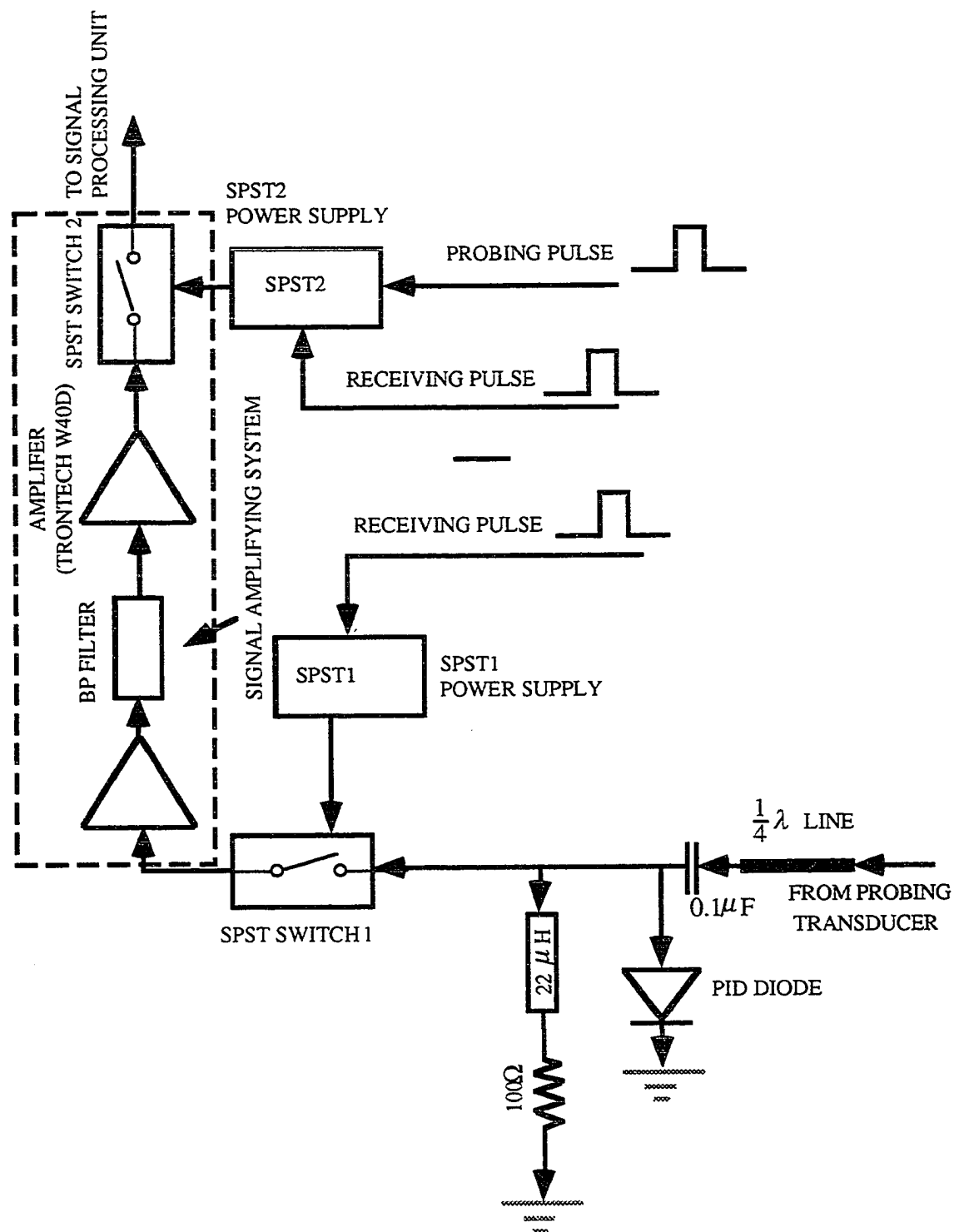


Figure. 3.11 Receiving module

In Fig. 3.9, the CAVITATION PULSE externally triggers the Hewlett-Packard signal generator (model 8116A) so that a tone burst from the signal generator will be sent to the power amplifier (model ENI 320L, 50 dB gain). Three different tone bursts of 5, 10 and 20 cycles are used in the experiments. The peak-peak voltage of the power amplifier output can be as high as 110 volts with 50 Ω termination, and this corresponds to the forward electrical power of 30 watts. The duty cycle of the system is about 1% (for different center frequencies and tone burst cycles, the duty cycles are a little bit different.), so the average forward electrical power is about 1.2 watts.

Fig. 3.10 shows the circuit of the probing module. The GENERAL RADIO oscillator outputs a continuous sine wave, precisely tuned to the frequency of 29.4 MHz, to the L-port of the 0° two way splitter (Model ZSC-2-2B, Mini-Circuits Lab., Brooklyn, NY). One output of the splitter is connected to the R-port of the double- balanced mixer (Model ZAD-1HB, Mini-Circuits Lab). After a delay of Dcp with the cavitation pulse as the reference, a probing pulse is fed into the I-port of the mixer. The output of the mixer L-port is a very fine tone burst, whose duration is the same as the probing pulse and center frequency is 29.4 MHz. The tone burst is amplified by the high gain power amplifier ENI Model 320L (55 dB gain, ENI Inc., Rochester, NY). The maximum peak-peak output is 45 volts terminated with 50 Ω . The amplified tone burst is fed to a forward biased PIN diode. When a transmitting pulse effectively shorts the diodes, the input impedance of the receiving circuitry is very high compared with the transducer impedance, and consequently, most electrical power from the ENI power amplifier will be applied to the probing transducer. A more detailed analysis is given by the following analysis. The input impedance of a transmission line of length l is

$$Z_i = Z_0 \frac{Z_L + jZ_0 \tan(kl)}{Z_0 + jZ_L \tan(kl)} \quad 3.3$$

where Z_i is the input impedance, Z_L the termination impedance, Z_0 the characteristic impedance of the transmission line and k the wave number. Typical PIN diode resistance is 0.75 when it is on and 40 K Ω when it is off. For quarter wavelength transmission line, the input impedance Z_i becomes

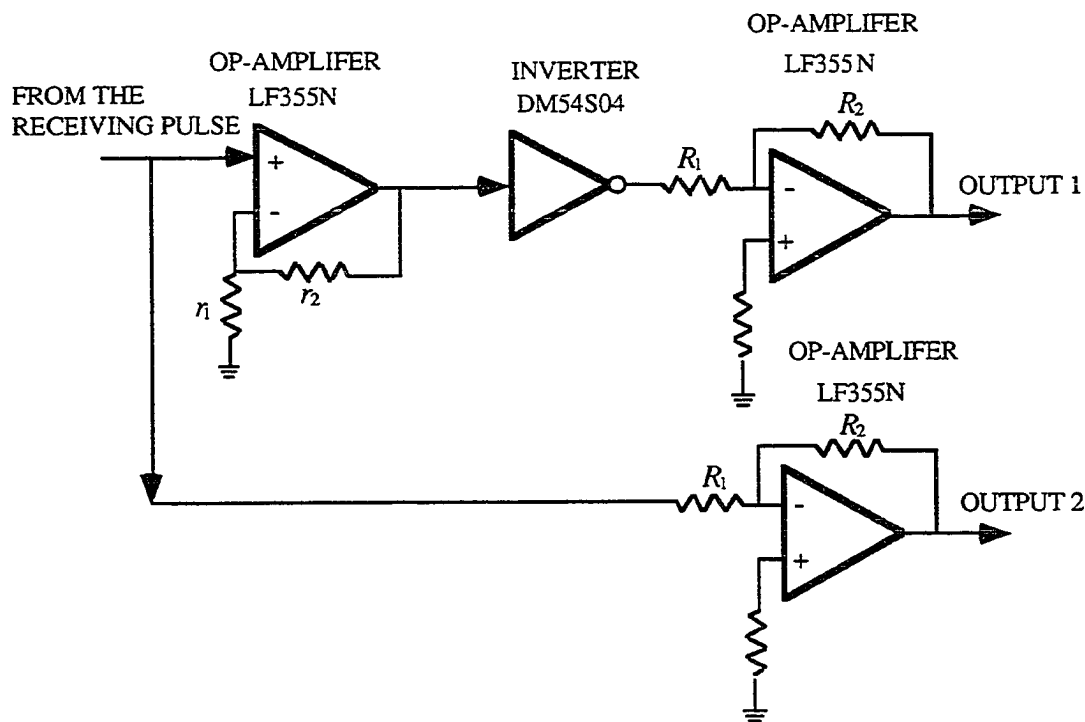
$$Z_i = \frac{Z_0^2}{Z_L} = \frac{50^2}{0.75} = 3.13 \text{ K}\Omega,$$

which is much higher than the transducer impedance of around 50 Ω .

When the transmitting pulse is in the "OFF" state, the diodes block the electrical noise leakage from the ENI power amplifier from passing through the receiving circuitry. The probing signal is backscattered by the cavitation generated in the confocal zone by the cavitating transducer, and the backscattered signal is received by the probing transducer. The backscattered signal will go to the receiving circuitry because the diodes are backward biased.

Fig. 3.11 shows the principles of the receiving circuitry. When the backscattered signal reaches the input-port of SPST 1 SWITCH (single-pole, single-throw, Mini-Circuits Lab., Brooklyn, NY), a receiving pulse triggers the SPST power supplier, and the SPST switch is in the "OPEN" state, which will let the backscattered signal pass through to the signal amplifying system. The amplifying system consists of a very low noise preamplifier (Model CLC 100, 20 dB gain, Colinear Co., Fort Collins, CO), a band-pass filter (3 dB bandwidth of 2.5 MHz, insertion loss of 8.0 dB. Wavetek RF Products Inc., Indianapolis, IN) and a high gain amplifier of 65 dB (Model W40D, Trontech Inc., Neptune, NJ). So, the net gain of the amplifying system is 79 dB. Experiments have shown that the whole detecting system is capable of sensing the existence of 0.25 μm polystyrene particle in its focal region when the system is operated with its highest sensitivity. Both SPSTs are high speed (speed time 3 ~ 7 ns) and high isolation (> 60 dB) GaAs switches. The SPST1 switch blocks the leakage of the main bang which excites the

probing transducer. Because of the high gain of the amplifying system, the original SPST1 video leakage of 25 mV will be amplified to about 1.0 V (please note that the video leakage is a wide band pulse, the band pass filter will filter out most of its frequency components which are not around its center frequency of 30 MHz.). The SPST2 blocks the video leakage from SPST1 by properly arranging the power supply which controls the switch the " OPEN" and "OFF" states. Both switches have a 0.8 dB insertion loss. Consequently, the receiving circuitry has a net gain of 77.4 dB for a narrow band signal with the center frequency of 29.4 MHz.



$$r_1 = 15\Omega, r_2 = 20\text{k}\Omega, R_1 = 10\text{k}\Omega, R_2 = 50\text{k}\Omega$$

Fig. 3.12 Circuitry generating the controlling signal for the switch SPST1.

Fig. 3.12 shows the circuit diagram of the power supply for switch SPST1. The RECEIVING PULSE is split into two ways fed into two channels. One channel only has a operation amplifier DM 355N. Another channel consists of two operation amplifiers, DM 355N, and an inverter, DM54S04. The outputs of the two channels are shown in Fig. 3.13.

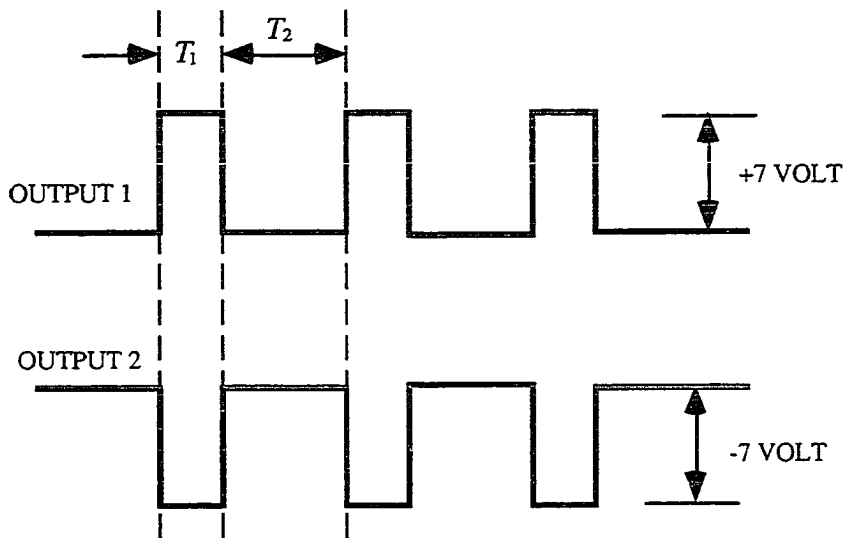
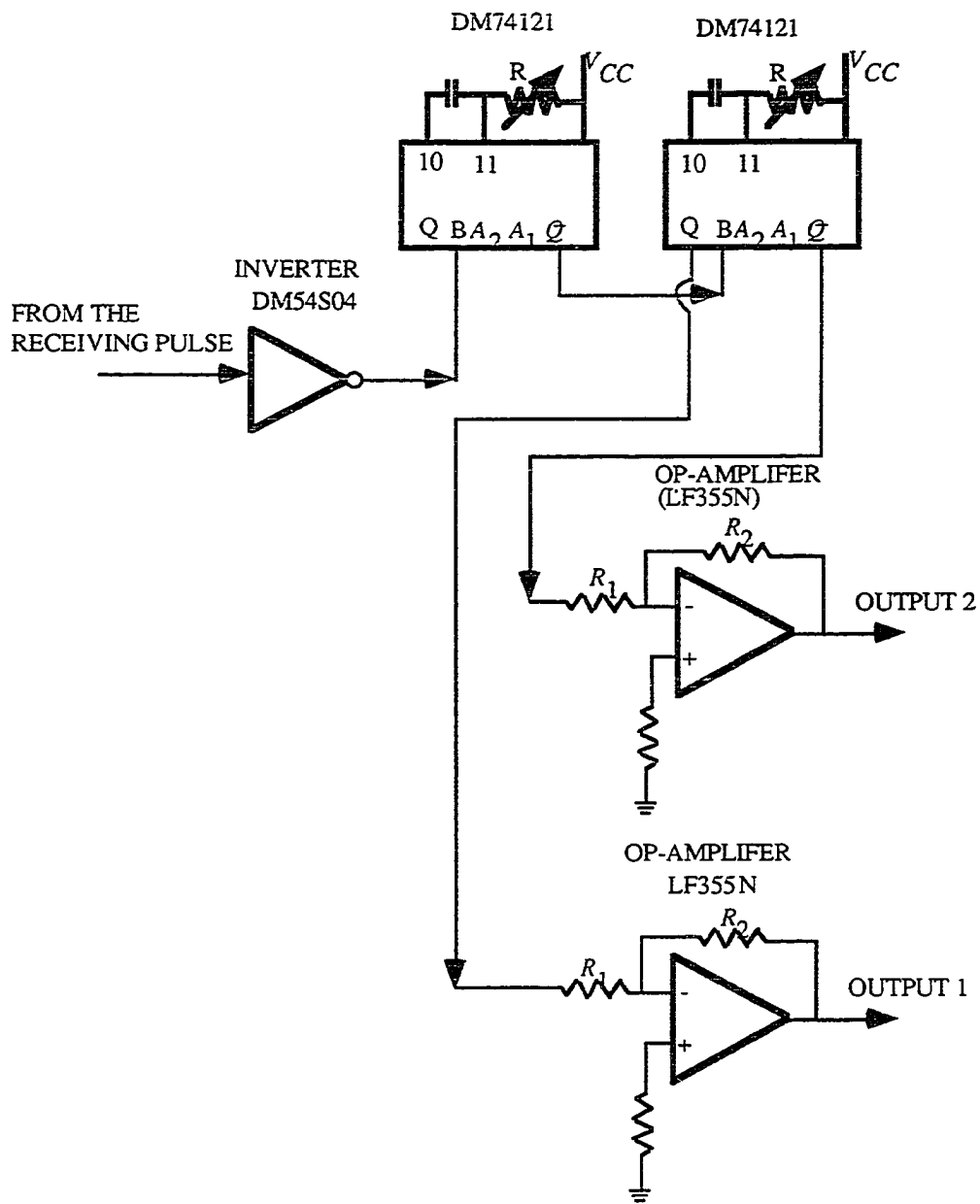


Figure 3.13. Waveform of the controlling signals for the switch SPST1.



$R_1=10 \text{ k}\Omega$, $R_2=50 \text{ k}\Omega$, $V_{cc}=5 \text{ V}$.

Fig. 3.14 Circuitry that generates the controlling signals for the switch SPST2.

Fig 3.14 is the circuit diagram of the power supply for SPST2. A TTL pulse from the RECEIVING PULSE is sent to the INVERTER DM54S04 (National Semiconductor

Co.), and its output is connected to the two seriesly combined monostable multivibrators (DM 54121, NATIONAL SEMICONDUCTOR Co.) which have complementary outputs. The pulse width of the output Q is essentially determined by the external timing components R and C, which can be expressed mathematically by

$$t_Q = 0.7RC, \quad 3.4$$

where R is the resistance in kilohms, C in picofarad and t_Q in nanoseconds. By adjusting the external resistance R, we can change both the positive pulse width t_1 and the negative pulse width t_2 .

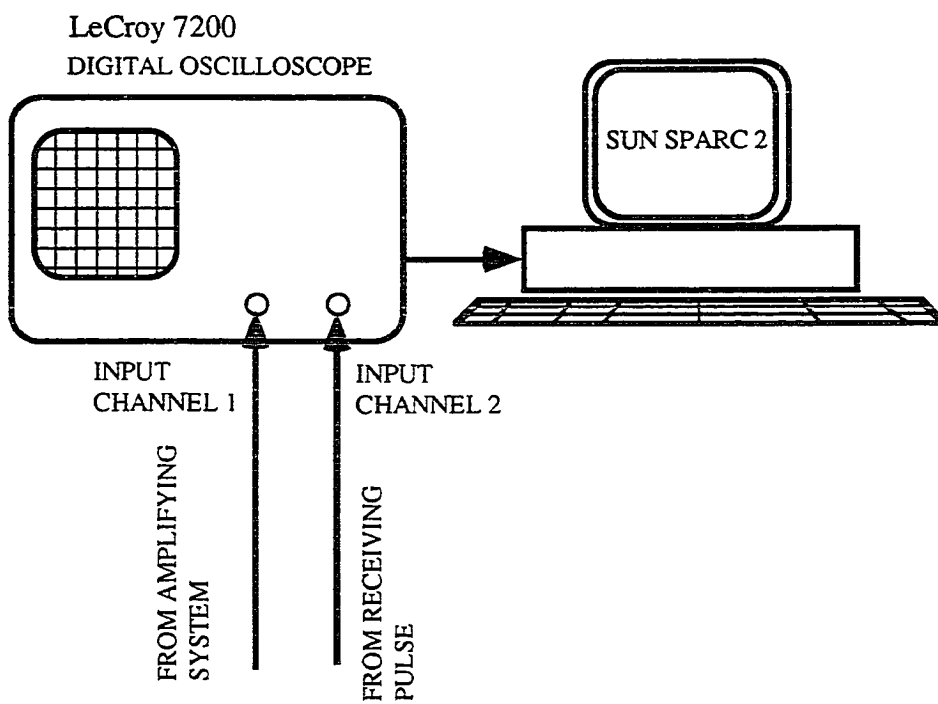


Fig. 3.15. The schematic for digital processing data .

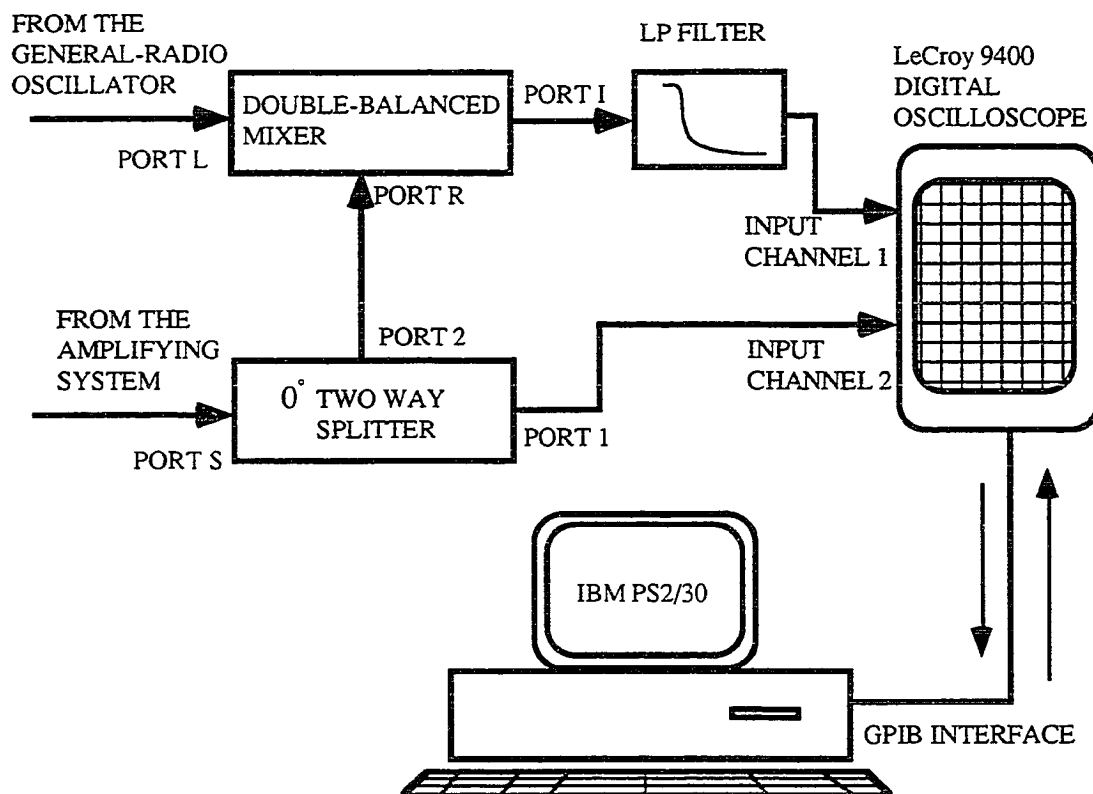


Fig. 3.16 Schematic for the analog method.

We have chosen two methods to process the backscattered signal: analog and digital. Fig. 3.15 shows the system for the digital method. The backscattered signal is digitized and recorded by the digital oscilloscope LeCroy 7200. The digitized data are transferred into the workstation SUN SPARC II for post processing, which will be discussed in the latter chapter. Fig. 3.16 is the schematic of the analog method. The backscattered signal is split into two paths, one of which is fed to the digital oscilloscope LeCroy 9400 and functions as a triggering source. Another is fed to the L-port of the double balanced mixer (Model ZAD-1HB, Mini-Circuit Lab.). The output of I-port is connected to the low-pass filter (Model NJ98, 3dB cut-off frequency = 8 MHz, TTE Inc.,

Los Angeles.). Here, the double balanced mixer functions as a multiplier; the I-port output is a product of the R-port input and L-port input.

Chapter 4 Transducer Calibration

To study transient microcavitation quantitatively, and to measure the cavitation threshold, we need to calibrate the transducers causing the cavitation. For transducers in the frequency range of diagnostic ultrasound applications, the force balance method is very convenient. In this chapter, we will concentrate on the force balance technique and two types of force balance targets: reflecting and absorbing.

4.1 Principles

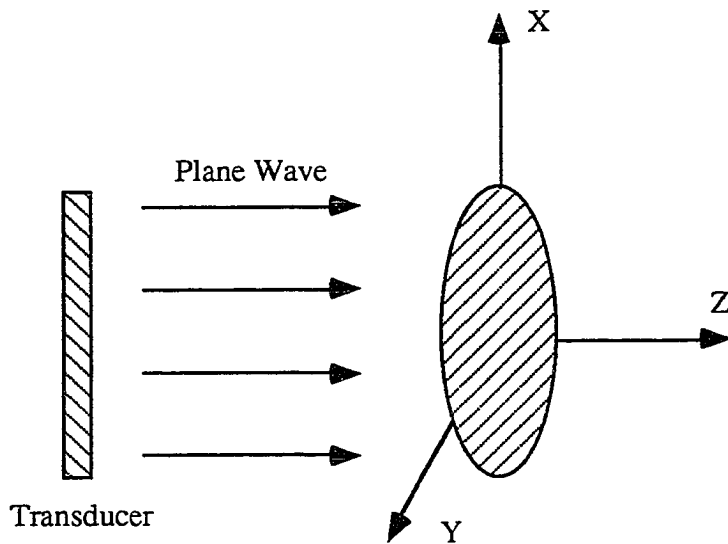


Figure 4.1. A simple model for the acoustic wave from the transducer

As shown in Fig. 4.1, we assume that the acoustic wave radiated by the transducer to be calibrated is a plane wave. The total acoustic power, w , associated with the transducer is [Madenshetty, S. I., 1989]

$$w = F_{hp} C, \quad 4.1$$

where c is sound speed, F_{hp} is the radiation force, and the subscript hp indicates the peak-to-peak output of the HP signal generator, which will be described later. The acoustic power is also equal to the integral of the acoustic intensity, $I(x, y, z)$, over the shaded area in x-y plane,

$$w = \int I_{hp}(x, y, z) ds = \int \frac{p_{hp}^2(x, y, z)}{\rho c} dx dy, \quad 4.2$$

where $p_{hp}(x, y, z)$ is the pressure distribution in space, and ρ the density of water. By assuming that the pressure distribution is symmetrical along the z-axis, we have

$$w = \int \frac{p_{hp}^2(y)}{2\rho c} 2\pi y dy, \quad 4.3$$

where $x=0$, and z = the focal length of the transducer to be calibrated.

To evaluate the integral in Eq 4.3, we use a hydrophone to scan the acoustic field along the y direction. The pressure and the hydrophone output, $V_{hp}(y)$, have a simple relationship as described below for a linear transducer,

$$p_{hp}(x, y, z) = k_{hp} V_{hp}(y) \quad 4.4$$

where k_{hp} is a constant to be determined, and the subscript indicates the level of HP output, which will be amplified and applied to the transducer. Consequently, Eq 4.3 becomes

$$w = k_{hp}^2 \frac{\pi}{\rho c} \int V_{hp}^2(y) y dy \quad 4.5$$

Solving the system equations of Eqs 4.1, and 4.5, we obtain k_{hp} to be

$$k_{hp} = c \sqrt{\frac{\rho F_{hp}}{\pi}} \frac{1}{\sqrt{\int V_{hp}^2(y) y dy}}. \quad 4.6$$

If the transducer operates in a tone burst mode with a duty cycle of D , Eq. 4.2 becomes

$$w = D \int I_{hp}(x, y, z) ds = D \int \frac{p_{hp}^2(x, y, z)}{\rho c} dx dy, \quad 4.7$$

and Eq. 4.6 becomes

$$k_{hp} = c \sqrt{\frac{\rho F_{hp}}{D \pi}} \frac{1}{\sqrt{\int V_{hp}^2(y) y dy}}. \quad 4.8$$

As shown later, we measure the transverse beam profile at a fixed output of the HP signal generator. For other outputs of the HP signal generator, we assume that the shape of the transverse beam profile does not change. Consequently, the transverse beam profile, $V(y)$, at the focal distance for any HP output, will be simply

$$V_{hp}(y) = \frac{V_{hp}(0)}{V_{ref}(0)} V_{ref}(y), \quad 4.9$$

where $V_{ref}(y)$ is the measured transverse beam profile, ref represents the value of the HP output. Eq. 4.8 becomes

$$k_{hp} = \frac{V_{ref}(0)}{V_{hp}(0)} c \sqrt{\frac{\rho F_{hp}}{\pi D \int V_{ref}^2(y) y dy}}, \quad 4.10$$

The peak acoustic pressure in the focal zone is

$$p_{hp}(0) = k_{hp} V_{hp}(0) = c V_{ref}(0) \sqrt{\frac{\rho F_{hp}}{\pi D \int V_{ref}^2(y) y dy}}. \quad 4.11$$

For numerical evaluation, the corresponding integral imbedded in Eq. 4.11 can be approximated by

$$IN_{ref} = \left(\frac{V_{ref}[i-1] + V_{ref}[i]}{2} \right)^2 \left(\frac{y[i-1] + y[i]}{2} \right) (y[i] - y[i-1]), \quad 4.12$$

where IN_{ref} is an intermediate parameter to simplify the expression Eq. 4.12.

4.2 Experimental Setup

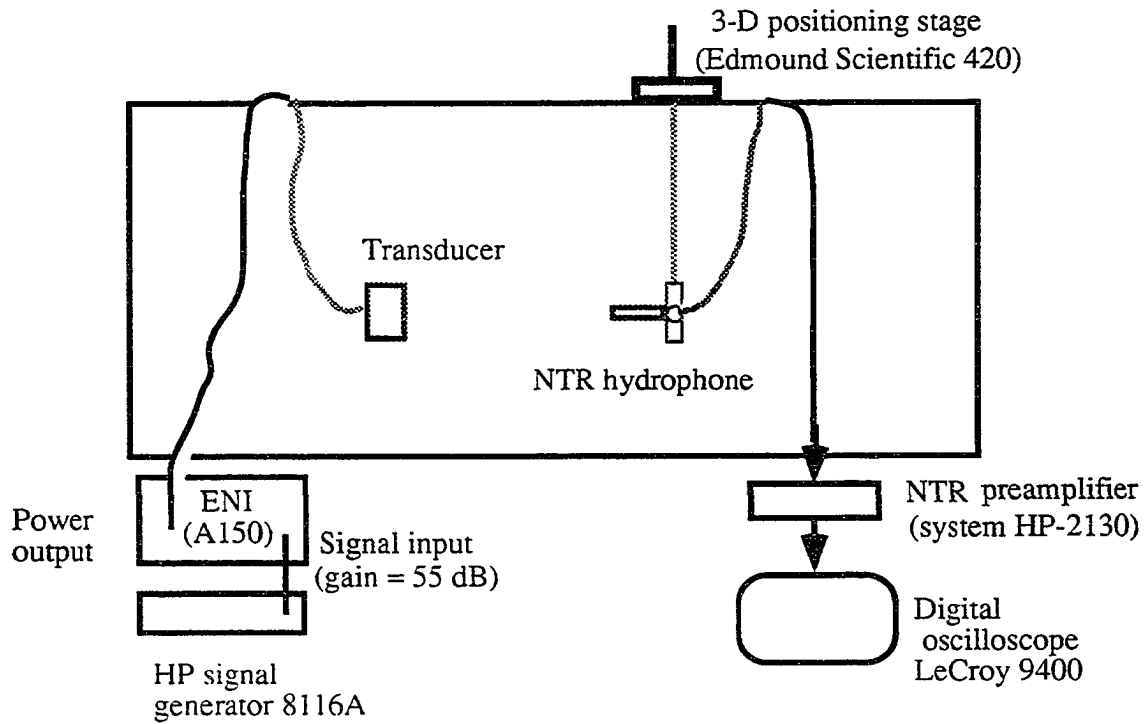


Figure 4.2. Setup for measuring the transverse pressure distribution of the transducer

Fig 4.2 is the schematic of the experimental setup. A power amplifier (ENI A150, ENI Inc., Rochester, NY) amplifies the output of a HP signal generator (model HP 8116A) and drives the transducer to be calibrated. The output of the HP signal generator is a tone burst with a center frequency equal to that of the transducer resonance frequency, and has a pulse repetition frequency (PRF) of 1 kHz. A hydrophone (NTR, Seattle, WA) is used to measure the relative acoustic pressure distribution along the y axis, or transverse beam profile. We measure the peak-to-peak amplitude of the preamplifier output by an oscilloscope (LeCroy 9400, LeCroy Co., Chestnut, NY) terminated with $50\ \Omega$. The transducer, hydrophone, and the target are immersed in distilled water. Both the hydrophone and the target are mounted on a 3-dimensional positioning stage with $20\ \mu\text{m}$ resolution (Edmund Scientific Co., Barrington NJ). The hydrophone is mounted in such way that it can also be rotated in any direction.

We calibrated the peak acoustic pressure in the focal region versus the output of the HP signal generator instead of the pressure versus the power amplifier output, because the same HP signal generator, power amplifier and transducer will be deployed for the cavitation experiment. Because of the high acoustic pressure and the limited dynamic range of the hydrophone, the output of the hydrophone becomes saturated when the HP output is over a certain value. To solve this problem, first we measure NTR output in the focal region when the HP output is below that value. Then, moving the NTR off the focal region along z-direction, we measure the NTR output when the HP output is over that value. The latter data will match the previous results by multiplying by a proper constant which is the ratio of the on-axis to off-axis pressures. Figs. 4.3 to 4.11 show the pressure distributions of the transducers at 516 KHz and 2.46 MHz, and the NTR output versus HP outputs. In Figs. 4.4, 4.5, 4.7, 4.10 and 4.11, when the HP output is set to zero, the fitted values converge to the acoustic noise levels which are not zero.

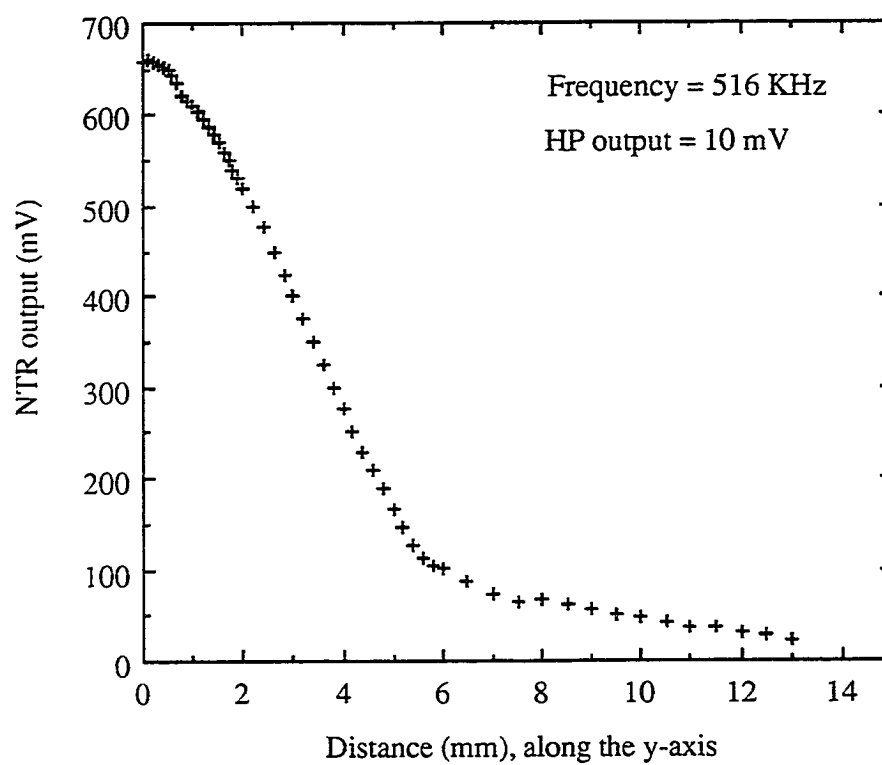


Figure 4.3. The transverse beam profile of the transducer at 516 KHz.

HP output = 10 mV.

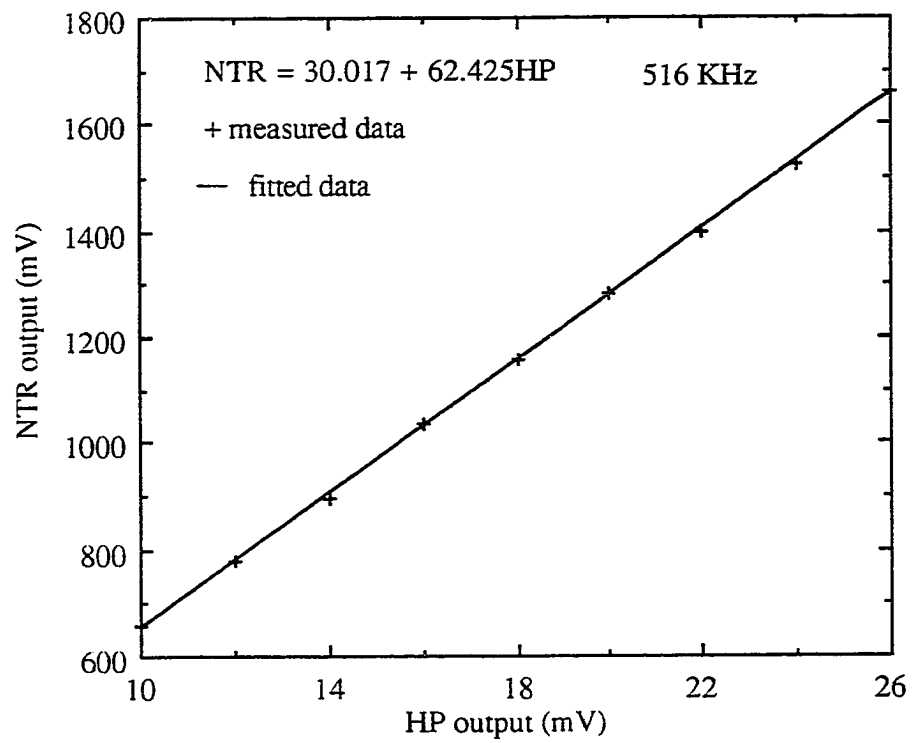


Figure 4.4. Hydrophone output versus HP output.

The data is obtained in the focal region

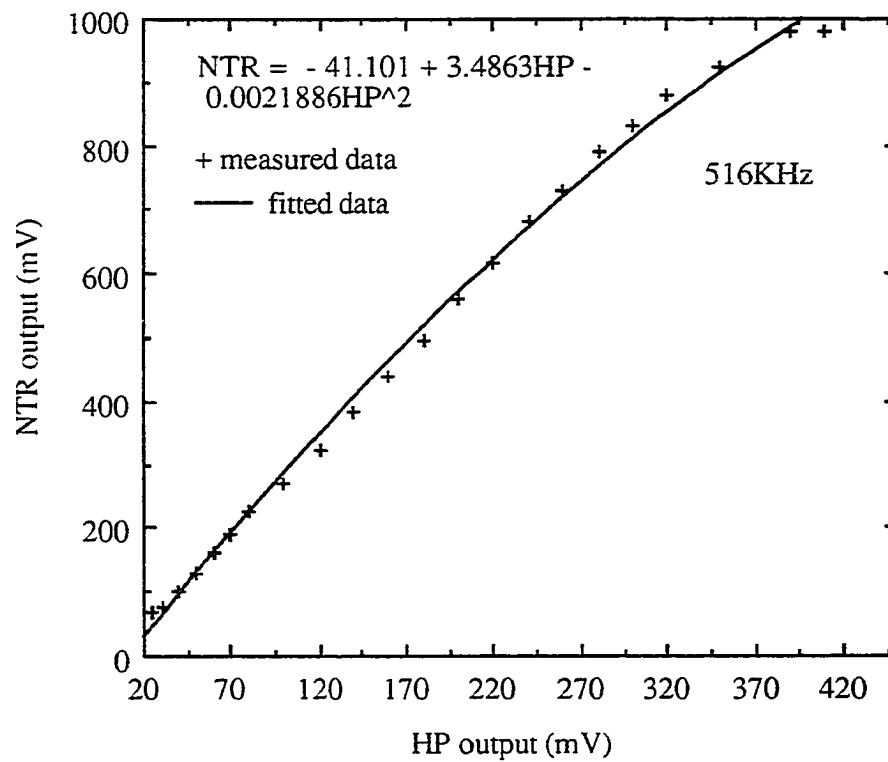


Figure 4.5. Hydrophone out versus HP output.

The data is obtained off the focal region.

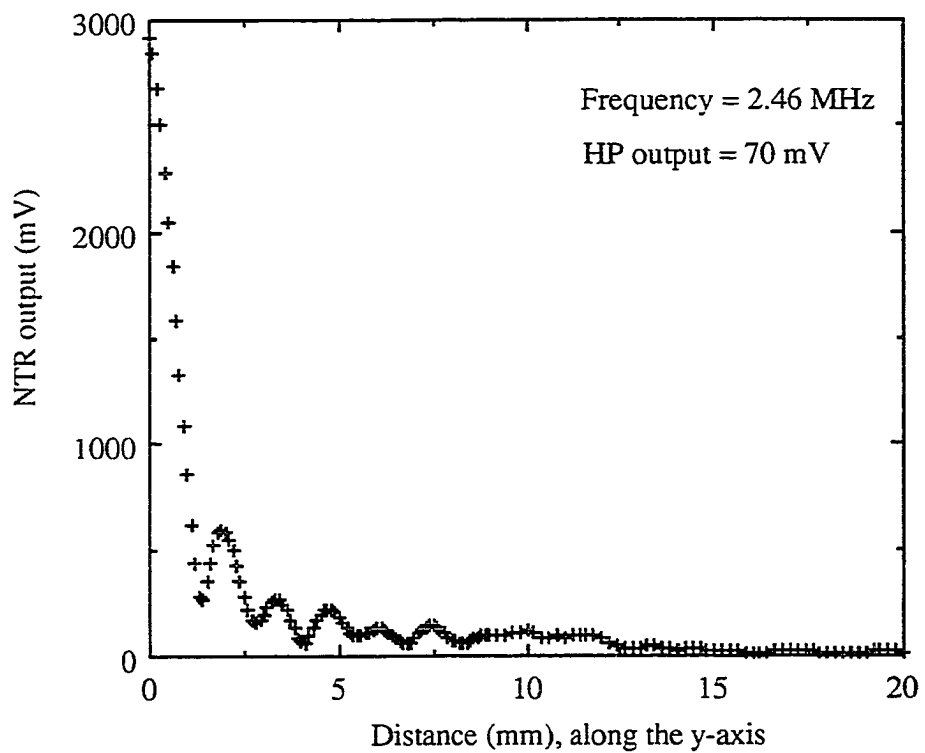


Figure 4.6. Transverse beam profile of the
transducer at 2.46 MHz. HP output = 70 mV.

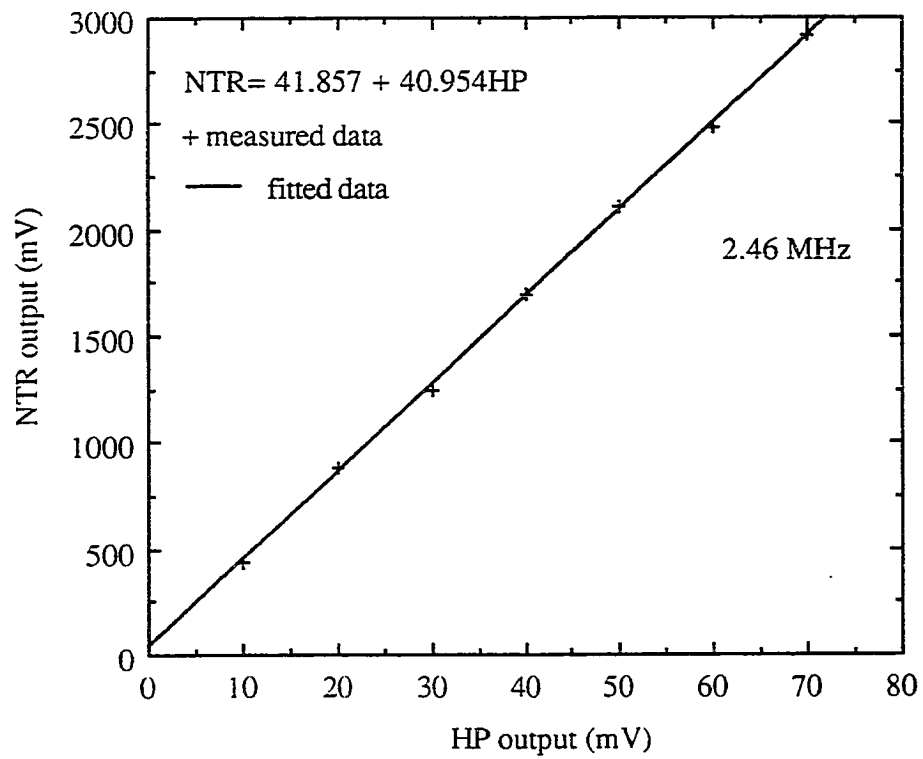


Figure 4.7. Hydrophone output versus HP output.

Data is obtained in the focal region.

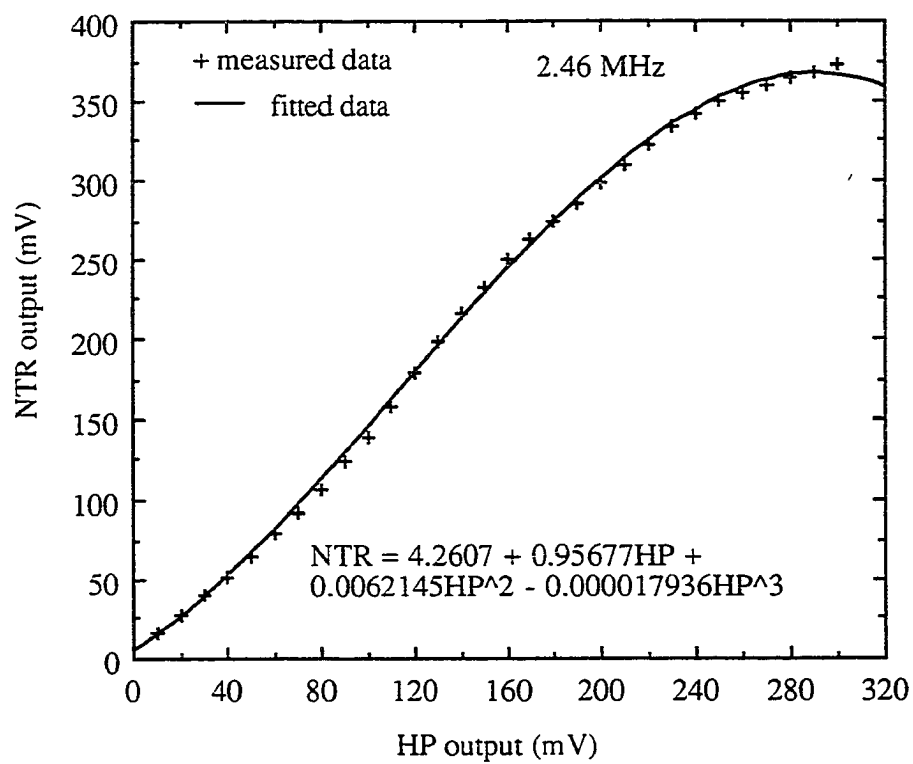


Figure 4.8. Hydrophone out vs HP output.

Data is obtained off the focal region.

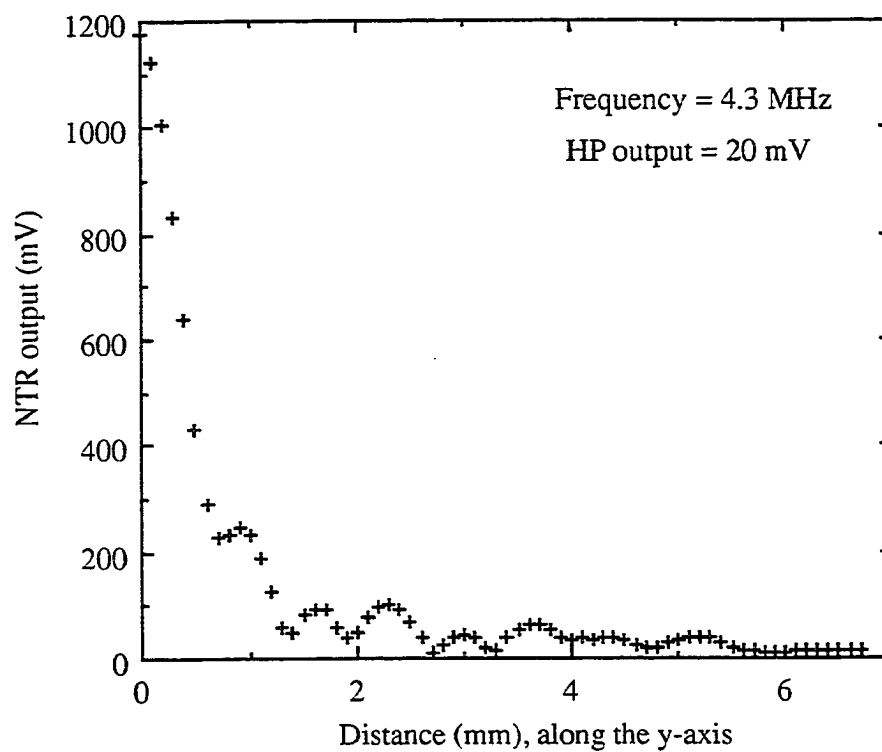


Figure 4.9. The transverse beam profile of the transducer at 4.3 MHz.

The HP output = 20 mV.

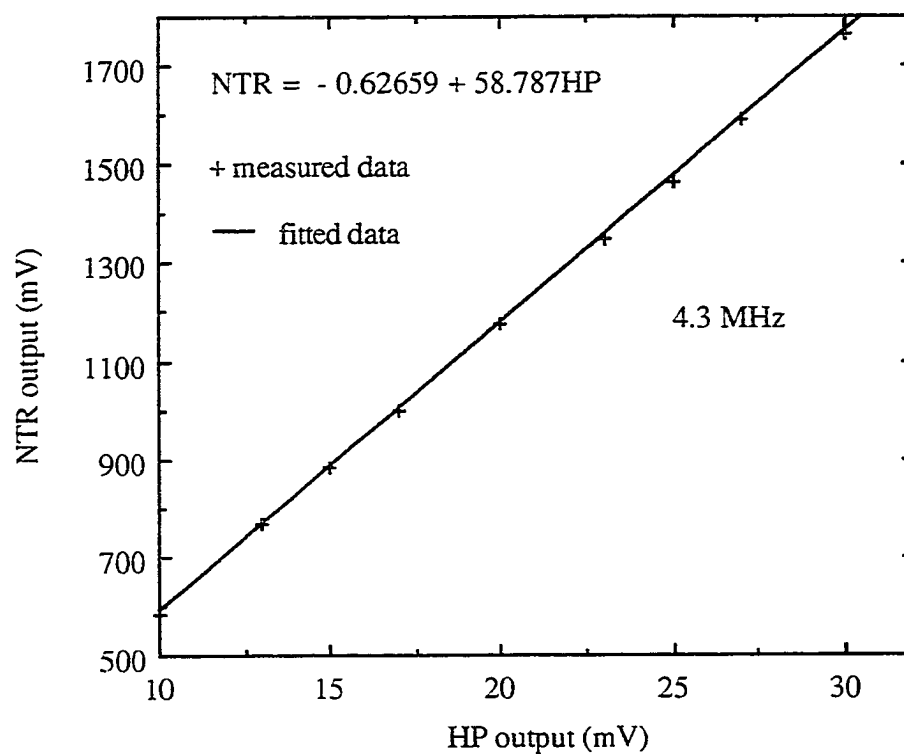


Figure 4.10. Hydrophone output versus HP output.

The data is obtained in the focal region.

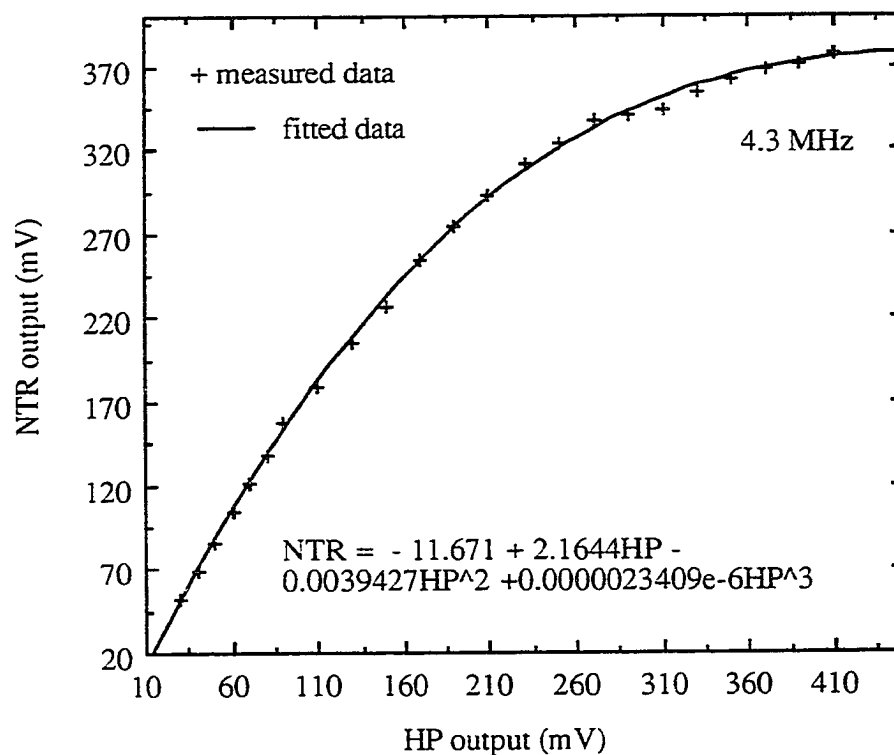


Figure 4.11. Hydrophone output versus HP output.

The data is obtained off the focal region.

4.3 The Measurement of Radiation Force

4.3.1 Measurements with a reflecting target

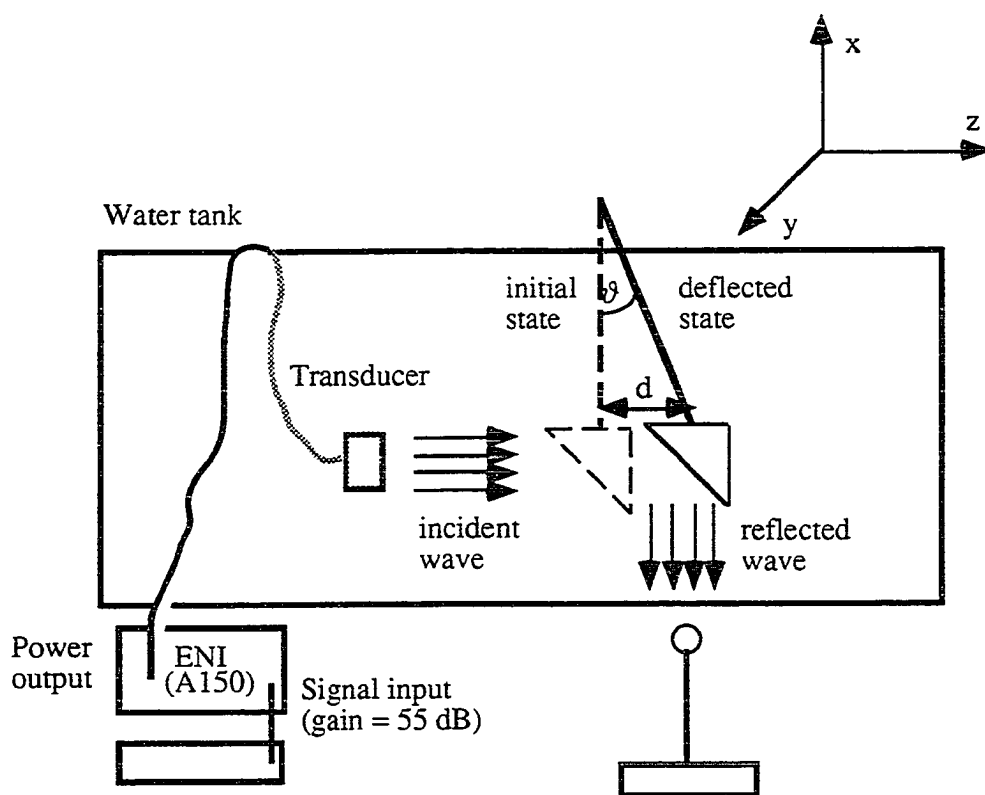


Figure 4.12. The schematic of the setup for measuring the radiation force with the reflecting target.

Fig. 4.12 shows the setup for measuring the radiation force with the reflecting target (Sameer Madenshetty, Ph.D dissertation, Yale University, 1989). The target measures 2.212" x 1.55" x 0.227" and weighs 21.708 g in air. The thread suspending the target is 15.9 cm long. The equipment for driving the transducer is the same as that shown in Fig.4.2 for measuring the transverse beam profile of the transducer. The target side facing the acoustic wave is made from stainless steel foil and the back side from Plexiglas. There is a cavity, filled with air, inside the target. Outside the water tank, a telescope with the resolution of 20 μm is used to measure the deflection of the target. Right after each pressure increase, the target would oscillate 1 to 5 minutes along the z-direction. We noted

the target position only when such oscillation displacement was below the resolution of the micrometer moving the telescope. For each HP output, we measured the target position three times and averaged them.

As shown in Fig. 4.12, under the action of an acoustic force, the target will deflect from its initial state shown, as a dash line, to the balanced state(the solid line). In the latter state, the radiation force acting on the target, F_{hp} , can be expressed in terms of the weight of the target;

$$F_{hp} = mg \tan(\vartheta) \approx \frac{d}{l} mg, \quad 4.12$$

where m is the mass of the target, ϑ the deflection angle, d the deflection, l the length of the thread hanging the target and g the acceleration due to gravity. Eq 4.11 can be rewritten as

$$p_{hp}(0) = k_{hp} V_{hp}(0) = c V_{ref}(0) \sqrt{\frac{\rho mg d}{\pi D l \int V_{ref}^2(y) y dy}}, \quad 4.13$$

where mg is the net weight of the target in water. The net weight is measured with Cahn electrical balance.

Two transducers, of 2.46 MHz and 4.3 MHz, are calibrated in this method. For the 2.46 MHz transducer, we obtain

$$\sqrt{IN_{70}} = 1.84886 \times 10^{-3} \text{ (V}_m\text{)}, \text{ and}$$

$$p_{hp} = k_{hp} V_{hp}(0) = c V_{70}(0) \sqrt{\frac{\rho mg}{D \pi l}} \frac{\sqrt{d_{hp}}}{\sqrt{\int V_{70}^2(y) y dy}} = 1.305 \times 10^8 \sqrt{d_{hp}} \text{ (Pa/m)}, \quad 4.14$$

where the following parameters have been used; $l=15.9$ cm, $mg=1.2446 \times 10^{-2}$ N(in water), $c=1498$ m/s, $\rho=1000$ kg / m³ , $D=0.00813$ and $V_{70}(0)=2.909$ Volt.

Figs. 4.13 and 4.14 show the measured deflection and pressure versus the HP output.

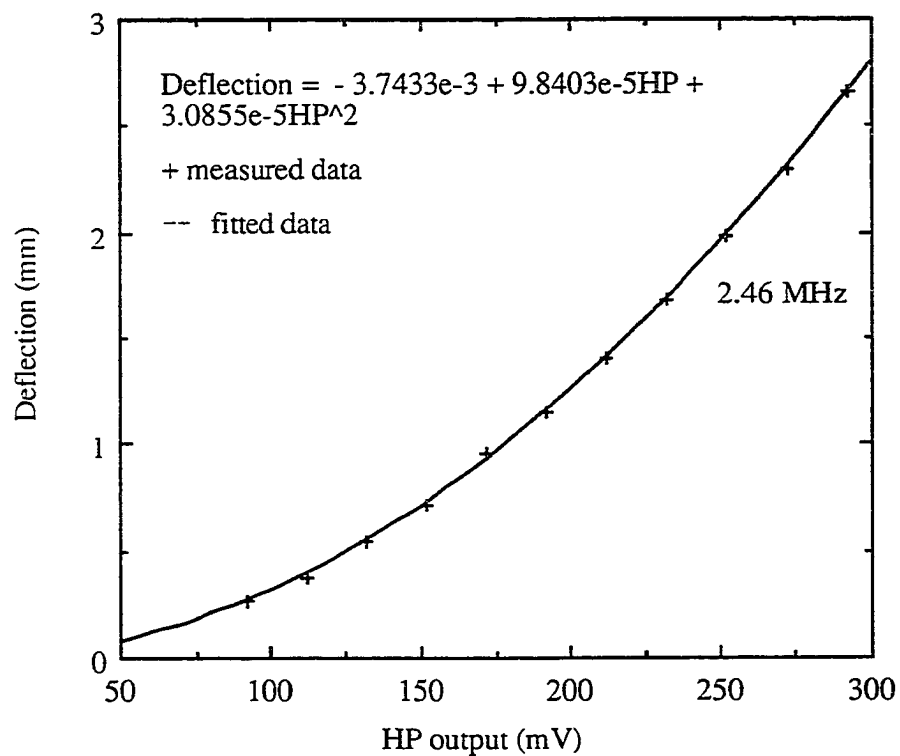


Figure 4.13. Target deflection vs the HP output.

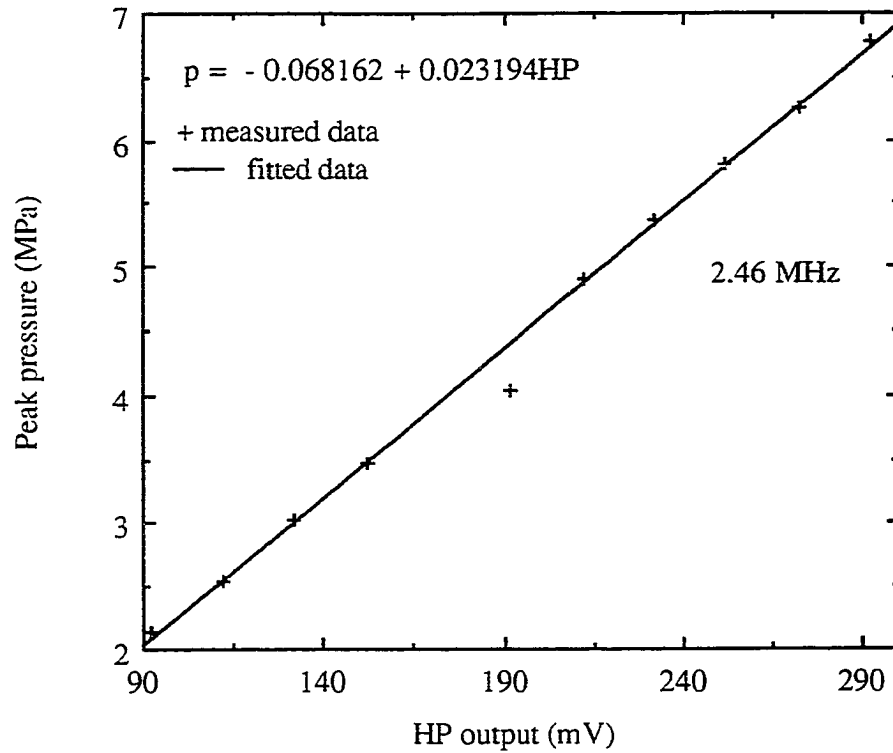


Figure 4.14. The peak acoustic pressure vs the HP output.

Since the acoustic pressure and the HP value are related by $p = -0.068162 + 0.023194HP$, therefore the pressure change, Δp , associated with HP change, ΔHP , is $\Delta p = 0.023194\Delta HP$. For example, the pressure change $\Delta p = 0.11597$ (MPa) given $\Delta HP = 5$ (mV).

For the 4.3 MHz transducer, we obtain the following result using the data in Fig. 4.8.

$$\sqrt{IN_{20}} = 4.4303 \times 10^{-4} \text{ (V}_m\text{)}, \text{ and}$$

$$p_{hp} = k_{hp} V_{hp}(0) = c V_{20}(0) \sqrt{\frac{\rho m g}{D \pi l}} \frac{\sqrt{d_{hp}}}{\sqrt{\int V_{20}^2(y) y dy}} = 1.2862 \times 10^8 \sqrt{d_{hp}} \text{ (Pa/m)}, \quad 4.15$$

where the following parameters have been used; $l=15.9$ cm, $mg=1.2446 \times 10^{-2}$ N(in water), $c=1498$ m/s, $\rho=1000$ kg / m^3 , $D=0.025$ and $V_{20}(0)=1.1175$ Volt.

Figs. 4.15 and 4.16 show the measured deflection and pressure versus the HP output.

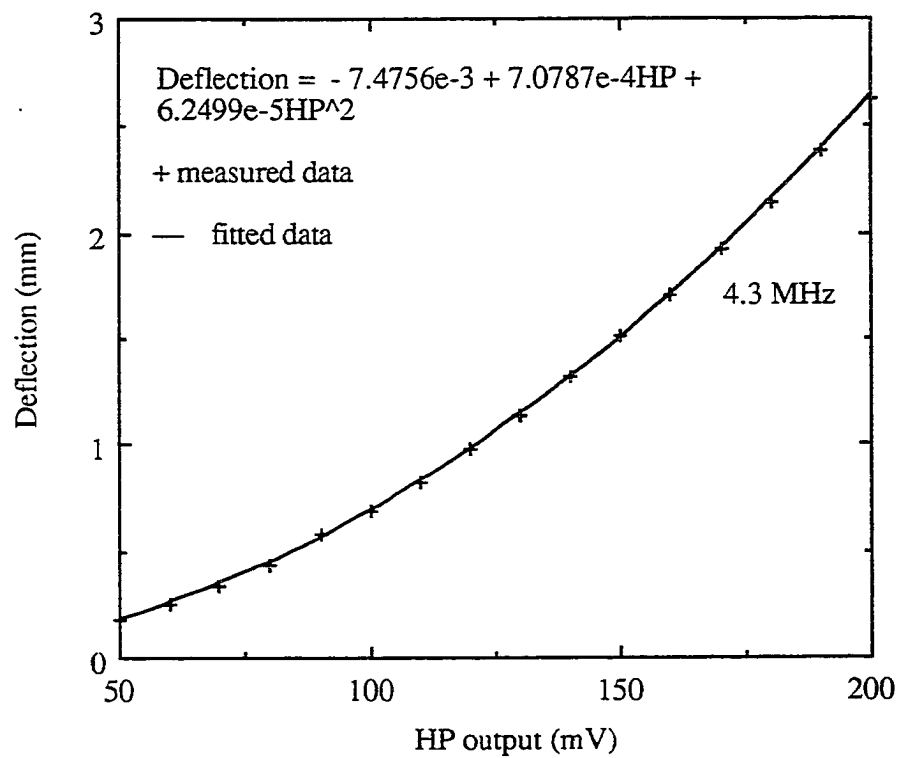


Figure 4.15. The target deflection vs the HP output.

The data was obtained on the focal region.

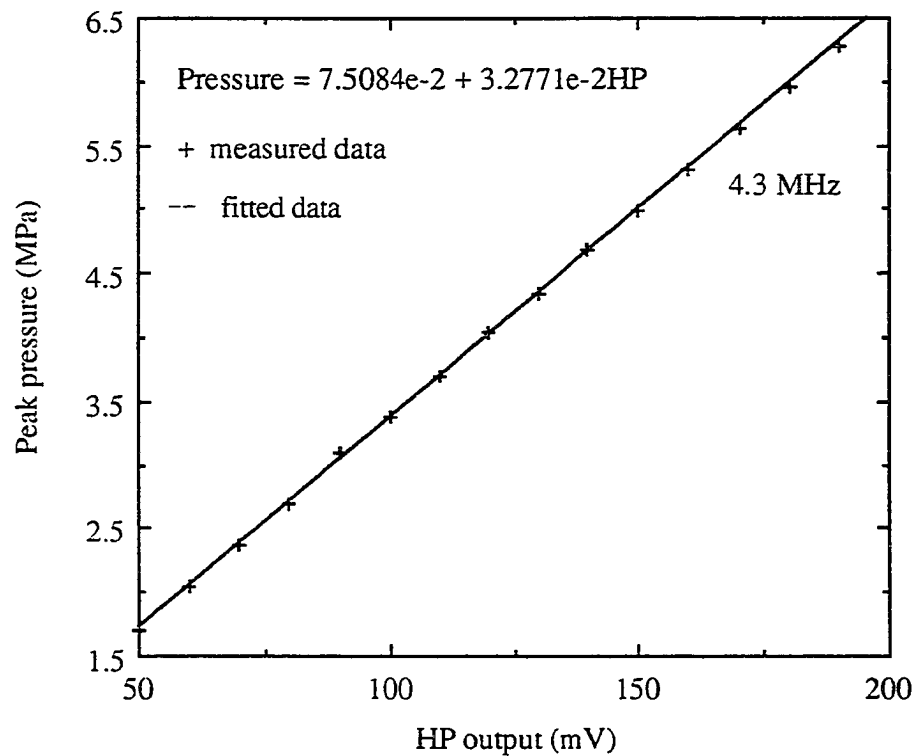


Figure 4.16. The peak acoustic pressure vs the HP output.

In Figs. 4.14 to 4.16, the fitted curves do not approach 0 when the HP output is 0. This is because of random and system errors during the measurement error, and also because of the noise level of NTR output. The noise is caused by the electrical noise and random acoustic disturbances in the water.

4.3.2 Measurement with an Absorbing Target

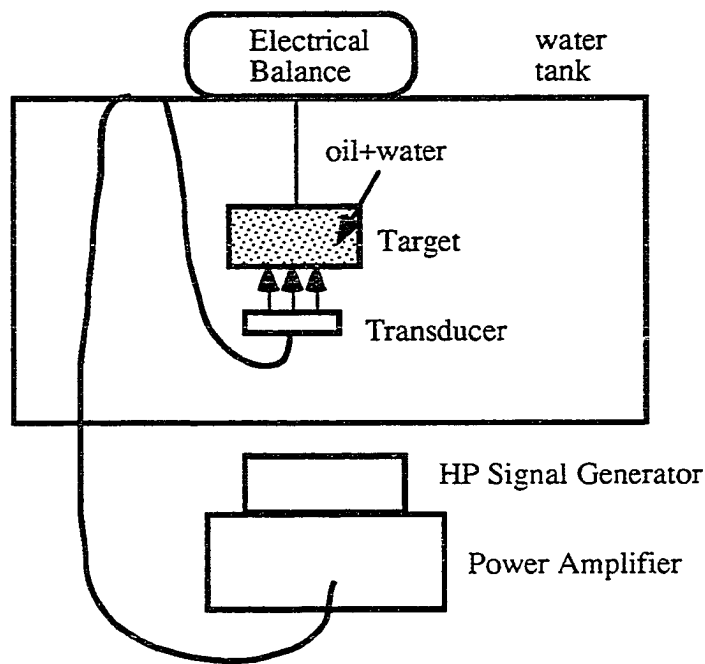


Figure 4.17. The schematic of the setup for measuring the radiation force with the absorbing target.

Fig. 4.17 is the setup for measuring the radiation force with the absorbing target. The equipment driving the transducer is the same as that in Fig. 4.11, calibrating the transducer with the reflecting target. The target (built by Mr. Thomas Stottlemyer) has a cylindrical shape, made from Plexiglas. The bottom of the target is sealed with a Tegaderm 1628(3M Co., MN) membrane. The inner and outer diameter of the side facing the acoustic wave are 3.26 cm and 3.7 cm, respectively. The target weighs 14.5 g in water. The emulsion filling the target consists of 10% olive oil, 89.5% water and 0.5% surfactant to keep the emulsion stable. By comparing the reflection from reflecting plate, which has the same location as the target does, with the reflection from the target, we estimate the absorption coefficient is 0.98. The electrical balance (Model TA450, Cahn Instruments,

Inc., Cerritos, CA) has a mass resolution of 0.001 gram, which corresponds to a force of 0.98 dyne.

Besides the two transducers calibrated earlier with the reflecting target, we also calibrated the 516 KHz transducer. For it, the following parameters are used; $D = 0.02$, $V_{70} = 0.657$ Volt and $\sqrt{IN_{70}} = 1.4555 \times 10^{-3}$ (Vm). Consequently, we have

$$p = 8.5303 \times 10^7 \sqrt{F_{hp}}.$$

Fig. 4.18 and 4.19 show the results for the 516 KHz transducer.

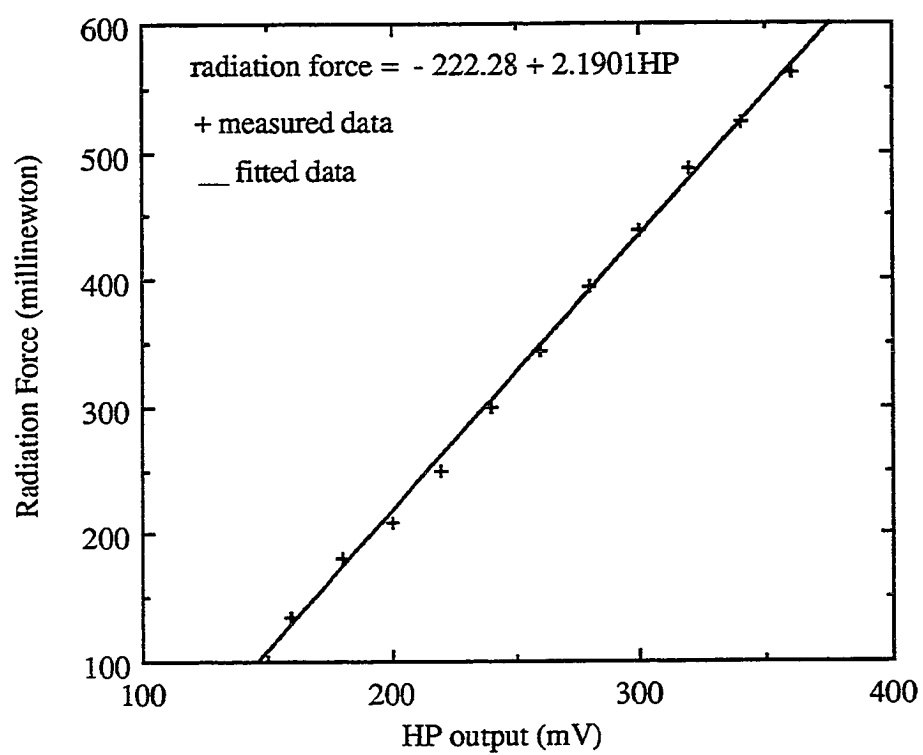


Figure 4.18. The radiation force vs the HP output.

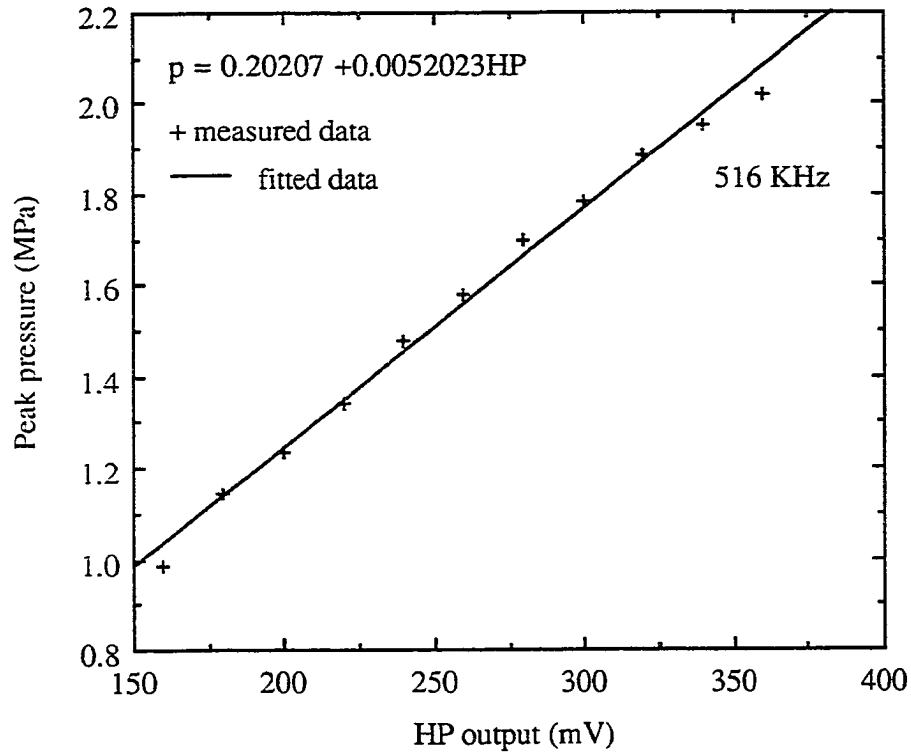


Figure 4.19. The peak pressure vs the HP output.

For the 2.46 MHz transducer, the following parameters are used; $D = 0.02$, $V_{70}(0) = 2.909$ Volt, $\sqrt{IN_{70}} = 1.84886 \times 10^{-3}$ (Vm). The peak pressure can be expressed by

$$p = 2.975 \times 10^8 \sqrt{F_{hp}}.$$

Fig. 4.20 and 4.21 show the results.

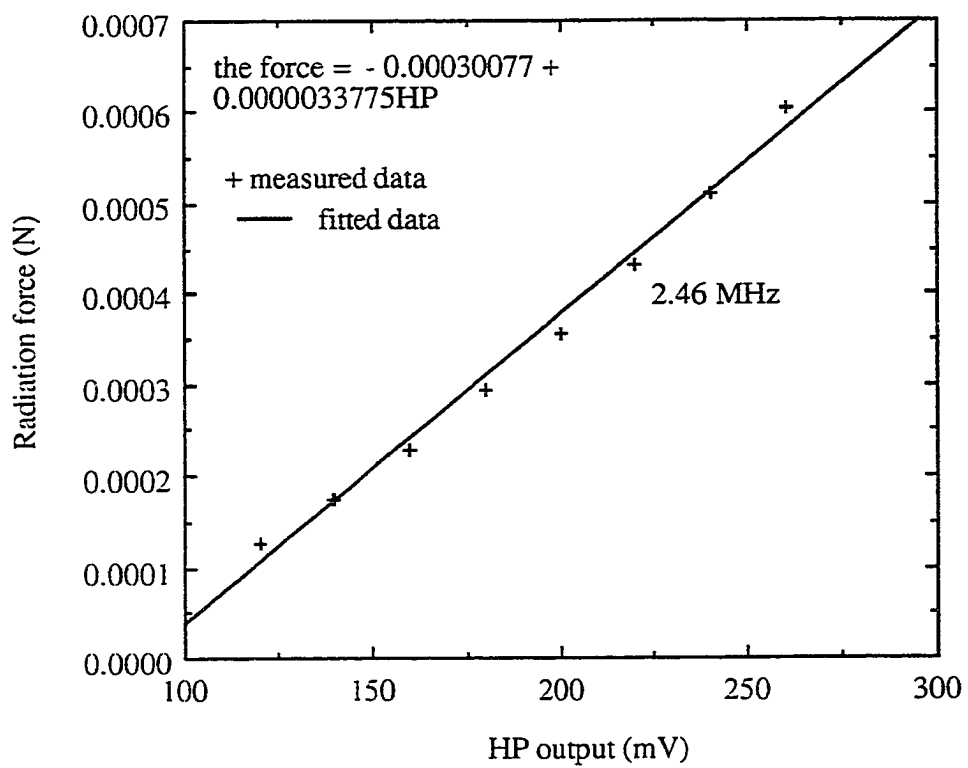


Figure 4.20. The radiation force vs the HP output.

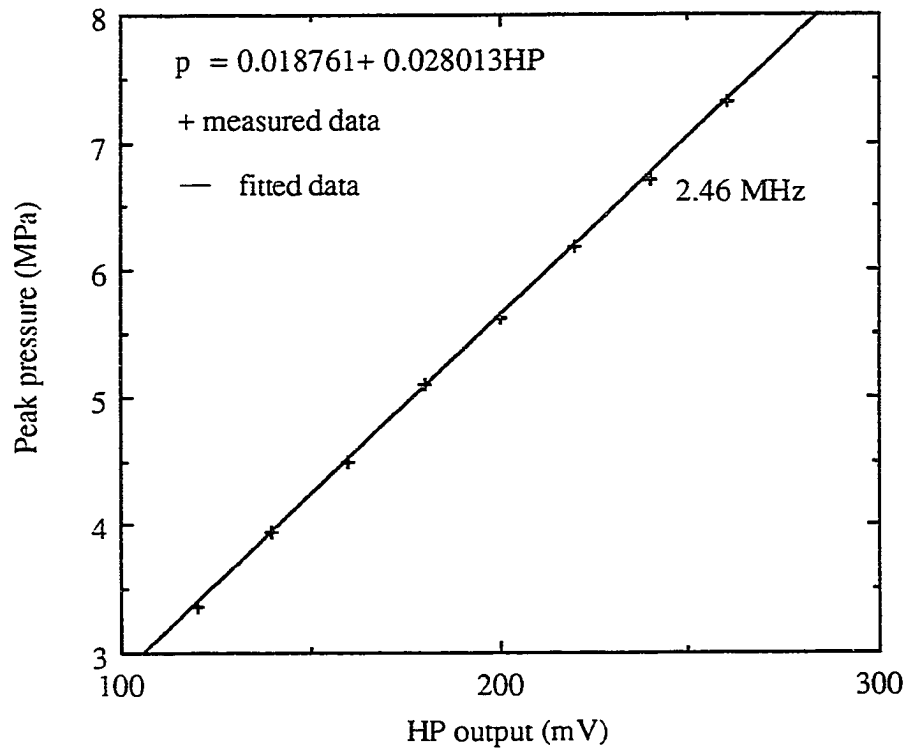


Figure 4.21. The peak pressure vs the HP output.

For the 4.3 MHz transducer, the following parameters are used; $D = 0.02$, $\sqrt{IN_{20}} = 4.38 \times 10^{-4}$ (V_m) and $V_{20}(0) = 1.1175$ Volt. Consequently, we have

$$p_{hp} = k_{hp} V_{hp}(0) = 5.07 \times 10^8 \sqrt{F_{hp}} \text{ (Pa/m)}, \quad 4.16$$

Figs.4.22 and 4.23 show the results.

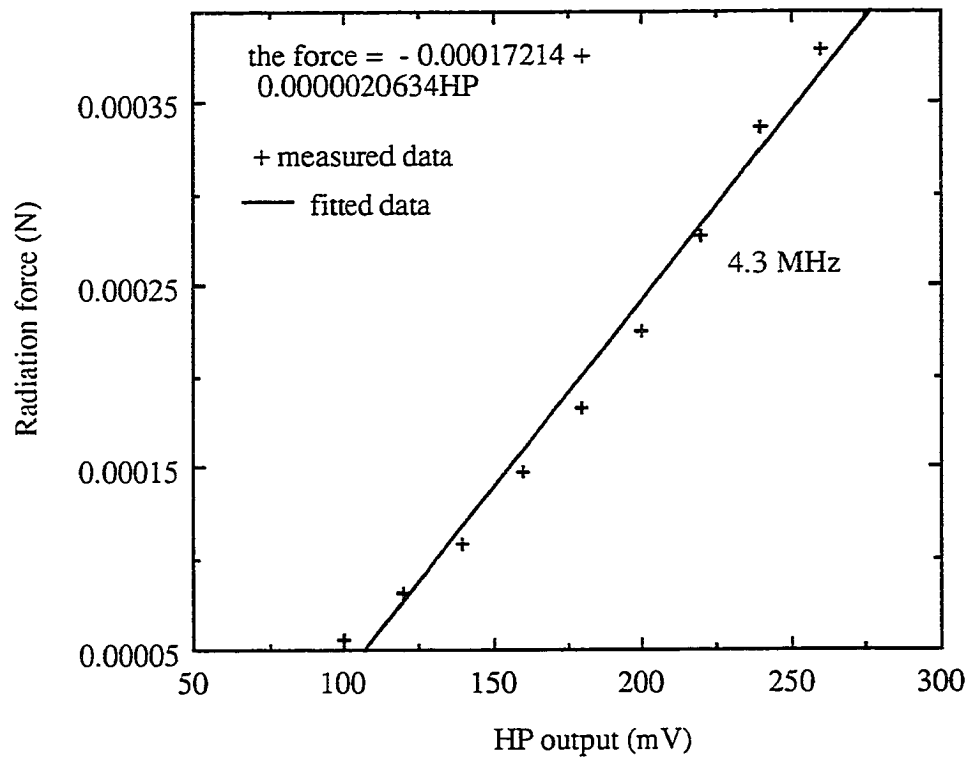


Figure 4.22. The radiation force vs the HP output.

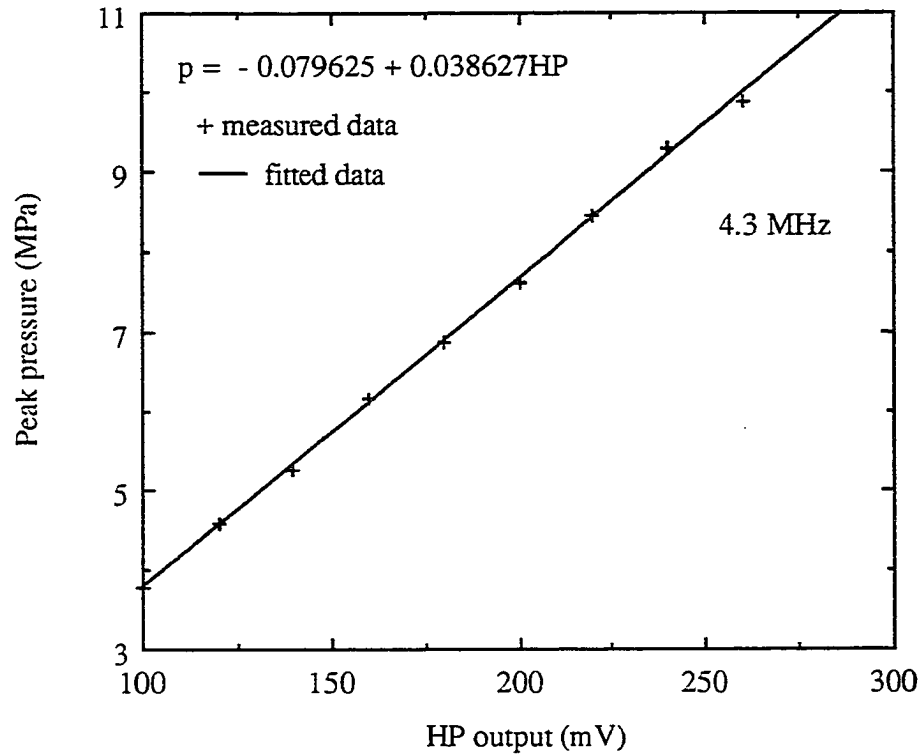


Figure 4.23. The peak pressure vs the HP output.

4.4 Comparisons of the Results by the Two Methods

Compared with the absorbing target method, the reflecting target provides better resolution for measuring the radiation force. As given in Eq 4.12, the radiation force for the reflecting target is

$$F_{hp} = mg \tan(\vartheta) \approx \frac{d}{l} mg = \frac{d}{0.159} 0.00124 \text{ (N)}.$$

The minimum deflection we can measure with the telescope is 5.0×10^{-6} m. Therefore, the minimum detectable force will be 3.8994×10^{-8} N, or 3.8994×10^{-3} dyne, which is much smaller than the 0.98 dyne, the minimum detectable force for the absorbing target. If the deflection angle, ϑ , is less than 1° , the reflecting target will give us more accurate measurement of the radiation force. The corresponding deflection and radiation force will be

$$d = l \frac{\pi}{180} = 0.002775 \text{ m},$$

and

$$F_{hp} = \frac{2.775 \times 10^{-3}}{0.159} \times 1.24 \times 10^{-3} = 2.1641 \times 10^{-5} \text{ N},$$

respectively.

Once the radiation force is strong enough to cause larger deflections, Eq. 4.12 will no longer be valid in estimating the radiation force, and the electrical balance will measure the radiation force more accurately, which means that the calibration using the absorbing target is more reliable.

Figs. 4.24 and 4.25 are the comparisons of the calibration curves with the two methods. At the higher end of the HP output, the two curves diverge because the reflecting target underestimates the radiation force. At the lower end of the HP output in Fig. 4.24, the absorbing underestimates the radiation force, and the two curves will intersect if we extrapolate the data of the absorbing target and reflective target.

We define the relative error, or discrepancy, as

$$e = \frac{|p_{\text{reflective}} - p_{\text{absorbing}}|}{p_{\text{absorbing}}}$$

By using the fitting functions given in Figs. 4.14, 4.20, 4.16 and 4.23, we calculate the relative discrepancy of the calibration data for the transducers of 2.46 MHz and 4.3 MHz. Fig. 4.24 and 4.25 show the results.

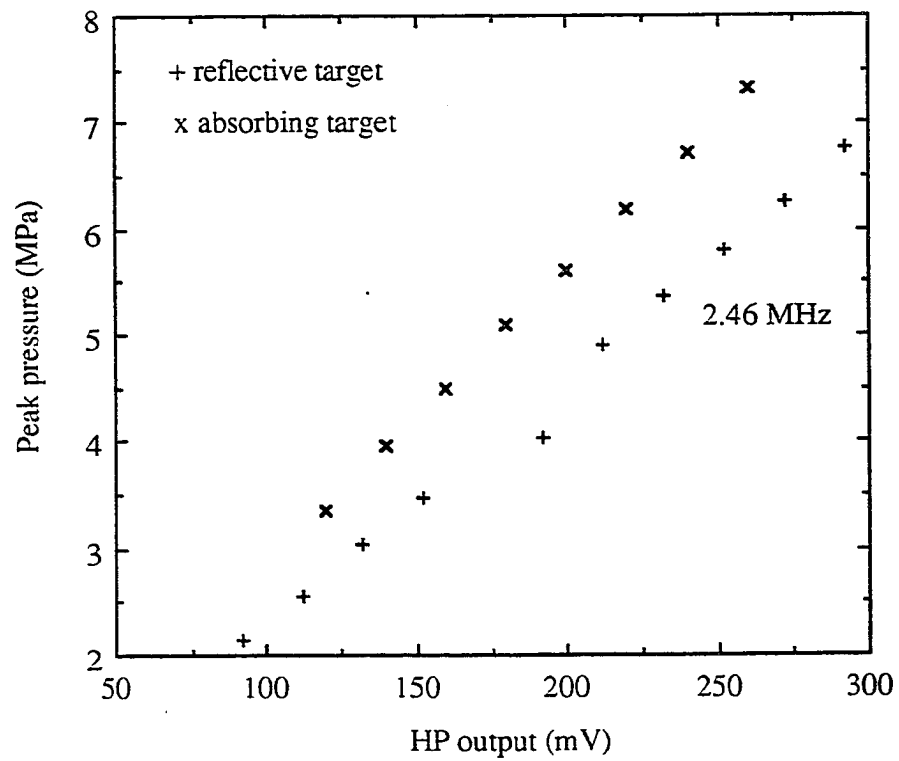


Figure 4.24. The comparison of Figure 4.14 and 4.20.

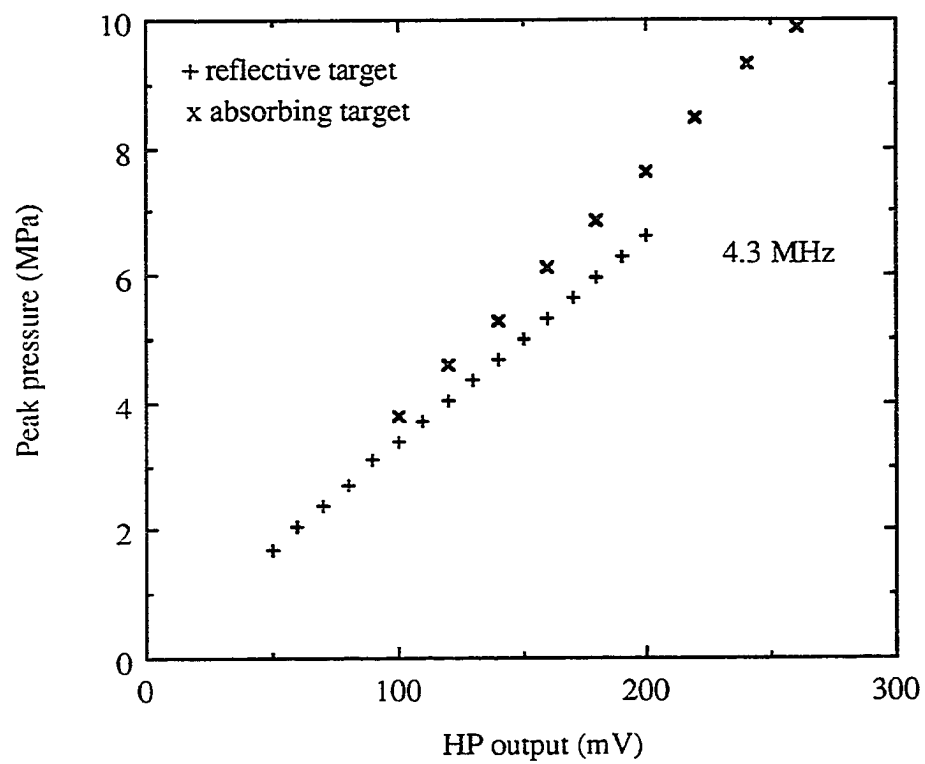


Figure 4.25. The comparison of Fig. 4.16 and 4.23.

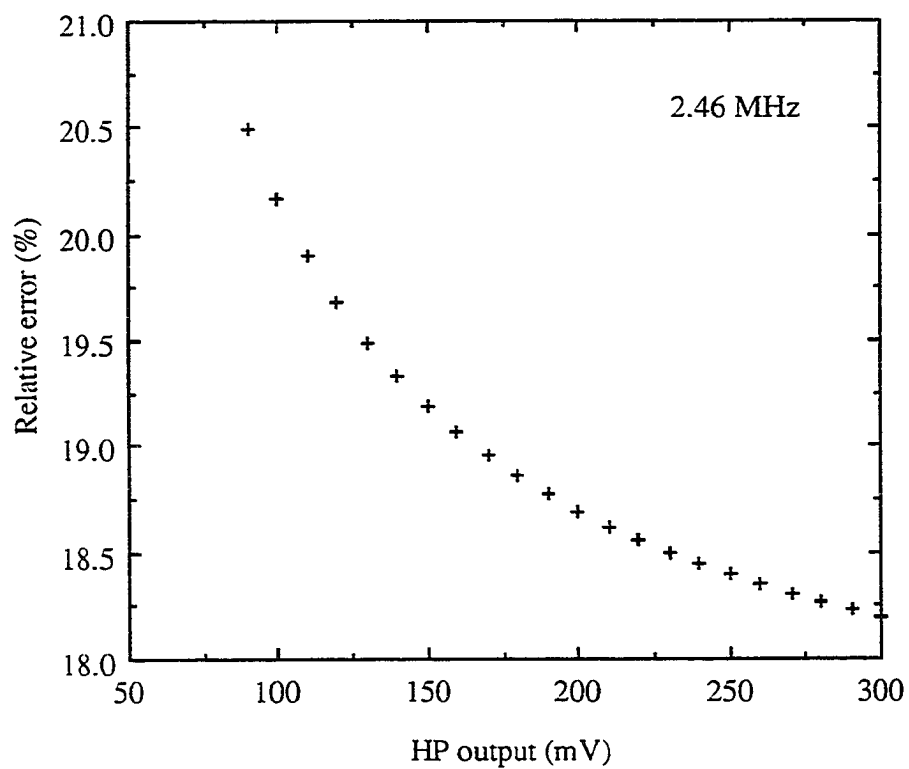


Figure 4.26. The discrepancy of the data from the two targets.

Frequency = 2.46 MHz

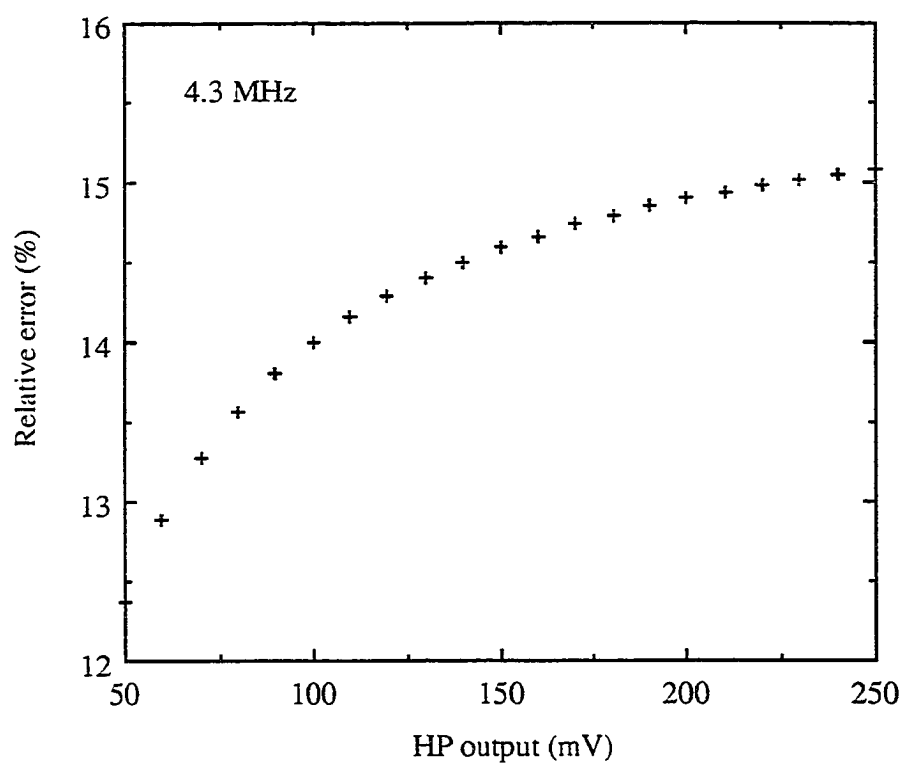


Figure 4.27. The discrepancy of the data from the two targets.

Frequency = 4.3 MHz

Chapter 5 Experimental Procedures

In the previous chapters, we have concentrated on the relevant theories about both acoustic scattering and cavitation, and the experimental apparatus deployed in this work. In this chapter, we will show how we integrate them to obtain experimental results.

5.1 Water Preparation

We first purify the water utilizing a Barnstead D2791 NANO Pure System. The resistance of the purified water is $16\text{M}\Omega/\text{cm}$. The processed water is poured into the main water reservoir, and is filtered through a closed circuit for about 1 hour before being pumped into the cavitation chamber through tygon tubing.

5.2 Transducer Alignments

Transducer alignment is extremely important for the measurement of cavitation thresholds. Misalignment of the transducers will lead to errors in the determination of the cavitation threshold. Each set of experiments requires careful alignment of both the cavitation transducer and the 30 MHz probing transducer to ensure that they are confocally positioned.

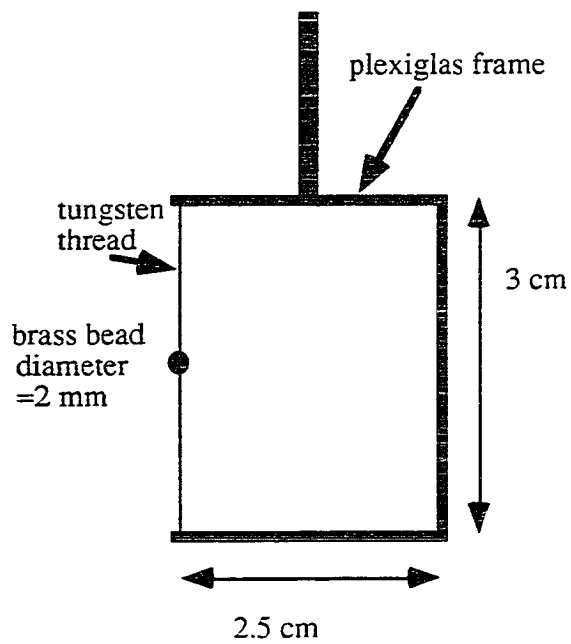


Fig. 5.1. Target used for aligning the cavitating and probing transducers.

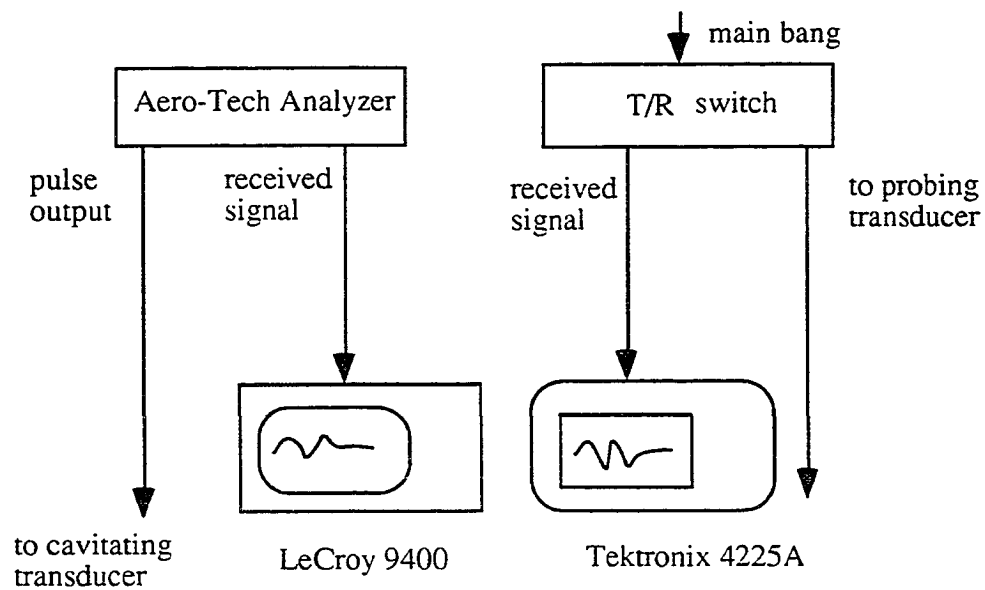


Fig. 5.2. Electronic apparatus for the alignment.

The container used for alignment is the same as that used for performing experiments. Before aligning the transducer, we fill the container with distilled water to about 70% of its capacity. Figs. 5.1 and 5.2 show the equipment used in aligning the transducers. The Aerotech analyzer(Aerotech Ultrasonic transducer analyzer Model UTA-3) sends the cavitating transducer a pulse which has a flat frequency spectrum up to 10 MHz. The cavitating transducer transmits a tone burst with center frequency at the resonance frequency of the transducer. The backscattered acoustic wave is filtered by an analog filter(Krohn-Hite, Model 3202). The filtered signal is shown on the oscilloscope(LeCroy 9400). Fig. 5.2 shows the target alignment. A brass bead of 2.3 mm diameter is adhered to a 50 μm diameter tungsten thread stretched across a Lucite fork. The fork is connected to a Plexiglas rod attached to a 3-D micropositioner. First, we place the target into the water, with the fork plane oriented 45° between cavitating transducer and probing transducer. The Aerotech analyzer drives the cavitating transducer. By adjusting the position of the cavitating transducer or the target, we maximize the backscattered signal from the brass bead. The focal length, L_c , of the transducer can be determined by

$$L_c = c \frac{t_c}{2}, \quad 5.1$$

where c is the sound speed in water, and t_c is the delay between the drive signal from the Aerotech analyzer and the received signal shown on the oscilloscope screen. A 30 MHz tone burst of 20 μs duration is used to excite the probing transducer. By adjusting the position of the probing transducer, we find the maximum backscattered signal from the brass bead. A typical backscattered signal from the brass bead is shown in Fig. 5.3. The focal length, L_p , of the probing transducer is similarly determined by

$$L_p = c \frac{t_p}{2}, \quad 5.2$$

where t_p is the delay between the tone burst exciting the probing transducer and the backscattered signal shown on the oscilloscope screen.

5.3 Setting of Time Base

After the two transducers are confocally positioned, we adjust the time base so that we can probe the microcavitation generated in the confocal zone. The duration of the probing pulse is set at $t_p = 20 \mu s$. The delay between the cavitation pulse and the probing pulse is given by

$$D_{cp} = \frac{t_c - t_p}{2}. \quad 5.3$$

The duration of the receiving pulse is adjusted at $40 \mu s$, and its delay relative to the probing pulse is set at

$$D_{pr} = t_p - 4 \mu s.$$

(See schematic diagram of the time base in Fig. 3.8.)

For the cavitation experiment in water, two oscilloscopes are used for the alignment, the Tektronix 2445A and LeCroy 9400. The 30 MHz backscattered signal is shown on the screen of Tektronix 2554A oscilloscope, and the focal length of the probing transducer can be easily determined by measuring the time delay between the transmitting signal and the backscattered signal. The received echo of the cavitating transducer is passed through the Krohn-Hite filter with the low-pass cut-off frequency set at 2.0 MHz; the output of the filter is displayed on the screen of the digital oscilloscope (LeCroy 9400).

The focal length of the cavitating transducer is determined from the time between the cavitating

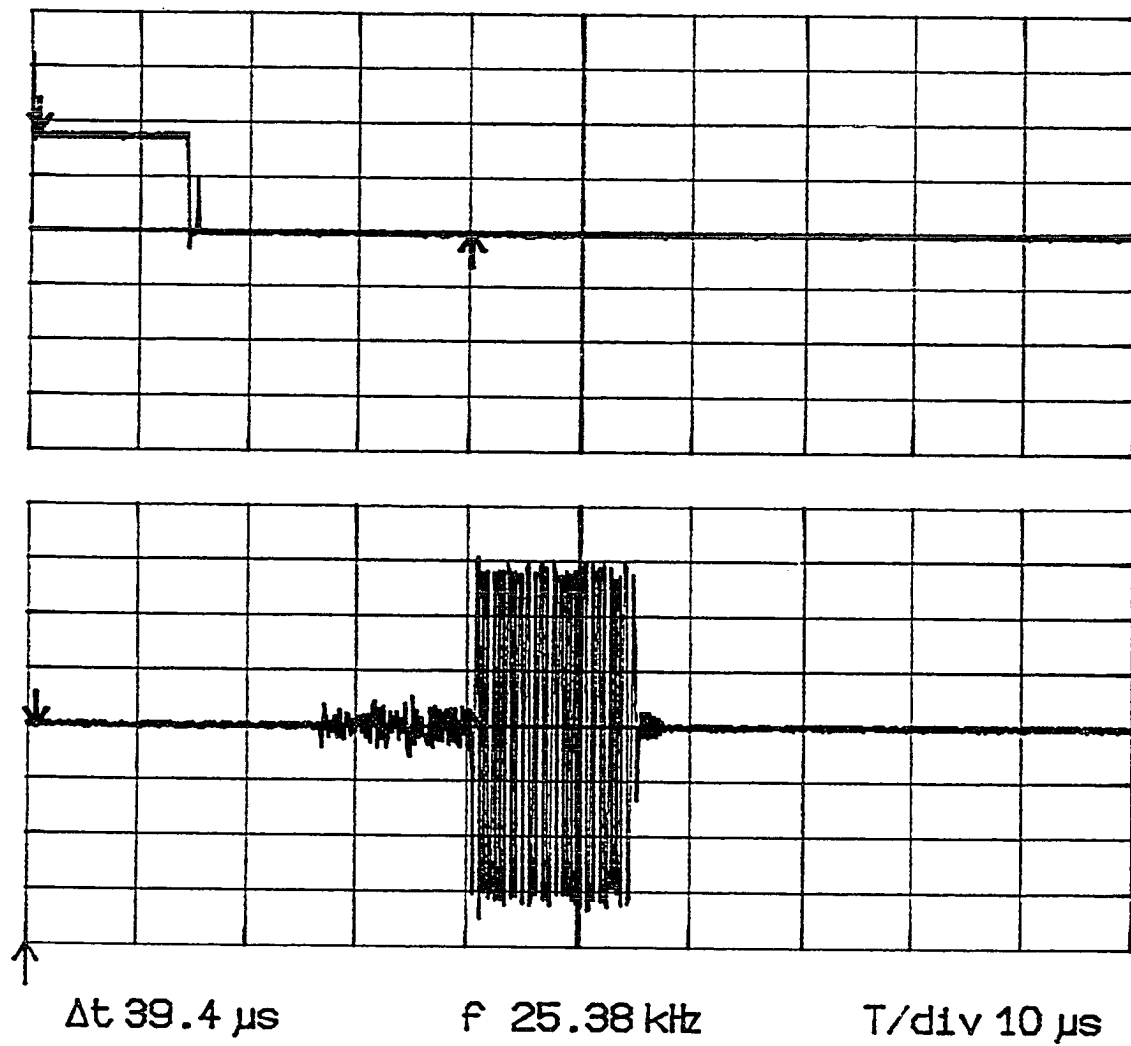


Fig. 5.3. Backscattered signal of the 30 MHz probing signal from the brass bead target

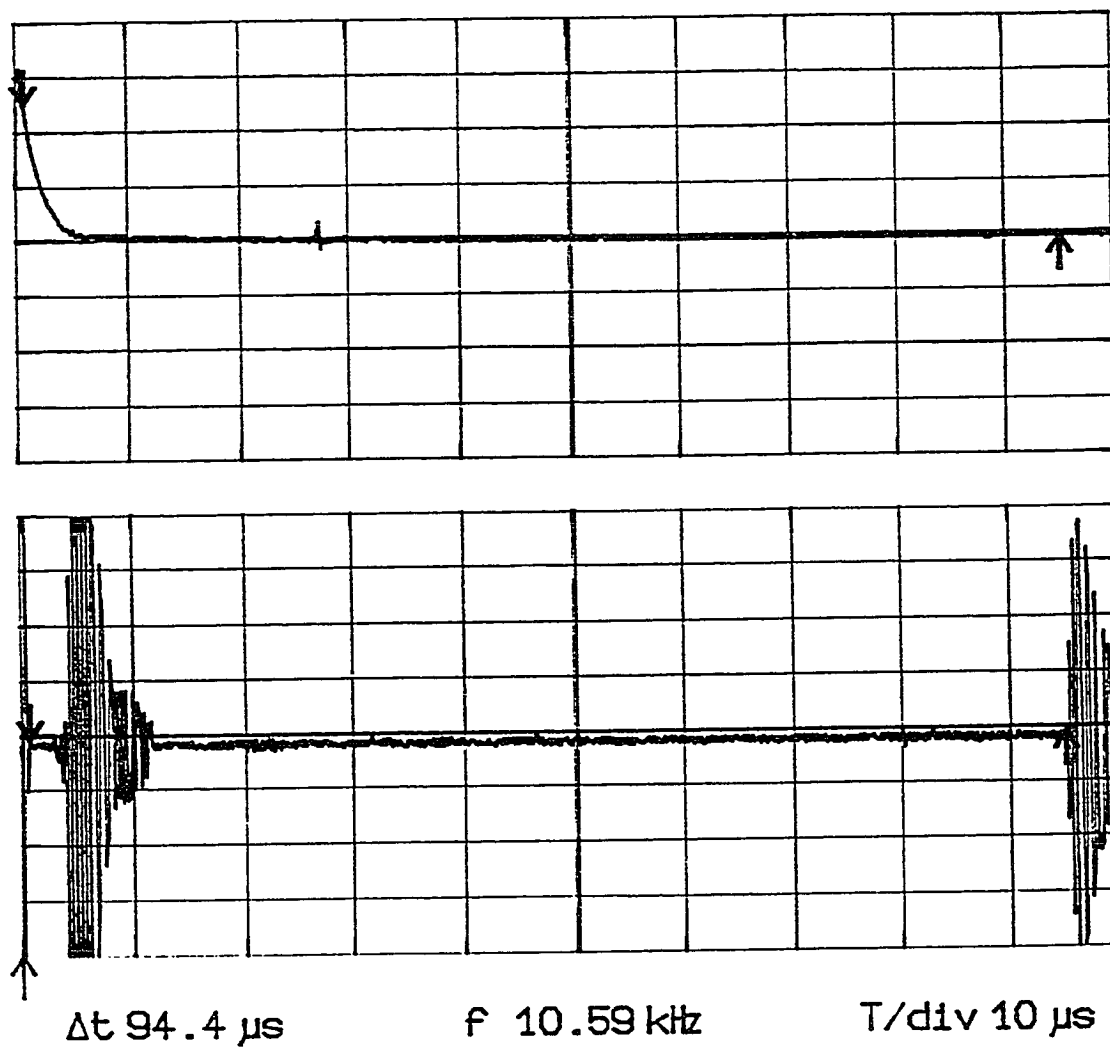


Fig. 5.4. Backscattered signal of the Ultrasonic Analyzer output from the brass bead target

and received signals. We adjust the delay of the probing pulse and the delay of receiving gate so that the probing signal and cavitating signal overlap in space. Once the delays are set, we gently take the target out of the water, disconnect the Aerotech Transducer Analyzer from the cavitating transducer, and connect the cavitating transducer to the ENI power amplifier.

5.4 Experiments in Water

Since the water is relative “clean,” a very high pressure is required to cause cavitation. But we are only interested in the minimum acoustic pressure required to generate the cavitation. So polystyrene particles (Duke Scientific Co., CA) are used as potential cavitation nuclei. Different sizes of particles yield different cavitation thresholds. Through a series of tests, it has been found that the polystyrene particles of 0.1 and 0.4 μm in diameter yield the lowest acoustic threshold. Therefore, we used the particles of these sizes as cavitation nuclei for our studies in water. For each set of experiments, about 3 to 5 droplets (each droplet contains about 10^6 particles.) are diluted with 60 cc distilled water, and then the water is contained in a syringe.

After injecting some polystyrene particles into the water, we gradually increase the output level of the HP signal generator until the LeCroy oscilloscope is triggered. For each output level, we count about 40 triggered events.

5.5 Experiments in Blood

To avoid the environmental noise problem encountered when performing experiments *in vivo* and to maximize the chances of observing cavitation, we chose human blood as the test biological material. The frequency of the signal we used to insonify the blood was 2.46 MHz, which is in the frequency range of clinical diagnostic ultrasound,

although it is lower than the center frequencies employed for most Doppler applications. The duty cycle in our work was 0.813%, which is close to that used in diagnostic ultrasound applications [Duck *et al.*, 1987]. The noise level and the propagation attenuation have been minimized by using distilled water and an acoustically transparent membrane, Tegaderm[®], to hold the blood samples. By diluting the blood samples, we measured the different cavitation thresholds versus the volume concentration of red blood cells (RBCs) to determine how the blood concentration affected the cavitation threshold. Much care was exercised during the experiment to prevent destruction of blood cells by nonacoustic forces and to prevent contamination of the samples.

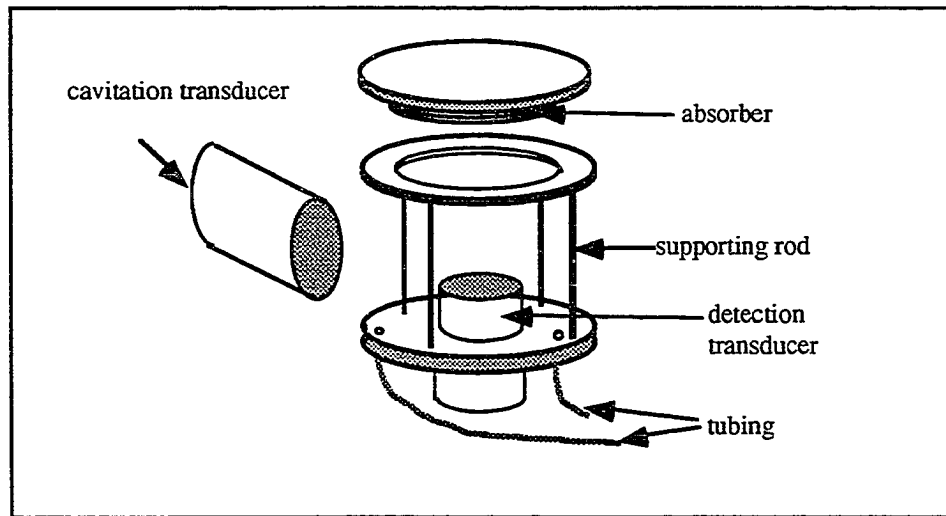


Fig. 5.5 The Schematic Diagram of Experiment Set-Up and the Cavitation Test Cell.

Center Frequency of Cavitation Transducer = 2.46 MHz,

Detection Transducer = 30.0 MHz.

Fig. 5.5 shows the experimental setup. The key parts of the system are the cavitation cell and two transducers which are immersed in distilled water. Both transducers are highly focused. The probing transducer, with center frequency of 30 MHz, has a focal

length of 2.5 cm, and the driving transducer, with center frequency of 2.46 MHz, has a focal length of 7.5 cm. The two transducers are carefully aligned so that the two focal regions overlapped. Microcavitation is caused in the cell by the driving transducer. The probing transducer detects the cavitation by transmitting a tone burst signal toward the region of cavitation and receiving the backscattered signal. The cell holds about 95 cc of liquid and has a cylindrical shape of 10.5 cm long with a inner diameter of 3.2 cm. An acoustical absorber is mounted at the end opposite the probing transducer. The detection system has approximately 70 dB gain and has sufficient sensitivity to detect individual polystyrene particles of 0.1 μm diameter in distilled water.

Tegaderm 1628 (3M), a sterile acoustic window used for transdermal ultrasound scanning applications, was used to seal the cavitation cell. Our test in distilled water has shown that Tegaderm 1628 has a pressure transmission coefficient of about 0.95 for a pulsed 2.46 MHz acoustic signal. For each experiment, the test sample was gently injected into the cavitation cell by a 60cc syringe through the bottom tubing. Once the experiment was finished, the test sample was withdrawn through the bottom tubing, and filtered air was pumped into the cell simultaneously through the top tubing. The cell was flushed with distilled water several times to ensure that the purity of each sample was not adulterated by the previous sample.

Samples of 50%, 25% and 12.5% blood were obtained by adding extra plasma to whole blood. The volume of the extra plasma needed was determined by

$$V_p = \frac{1-p}{p} V_b, \quad 5.4$$

where V_p is the volume of the extra plasma and V_b the volume of blood. Neither the extra plasma nor whole blood were degassed.

The backscattered signal from the test sample was regarded as noise during the experiment. The noise level from each sample was different from the others. Two

parameters were used to determine whether the received signal indicated the presence of cavitation. The detection criteria included 1) the amplitude of the scattered tone burst had to exceed a pre-defined noise level and 2) the duration of the received signal had to be less than the duration of the driving signal, which was $8.1 \mu s$. These criteria had been used to indicate cavitation in earlier threshold measurements in water seeded with polystyrene particles [Holland, *et al.*, 1992].

The amplitude of the backscattered signal is directly related to the strength of the acoustic signal transmitted by the probing transducer. Therefore, the detection sensitivity of the system depends on the power which excites the probing transducer. On the other hand, too much power applied to the probing transducer can cause the acoustic noise level to be extremely high, which sometimes obtuses the cavitation signal. For a specified driving power of the probing transducer, the sensitivity can be preserved while the acoustic noise is minimized. Throughout the experiment, electrical noise was mainly caused by the leakage from the RF mixer in the T/R switch, which is part of the controlling circuit, and remained constant at a level of 135 mV for the experiments. In the following discussion, all the noise levels to which we refer are due to acoustic noise level. Such noise levels were dependent on the test sample and the output of the detector.

The insonification and detection system was automatically controlled by an IBM/PS/2/30 through a GPIB interface board (National Instrument Inc., Austin, TX). During the experiments, the triggering level for the digital oscilloscope (LeCroy 9400, LeCroy Co., Chestnut Ridge, NY) was preset just above the noise level. The computer continuously monitored the triggering status of the oscilloscope. If the computer was not triggered for 3 minutes, the computer would signal the HP generator to increase the output power to the driving transducer. When the oscilloscope was triggered, a status signal was sent to the computer through the status bus line, and the backscattered signal was transferred to the computer. The computer would then check the duration of the backscattered signal. If the duration of the signal was less than $8.1 \mu s$, the amplitude of

the driving signal was noted, and the computer would signal that cavitation had been generated by the driving transducer. Otherwise, the computer would signal that there was no cavitation and would keep monitoring the triggering status of the oscilloscope.

Chapter 6 The Results of Cavitation Threshold Experiments and Discussions

In the previous chapters, we have described the experimental setup and the theoretical background. In this chapter, we will present and interpret the experimental results. The experiments are classified into two groups: cavitation in water and cavitation in blood. The details for running the experiments in water have been described in detail in the previous chapters, but we will present the apparatus and experimental procedures which are not covered in Chapter 5.

6.1 Cavitation in Water

6.1.1 Cavitation Thresholds and Mechanical Index for the 2.46 MHz Transducer

Fig. 6.1 is a typical backscattered signal from cavitation. There are two interesting features about this figure. First, the duration of the cavitation signal is shorter than the probing signal duration of $8.13\mu s$. Second, the envelope of the backscattered signal clearly indicates the bubble growth and collapse process. Those two features help us distinguish cavitation from a polystyrene particle. The following figures and tables show the cavitation thresholds.

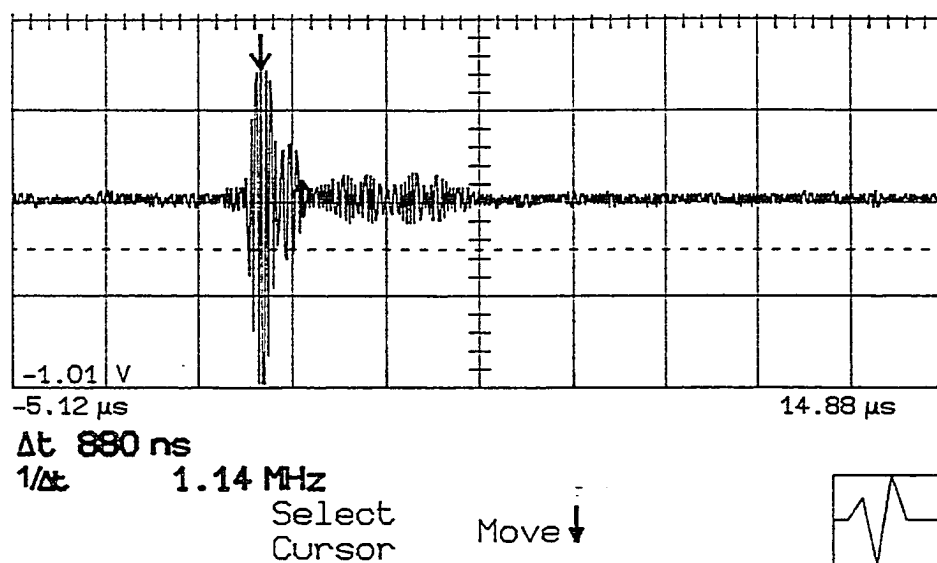


Fig. 6.1. Transient microcavitation of single bubble event in distilled water seeded with 0.1 micron polystyrene particles. The frequency of the acoustic signal causing cavitation = 2.46 MHz.

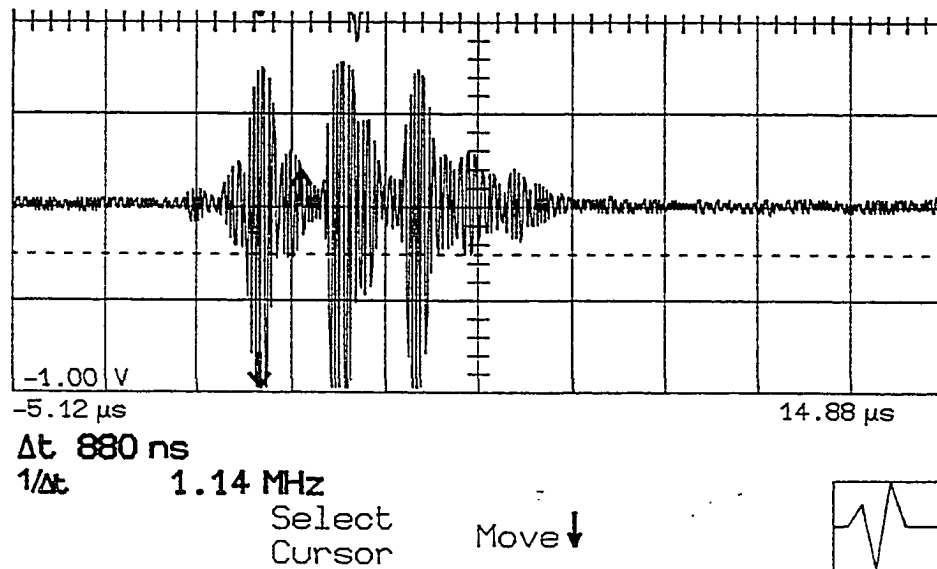


Fig. 6.2. Transient microcavitation of multiple bubble events in distilled water seeded with 0.1 micron polystyrene particles. The frequency of the acoustic signal causing cavitation = 2.46 MHz.

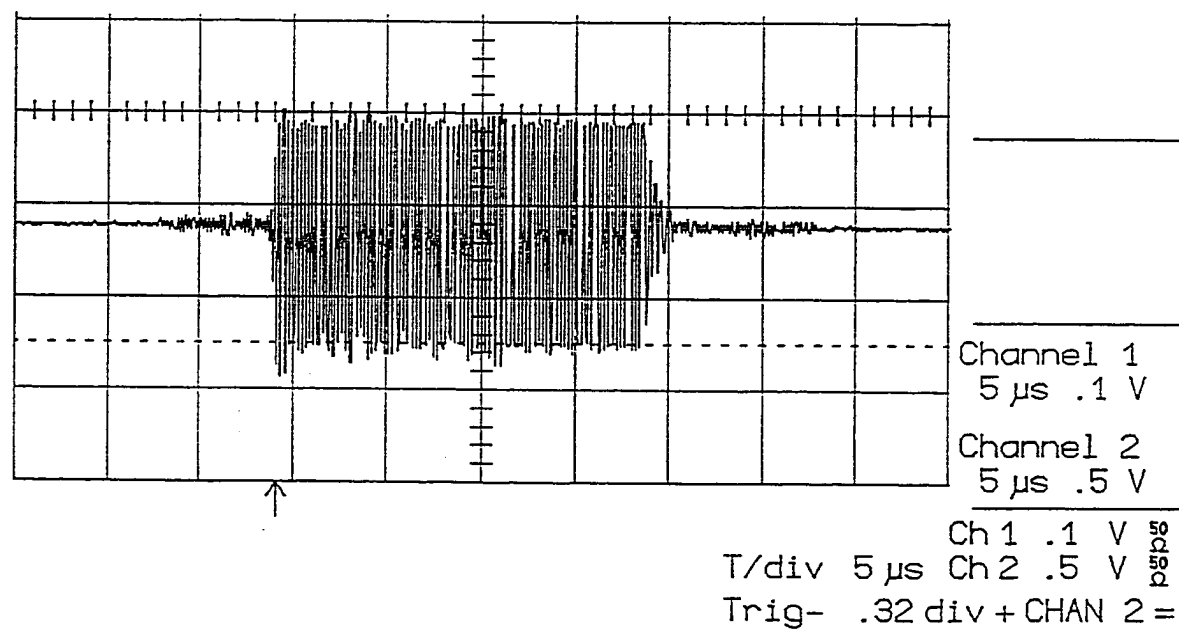


Fig. 6.3. Backscattered signal of 30 MHz from a polystyrene particle.

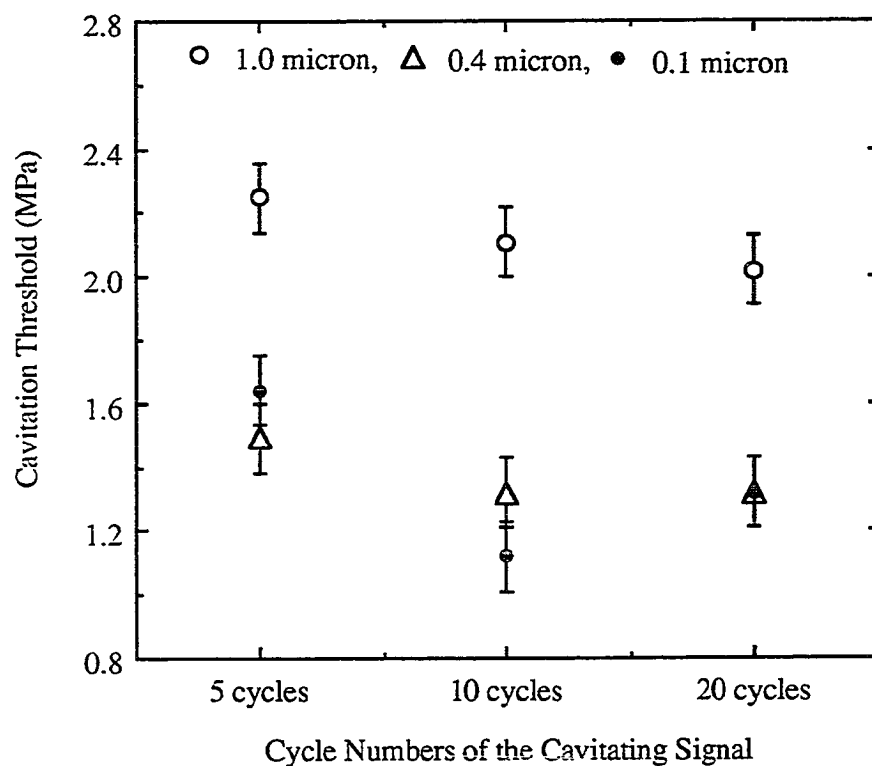


Fig. 6.4. Cavitation threshold vs the cycle numbers of the cavitating signal. The frequency of the cavitating signal = 2.46 MHz.

Cycle Number	HP output (mV)	Threshold (MPa)	Error (MPa)
5 cycles	150	2.25	± 0.11
10 cycles	110	2.11	± 0.11
20 cycles	90	2.02	± 0.11

Table 6.1. Cavitation thresholds vs the cycle numbers of the cavitating signal.

Frequency = 2.46 MHz, polystyrene particle size = 1.0 micron.

Cycle Number	HP output (mV)	Threshold (MPa)	Error(MPa)
5 cycles	100	1.49	± 0.11
10 cycles	70	1.32	± 0.11
20 cycles	60	1.32	± 0.11

Table 6.2. Cavitation thresholds vs the cycle numbers of the cavitating signal.

Frequency = 2.46 MHz, polystyrene particle size = 0.4 micron.

Cycle Number	HP output (mV)	Threshold (MPa)	Error(MPa)
5 cycles	110	1.64	± 0.11
10 cycles	60	1.12	± 0.11
20 cycles	60	1.32	± 0.11

Table 6.3. Cavitation thresholds vs the cycle numbers of the cavitating signal.

Frequency = 2.46 MHz, polystyrene particle size = 0.1 micron.

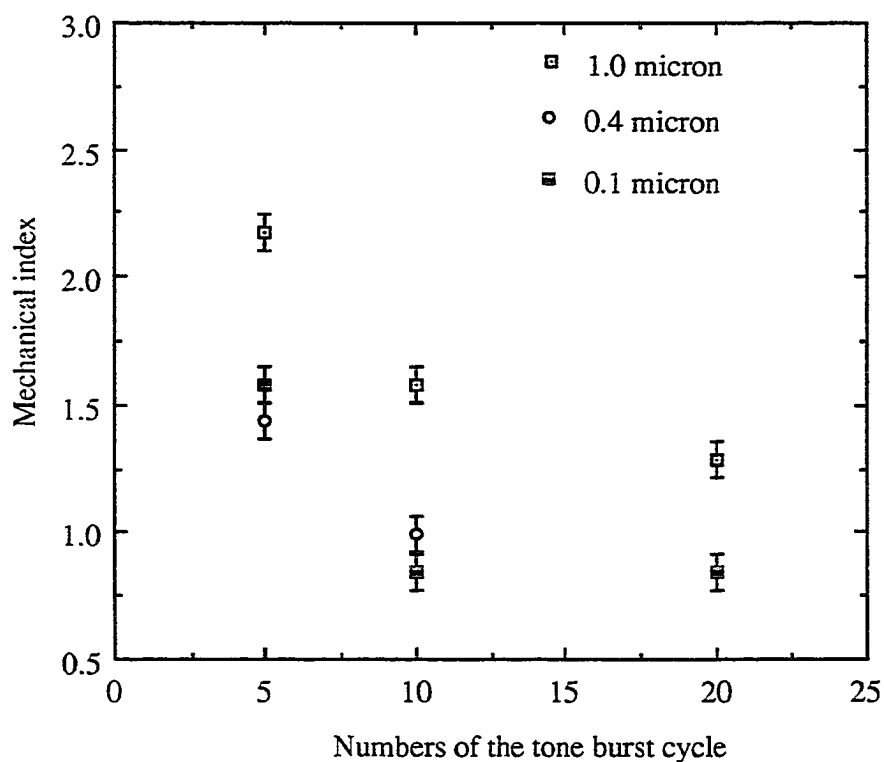


Fig 6.5. The mechanical index as a function of the tone burst cycles. The frequency = 2.46 MHz

PRF (KHz)	HP output (mV)	Threshold (MPa)	Error (MPa)
0.667	60	1.12	± 0.11
1	60	1.12	± 0.11
2	60	1.12	± 0.11

Table 6.4. Cavitation threshold vs the pulse repetition frequency (PRF) of the cavitating signal (PRF). The cavitating frequency = 2.46 MHz, polystyrene particle size = 0.1 micron, the width of probing signal = $20\mu s$.

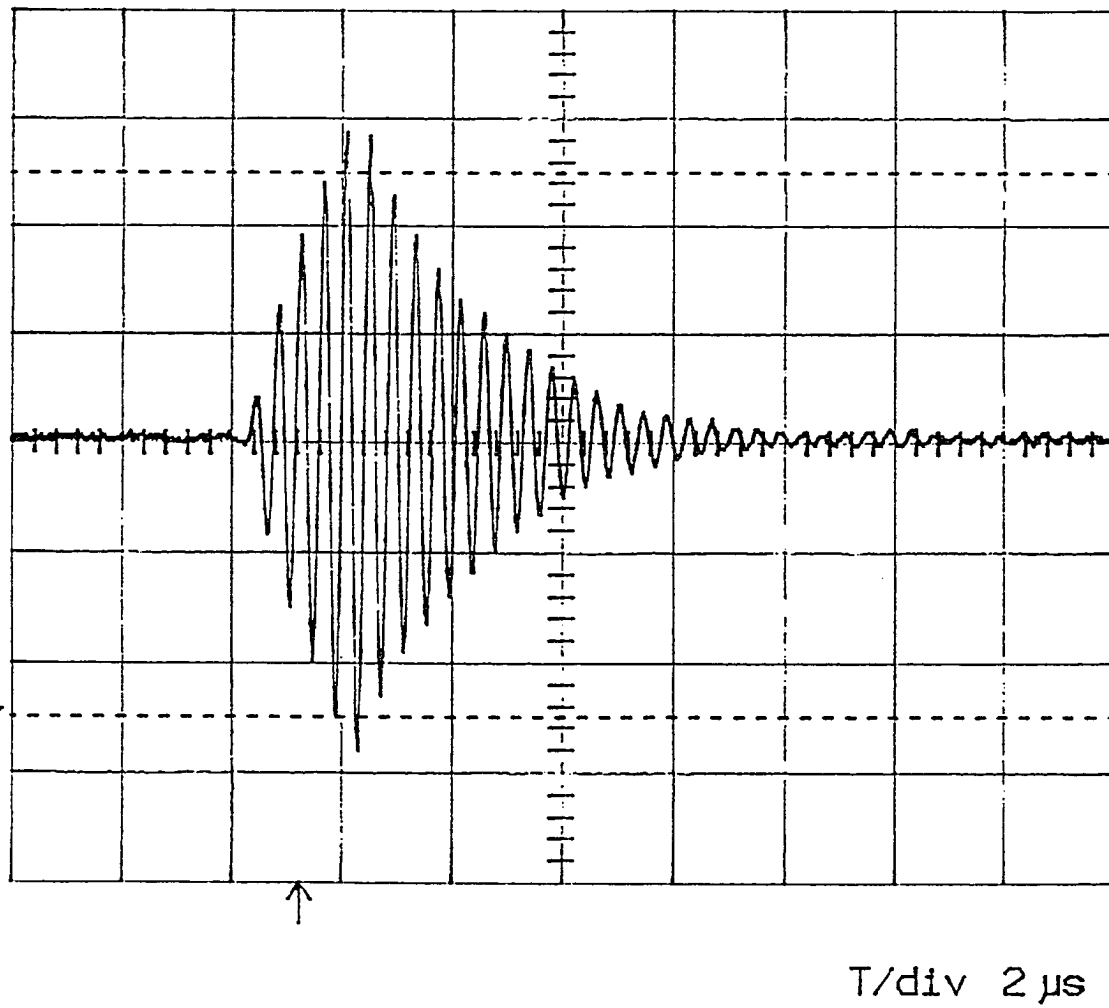


Fig. 6.6. The pressure waveform of a 5 cycles signal, 2.46 MHz.

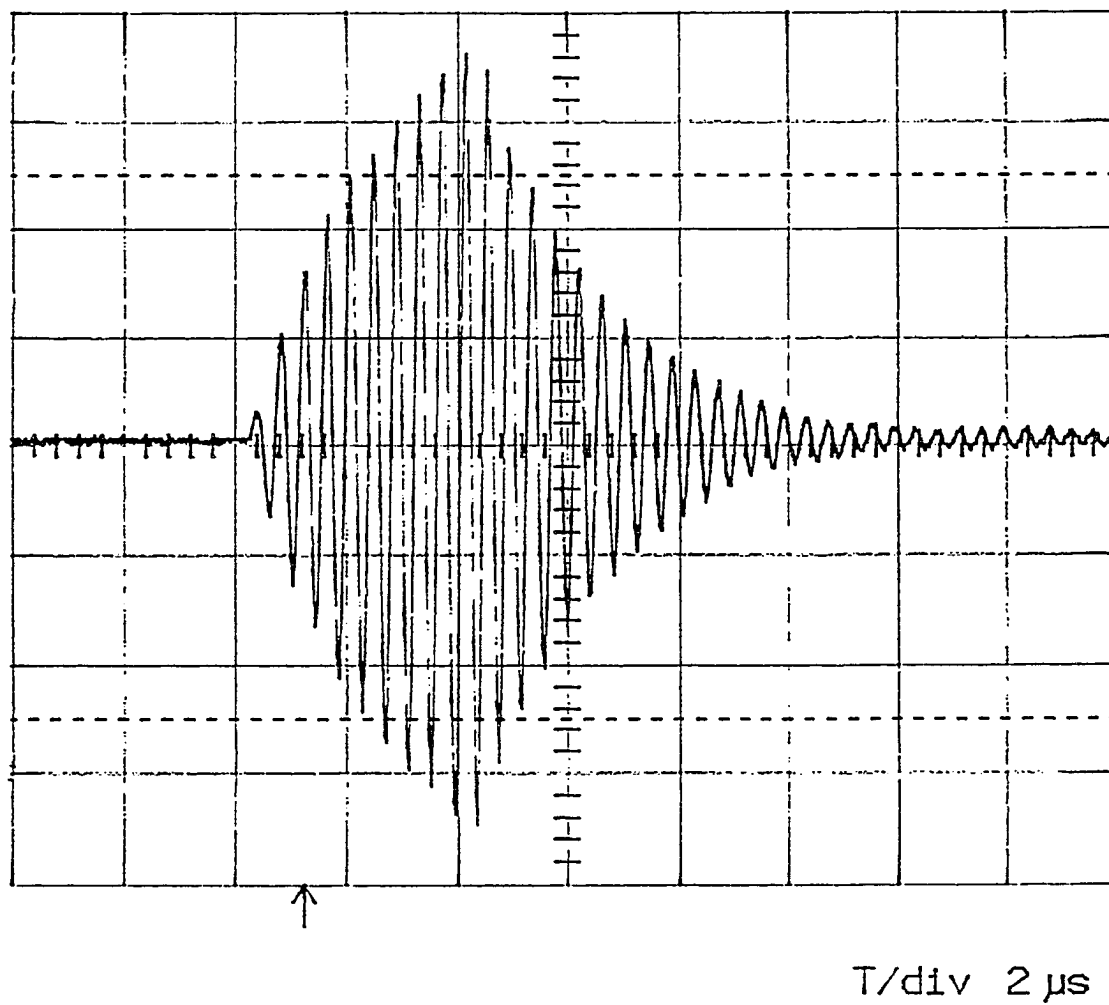


Fig. 6.7. The pressure waveform of a 10 cycles signal, 2.46 MHz.

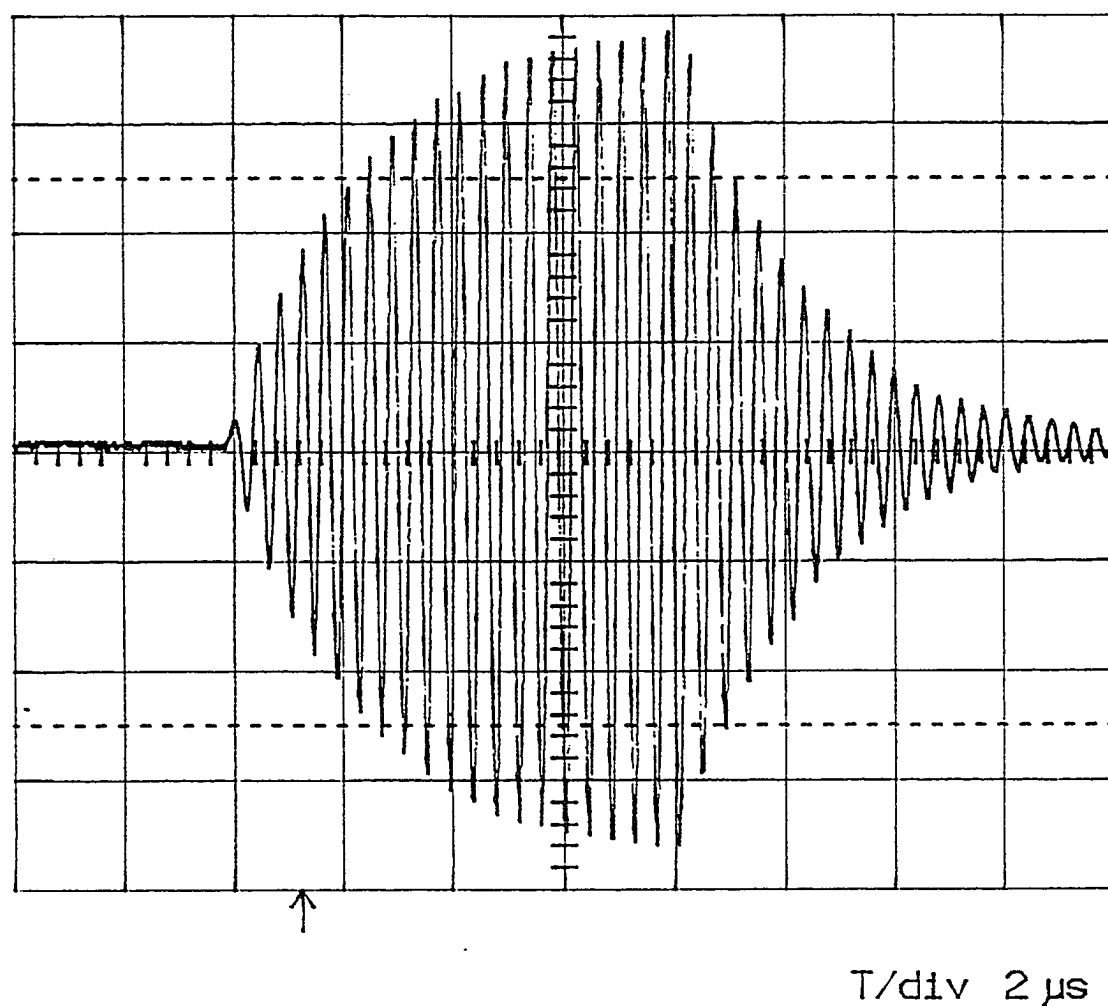


Fig. 6.8. The pressure waveform of a 20 cycles signal, 2.46 MHz.

In Fig. 6.1, a single cavitation bubble experiences a growth and collapse process. In Fig. 6.2, there is a multiple-cavitation event, the cavitation bubble grows, collapses, and rebounds several times. Acoustic pressure, host liquid, and initial nuclei size of cavitation determine cavitation behavior. In our case, we add polystyrene particles into the water. Those particles may not have the exact size as that of the initial nuclei. Therefore, the chance for cavitation to become single or multiple events is random in our case, because we can not determine the exact size of initial nucleus size. Figure 6.3 shows the backscattered signal from a polystyrene particle. There are two features which are different from that in Figs. 6.1 and 6.2: (1) the duration of backscattered signal is the same as that of the probing signal. (2) the amplitude of the signal remains constant.

Table 1 is the cavitation threshold vs the cycle number of the cavitating signal. As the cycle number increases, the cavitation threshold decreases. Since the center frequency is the same for the above data, we can easily convert the cycle number into time, in microseconds. The 5 cycle pulse corresponds to a duration of $\frac{5}{2.46\text{MHz}} = 2.03\mu\text{s}$, the 10 cycle pulse to $4.06\mu\text{s}$ and the 20 cycle pulse to $8.13\mu\text{s}$. Figs. 6.6, 6.7 and 6.8 show the pressure waveforms of the cavitating transducer at the focal region. These waveforms were received by an NTR probe, which has a frequency response to 10 MHz.

The amplitudes of the pressure signals in Figs. 6.6, 6.7 and 6.8 fluctuate dramatically even though all have the same amplitude input. This fluctuation is mainly because the cavitating transducer is air backed, narrow band and has a high Q. To use the calibration curve given in the previous chapter to estimate the acoustic pressure amplitude from the HP input, we will calculate the effective HP input, HP_{eff} , as follows: $HP_{eff} = \frac{86.2}{131.2} \text{HP} = 0.66\text{HP}$ for the 5 cycles, $HP_{eff} = \frac{110.2}{131.2} \text{HP} = 0.84\text{HP}$ for the 10 cycles and $HP_{eff} = \text{HP}$ for the 20 cycles, where the HPs are the outputs of the HP signal generator, which are listed in the second columns of Table 6.1, 6.2 and 6.3. The thresholds in the third columns are obtained by using the effective HP input based on the calibration discussed in the previous chapter.

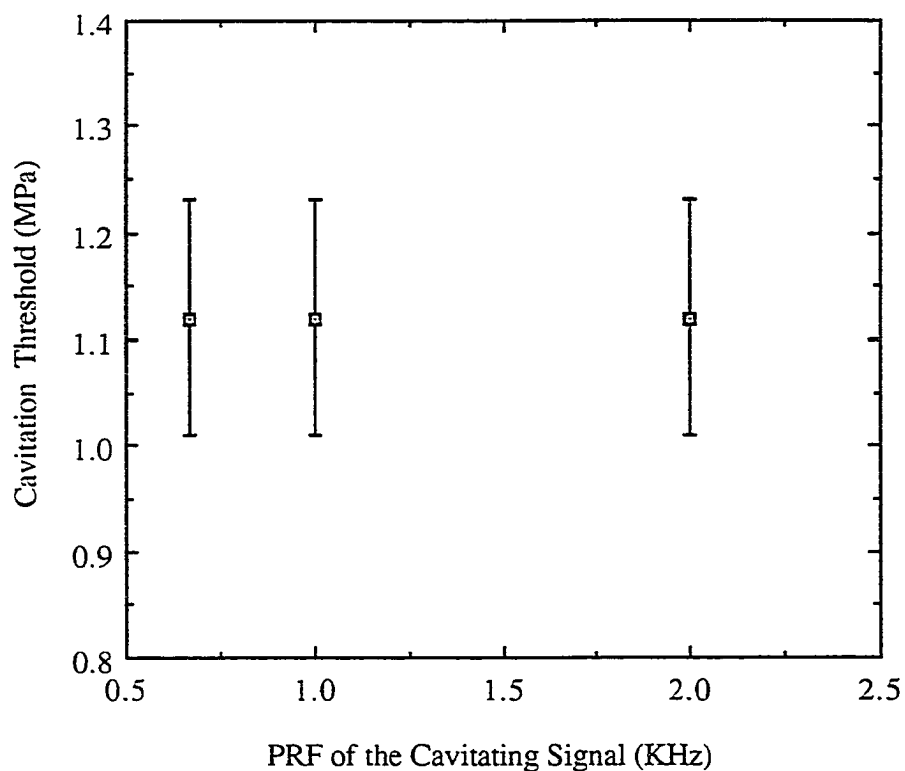


Fig. 6.9. The cavitation threshold vs the pulse repetition frequency (PRF) of the cavitating signal. Frequency = 2.46 MHz, cycle number = 10, polystyrene particle size = 0.1 micron.

Fig. 6.9 shows the results when we change the PRF, but not the frequency of the cavitating signal and polystyrene particle size. There is no clear indication that the PRF affects the cavitation threshold.

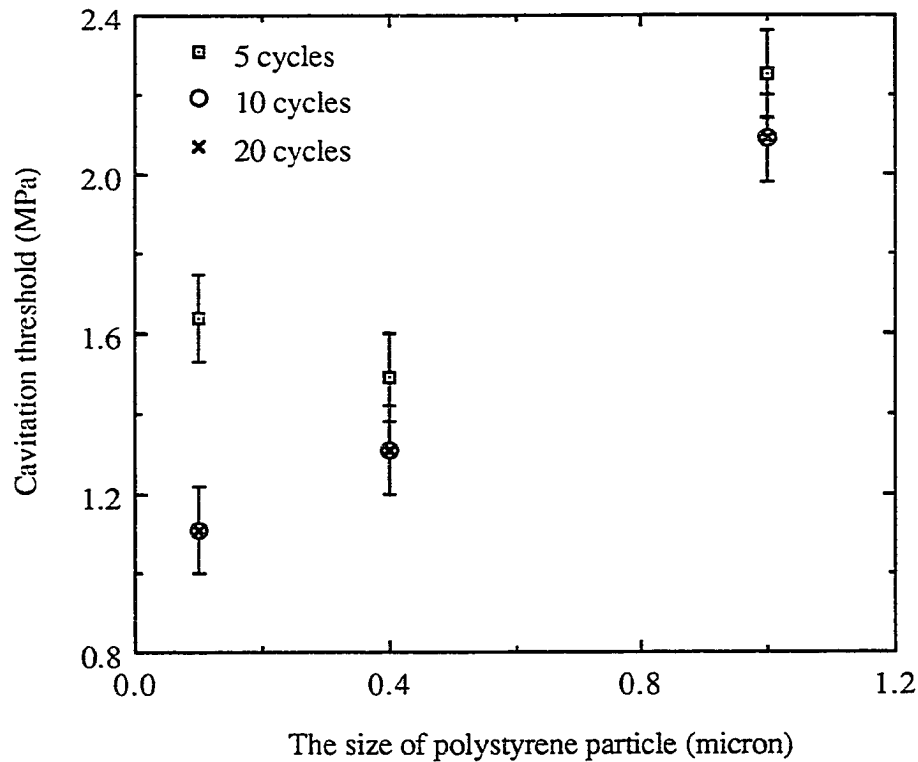


Fig. 6.10. The cavitation threshold vs the size of polystyrene particle added in the water.

Frequency = 2.46 MHz.

Fig. 6.10 shows the changes in cavitation threshold as the size of the polystyrene particle added in the water changes. Changing the polystyrene particle size means a change in the initial size of the cavitation nuclei. According to the approximate theory proposed by Apfel [Apfel, 1986] and modified by Holland [Holland and Apfel, 1989], the nuclei size, R_0 , and the amplitude, P_A , of the acoustic pressure causing the transient cavitation has the following relationship,

$$f = \frac{\frac{1}{3\pi R_0} \sqrt{\frac{P_0 \xi}{\rho}} \left(\sqrt{\frac{p-1}{p}} + \sqrt{\frac{p-p_b}{p}} \right)}{\left(\frac{T'}{T_0(\gamma-1)(\frac{\xi}{3}+1)} \right)^{1/3} - 0.46 + \frac{4\mu}{R_0} \sqrt{\frac{2}{\rho P_0 \xi}} + \frac{2\sigma}{P_0 R_0} \left(\frac{1}{p-p_0} \right)^{1.5} \sqrt{\frac{\xi}{3}}}, \quad 6.1$$

where p_b is the normalized Blake threshold pressure given by

$$p_b \equiv \frac{P_B}{P_0} = 1 + \frac{4}{9} X_B \sqrt{\frac{3X_B}{4(1+X_B)}}, \quad 6.2$$

where $X_B = \frac{2\sigma}{R_0 P_0}$, P_0 is the ambient pressure, $p = \frac{P_A}{P_0}$, $\xi = p + p_b - 2 + \sqrt{(p-1)(p-p_b)}$, T_0 is the initial ambient temperature, T' is the maximum cavitation collapse temperature, γ is the ratio of the specific heat, ρ , μ and σ are the mechanical properties of the host liquid (water in our case)-density, viscosity and surface tension.

Eq. 6.1 has two parameters to be solved for: the maximum cavitation collapse temperature, T' , and cavitation pressure threshold, p . Following the same procedures in [Holland and Apfel, 1989], we set $T' = 5000\text{K}$. Eq. 6.1 is an implicit function of p , and cannot be analytically solved for without approximation, but can easily be solved by numerical method. Using the bisection method to solve the Eq. 6.1, we calculate the cavitation pressure threshold, $P_A = pP_0$, in water for different initial bubble radius. The results are shown in Fig. 6.11.

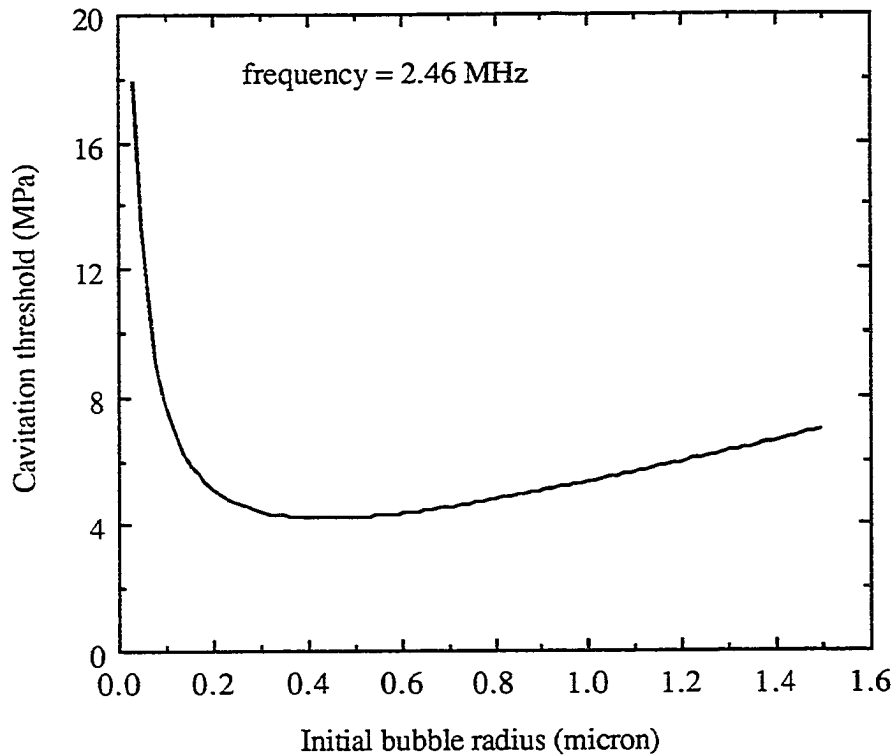


Fig. 6.11. The calculated cavitation threshold, P_A , in water as the function of the initial bubble radius with the following parameters- $\rho = 1000.0 \text{ kg/m}^3$, $\mu = 0.0010 \text{ kg/m-s}$, $T_0 = 300.0 \text{ K}$, $T^* = 5000.0 \text{ K}$, $\gamma = 1.4$ and $P_0 = 1.0 \times 10^5 \text{ Pa}$.

Fig. 6.10 shows that the cavitation thresholds increase as the sizes of the polystyrene particle become larger when the initial nuclei size is larger than a certain value. This trend agrees with the theoretical prediction shown in Fig. 6.11 that cavitation threshold increases as the initial bubble radius increases. Such an observation is consistent with the assumption that liquid inertial effect becomes important and therefore it is actually easier to nucleate transient cavitation from smaller bubbles.

6.1.2 The Cavitation Thresholds and Mechanical Index for the 4.3 MHz Transducer

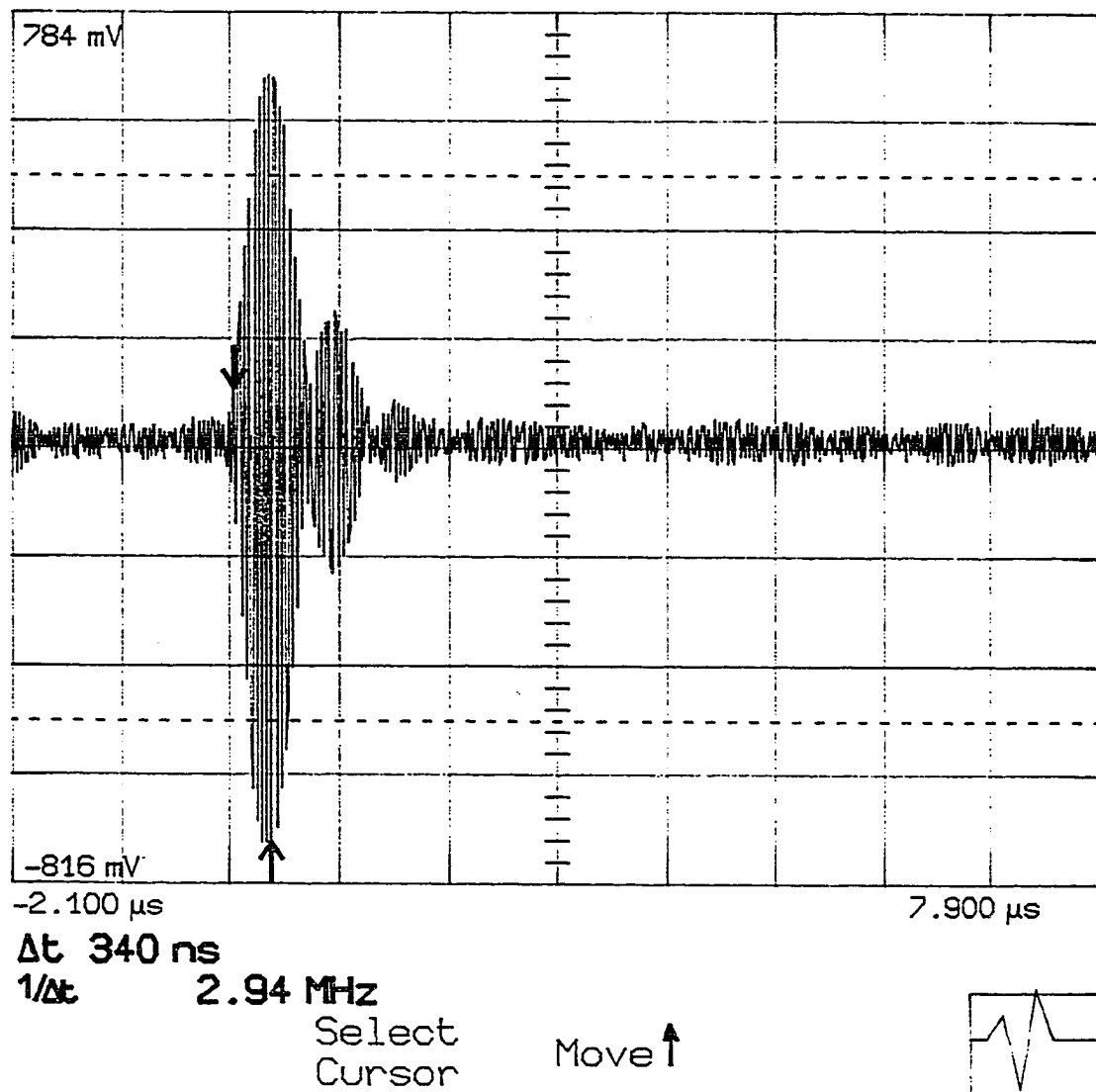


Fig. 6.12. Transient microcavitation of single event in distilled water seeded with 0.1 micron polystyrene particles. The frequency of the acoustic signal causing cavitation = 4.3 MHz.

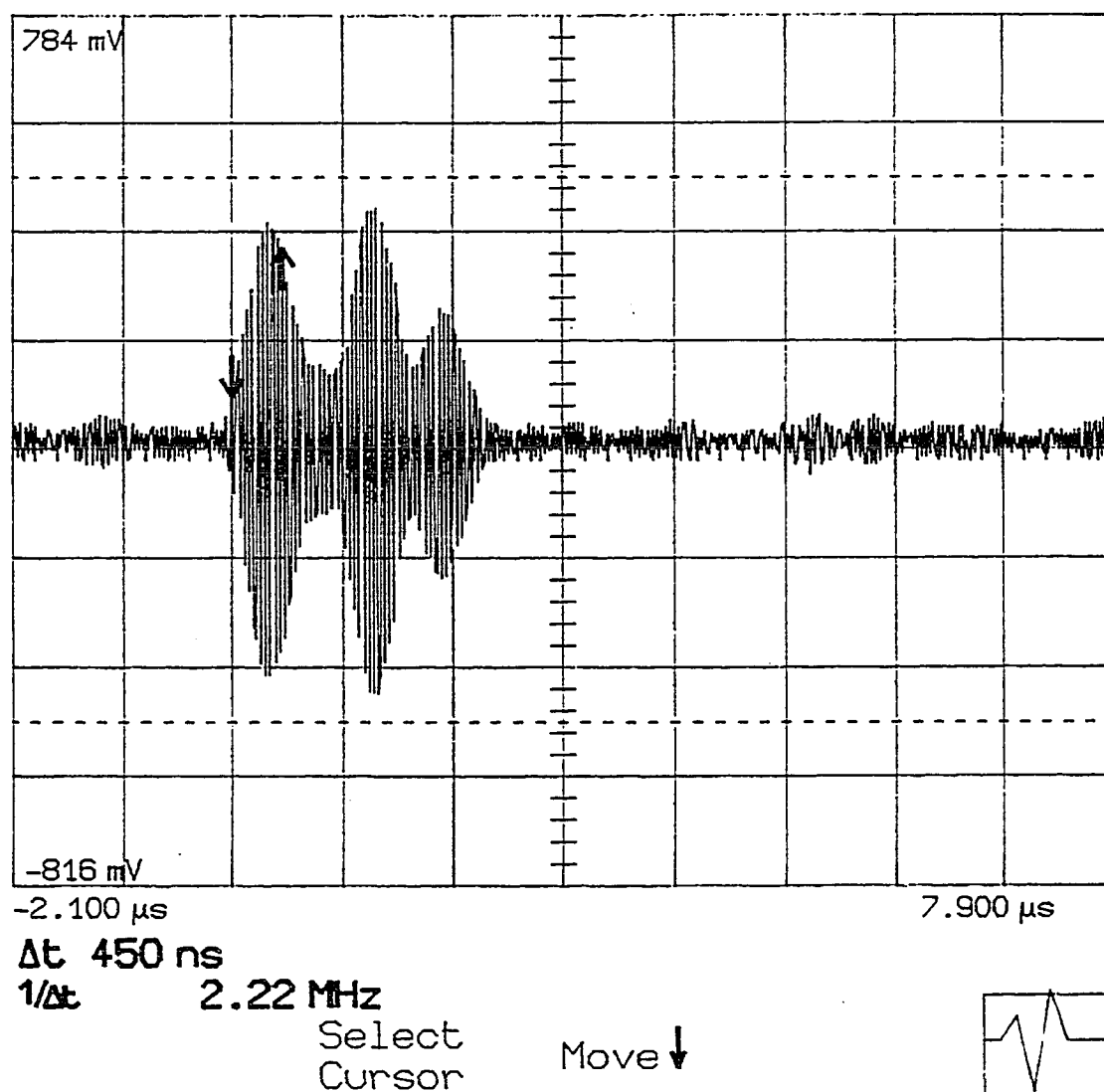


Fig. 6.13. Transient microcavitation of multiple events in distilled water seeded with 0.1 micron polystyrene particles. The frequency of the acoustic signal causing cavitation = 4.3 MHz.

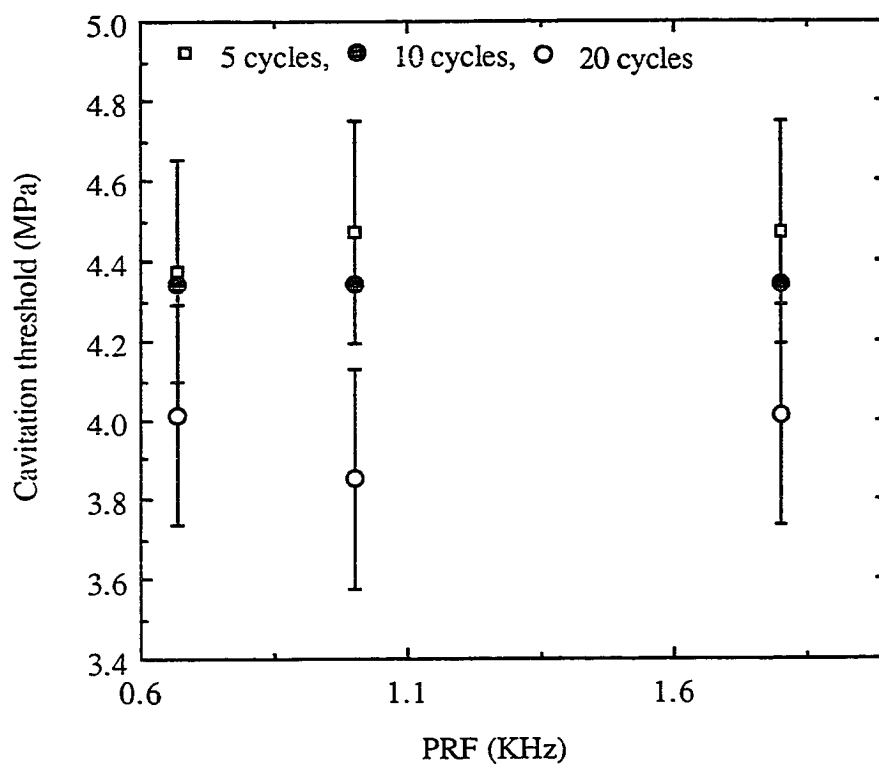


Fig. 6.14. Cavitation thresholds as a function of PRF.

Center frequency of the cavitating signal = 4.3 MHz.

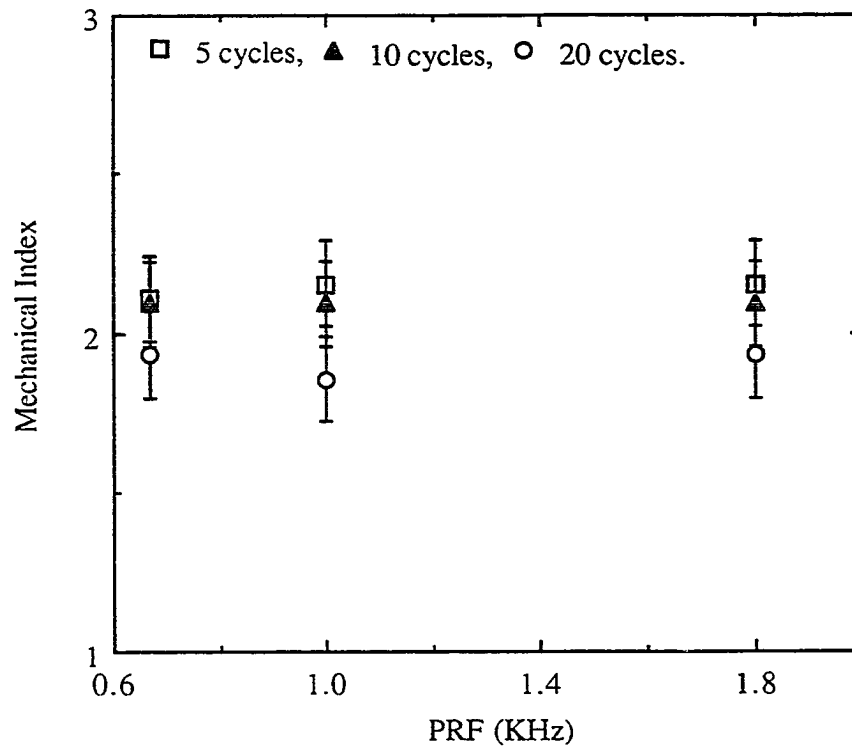


Fig. 6.15. Mechanical indexes as a function of PRF.

Center frequency of the cavitating signal = 4.3 MHz

The envelopes of the backscattered signals from the cavitation caused by the 4.3 MHz acoustic pressure are similar to that by the 2.46 MHz acoustic pressure. In Fig. 6.11, the microcavitation experiences a single growth and collapse process. The duration of the microcavitation is about $0.4\mu s$, which is longer than the cavitating signal period of $\frac{1}{4.3MHz} = 0.23\mu s$, but is on the same order as the period. Fig. 6.12 shows microcavitation exhibiting multiple events. The cavitation experiences several growth and collapse cycles. Figures 6.14 and 6.15 give cavitation thresholds and mechanical indices

vs. PRF (pulse repetition frequency) for different cavitating pulse widths. We see that the cavitation thresholds remain constant for different PRFs. The cavitation threshold decreases as the pulse width increases.

6.1.3 Some Experimental Study on the Cavitation Scattering

For our present work, the cavitation bubble size is assumed to be less than $10 \mu\text{m}$. During the bubble growth period, the bubble is spherically symmetrical and behaves like a soft sphere to the probing wave. For this reason, during the growth period, a fluid sphere is well suited for modeling the scattering process. The collapse time is much shorter than the growth time, and the bubble may become asymmetric. Since the operating frequency for the probing transducer is 30 MHz, and the wavelength of the probing wave in distilled water is about 50 microns in water, we can use a long wavelength approximation which is not sensitive to bubble shape. Although there is no direct way to measure precisely the mechanical properties of the material inside the cavitation bubble, we assume that both gas and vapor fill the cavitation bubble.

6.1.3.1 Estimation of Bubble Radius vs Time

By assuming the probing wave is a plane wave of the form $p_0 e^{i(\vec{k} \cdot \vec{r} - \omega t)}$ and the microcavitation bubble is a soft sphere of radius a , and by using the theory of long wavelength scattering discussed in chapter 2, the backscattered wave p_{sc} can be expressed as follows after omitting the time factor for convenience,

$$p_{sc} = p_0 \frac{[ka(t)]^3}{3kr} \left[\frac{\kappa_{in}(t) - \kappa}{\kappa} - \frac{3\rho_{in}(t) - 3\rho}{2\rho_{in}(t) + \rho} \right] = w(t)a^3(t), \quad 6.2$$

where κ is the compressibility and

$$w(t) = p_0 \frac{k^3}{3kr} \left[\frac{\kappa_{in}(t) - \kappa}{\kappa} - \frac{3\rho_{in}(t) - 3\rho}{2\rho_{in}(t) + \rho} \right]. \quad 6.3$$

where $w(t)$ is an intermediate term.

If both the position and the driving frequency of the transducer are fixed, k and kr will be constants. During bubble growth and collapse, both compressibility $\kappa(t)$ and density $\rho(t)$ are slowly varying quantities to be considered as constants. Therefore, the quantity $w(t)$ describing the contrasts for both compressibility and density will not change much during the period of cavitation. The backscattering strength and the bubble radius will have the following simple relationship,

$$p_{sc} = da^3(t), \quad 6.4$$

where $d = \frac{p_0}{3kr} \left[\frac{\kappa_e - \kappa}{\kappa} - \frac{3\rho_e - 3\rho}{2\rho_e + \rho} \right]$ and is constant

or

$$a(t) = \left[\frac{p_{sc}(t)}{d} \right]^{\frac{1}{3}}, \quad 6.5$$

Since the frequency of the acoustic pressure which causes the cavitation is in the range of megahertz, the size of the cavitation bubble will be in the range of microns, the corresponding lifetime will be in the range of microseconds, and the bubble grows and collapses so rapidly that there is almost no heat transfer or diffusion.

6.1.3.2. Demodulation of Backscattered Signal

The envelope of the backscattered signal represents the scattering strength, and its fluctuation with time reflects the changes in scatterer size. Through our theoretical analysis, we see that by obtaining the envelope of the backscattered signal, a normalized bubble radius $a(t)$ imbedded in the scattered wave can be achieved. The signals for the cavitating transducer were tone bursts with different cycles ranging from 5 to 12 cycles which corresponds to $2 \mu s$ to $6 \mu s$. Therefore, the probing wave with duration of $20 \mu s$ is long enough to interrogate the microcavitation process.

Fig. 6.17a is a backscattered signal from microcavitation exhibiting multiple events. We see it is an amplitude modulated wave (AM) with the carrier frequency 30 MHz. The envelope of this AM wave represents the expanding and collapsing process of the transient bubble.

An amplitude modulated wave can be expressed as

$$x(t) = p_{sc}(t) \cos(\omega t + \varphi),$$

where ω is the frequency of carrier wave, φ the initial phase and $p_{sc}(t)$ is the modulating wave which represents the envelope of $x(t)$. The extraction of $p_{sc}(t)$ from $x(t)$ can be completed by using Hilbert transform defined as follows,

$$\hat{x}(t) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{x(\tau)}{t - \tau} d\tau = x(t) \otimes h(t), \quad 6.6$$

where $\otimes$ stands for convolution and $h(t)$ is

$$h(t) = \frac{1}{\pi t}. \quad 6.7$$

Since the frequency of probing signal is higher than that of the driving signal, we have (see Stark, H *et al.*)

$$\hat{x}(t) = p_{sc}(t) \sin(\omega t + \varphi). \quad 6.8$$

The envelope $p_{sc}(t)$ is

$$|p(t)| = \sqrt{x(t)^2 + \hat{x}(t)^2}. \quad 6.9$$

In our work, all the data are discrete and the Hilbert transform is implemented by the following convolution,

$$\hat{x}[i] = \sum_{k=0}^M x[i-k] h[k], i = 0, \dots, N, \text{ where}$$

$h[k]$, $k=0, \dots, M$, is given by

$$h[k] = \frac{I_0 \left\{ \beta (1 - [(k - M_d) / M_d]^2)^{1/2} \right\}}{I_0(\beta)} \left\{ \frac{2 \sin^2[\pi(k - M_d) / 2]}{\pi (k - M_d)} \right\}, \quad 6.10$$

where I_0 is the modified Bessel function of zero order, $\beta=2.629$, $M=30$, $M_d=15$ and $k=0, \dots, M$. We notice that $h[k]$ gives an exact 90° phase shift for the carrying wave plus a delay of 15 samples.

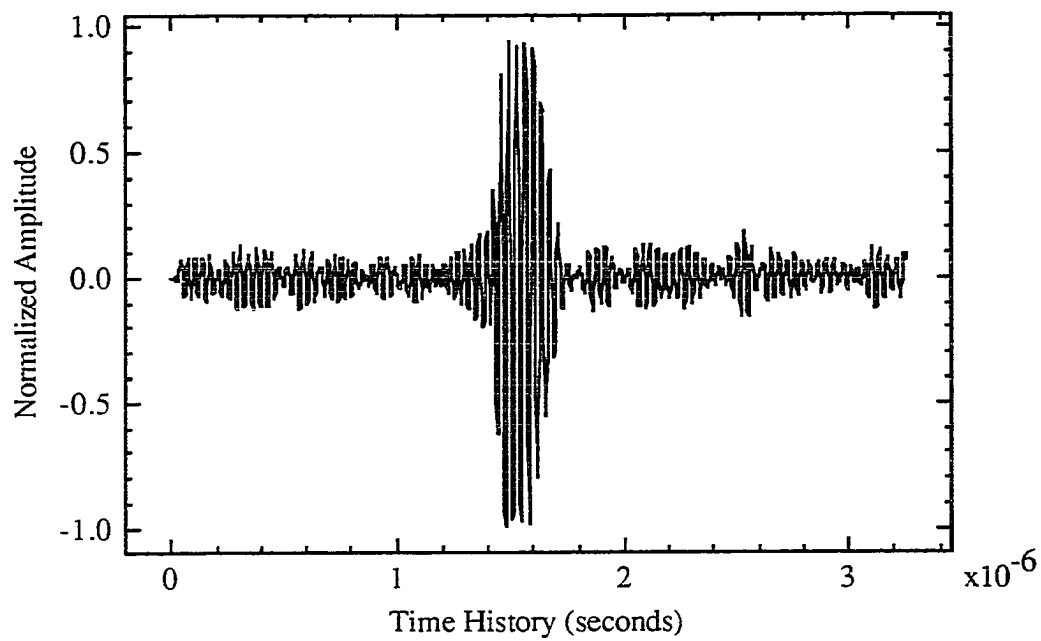


Fig. 6.16a. Backscattered signal from a transient microcavitation generated by 2.46 MHz acoustic wave.

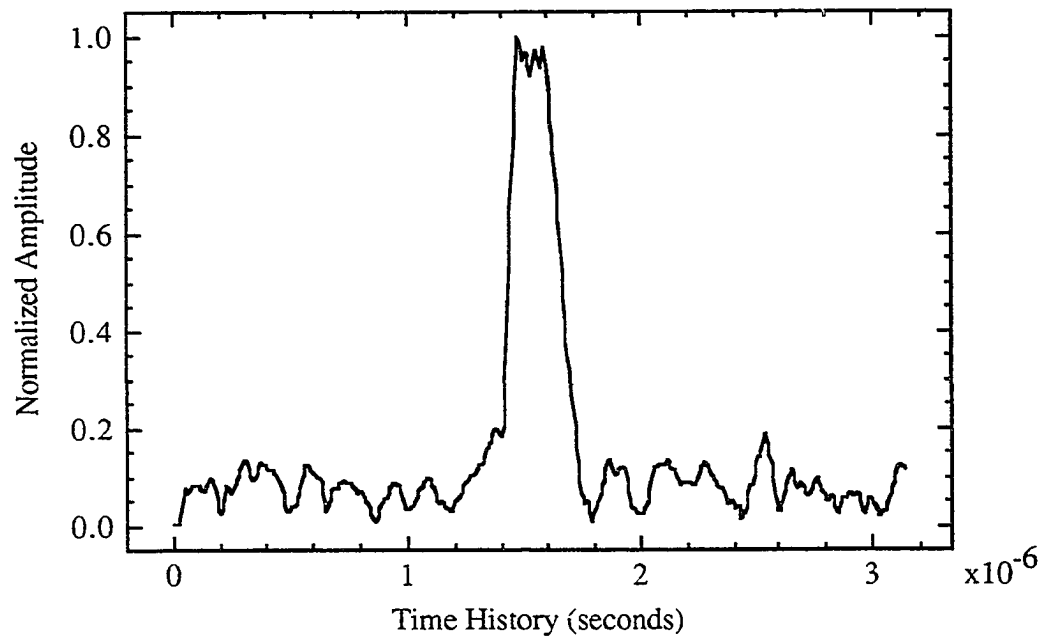


Fig. 6.16b. Envelope of the backscattered signal in Fig. 6.16a.

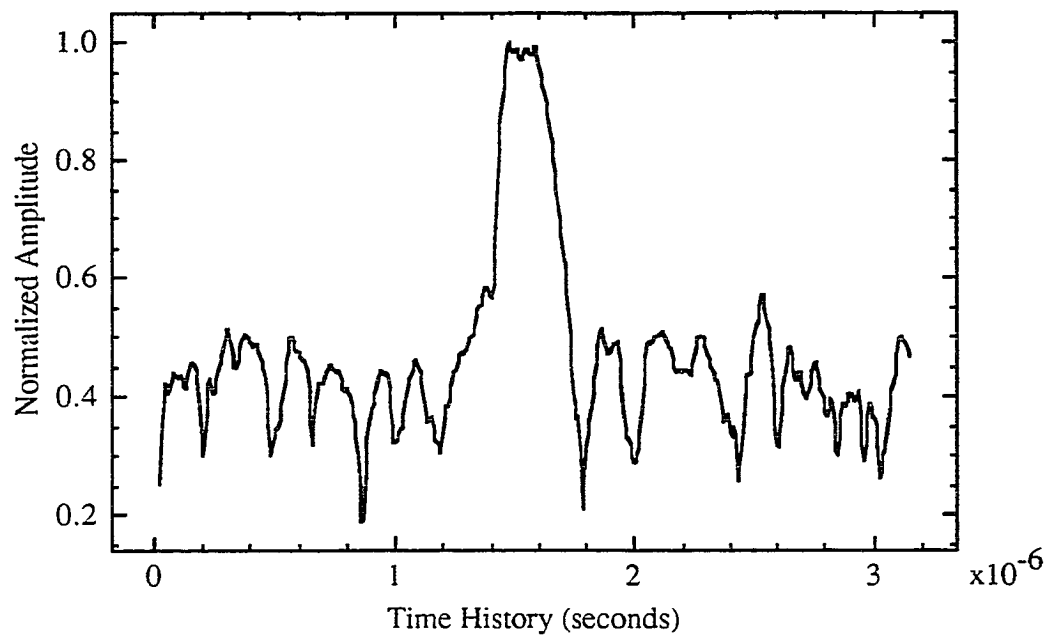


Fig. 6.16c, The curve of bubble radius versus time for the transient microcavitation in Fig. 6.16a.

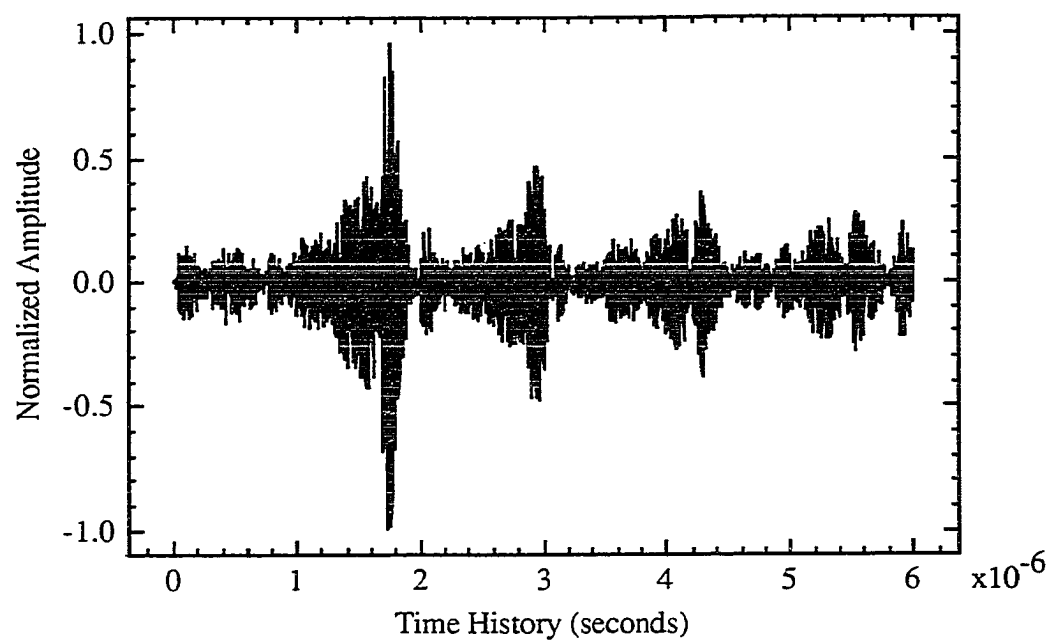


Fig. 6.17a. Backscattered signal from the microcavitation generated by 2.46 MHz acoustic wave.

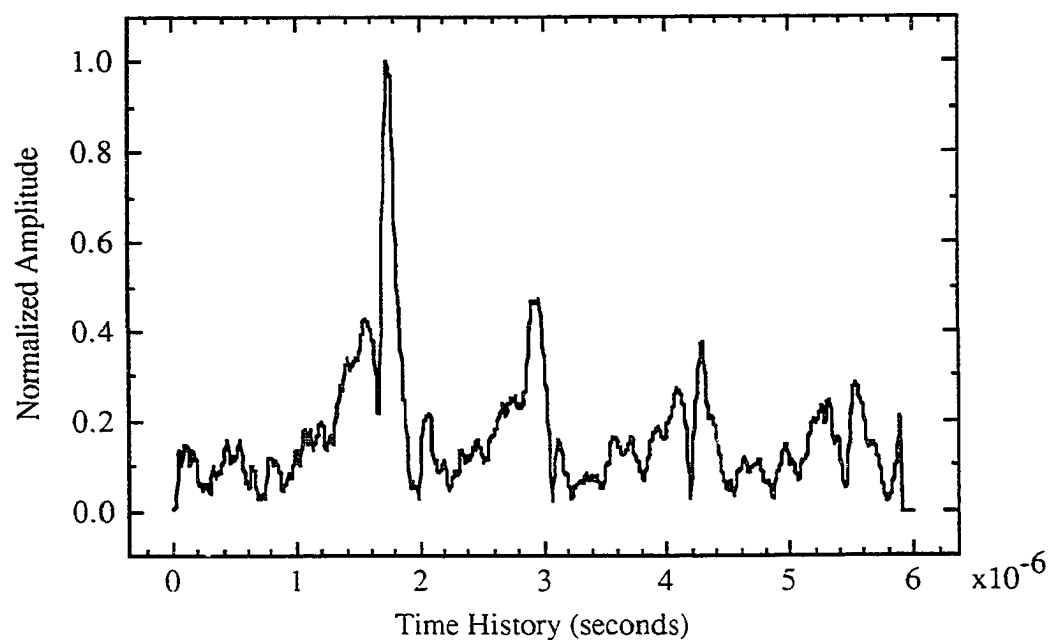


Fig. 6.17b. Envelope of the backscattered signal in Fig. 6.17a.

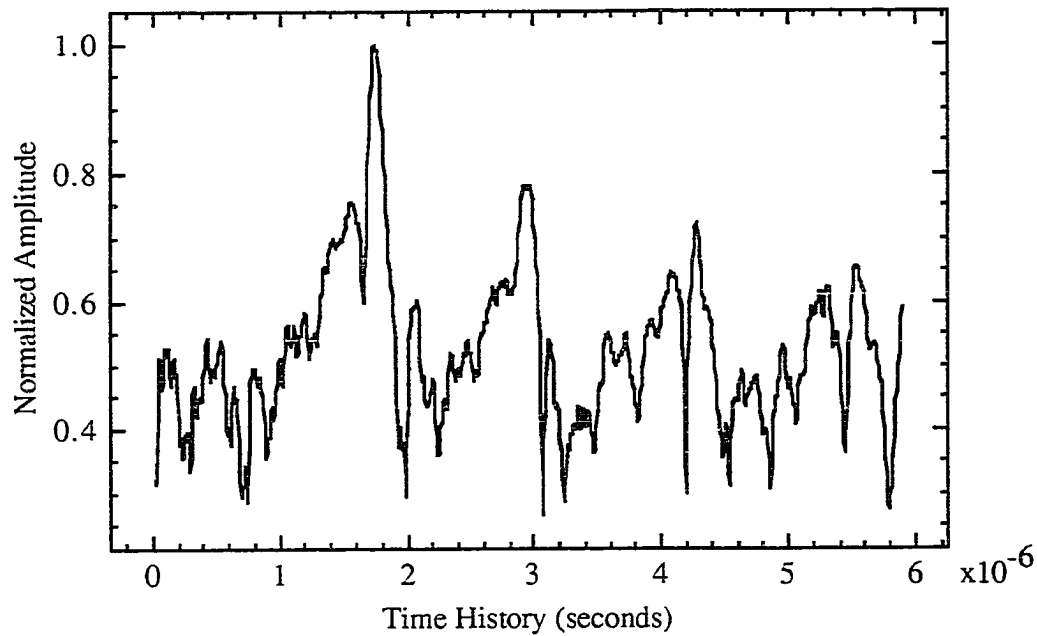


Fig. 6.17c. The curve of bubble radius versus time for the microcavitation in Fig. 6.17a.

Figs. 6.16a and 6.17a are the backscattered signals from the microcavitation caused by the cavitating transducer. Figs 6.16b and 6.17b are the corresponding envelopes. Fig. 6.16c and 6.17c are the curves of bubble radius versus time. In Fig. 6.16a, the cavitation bubble undergoes a single growth and collapse process. With a polystyrene particle as a nuclei driven by the 2.46 MHz acoustic pressure, a tiny bubble grows slowly until it reaches the maximum radius, then suddenly collapses. The complete event happens in a very short time and the bubble does not have time to resonate. The duration of the transient microcavitation is about $T_b = 0.60 \mu s$, and the period of the driving signal is $T_d = 0.41 \mu s$. We have, therefore,

$$T_b = 1.46 T_d.$$

In Fig. 6.17a, we see the cavitation bubble undergoes multiple growth and collapse cycles because the driving signal is a tone burst with several cycles. The backscattered signals are pseudo-periodic over the period of the driving signal. Estimating the whole duration of cavitation and dividing it by the number of bubble growth and collapse processes, the average cavitation period is approximately 2.5 MHz. For the first event, a polystyrene particle functions as a nuclei, and for the rest, it is the tiny gas bubble that serves as a nuclei since the bubble may not completely collapse.

6.1.3.3. Estimation of Maximum Bubble Radius

As shown in Fig. 6.16a, the transient microcavitation happens in such a short time that it is believed that there is no energy and mass transfer during this period, and the bubble collapses like a void. The maximum bubble radius $R_{\max}$ is proportional to T_b and can be estimated from the Rayleigh formula,

$$T_b = 0.915 R_{\max} \sqrt{\frac{\rho}{p_{\text{stat}} - p_v}}, \quad 6.9$$

where T_b is bubble collapse time, ρ the density of surrounding liquid, p_{stat} the static pressure and p_v the vapor pressure of the liquid.

Letting $p_{\text{stat}}=100$ kPa, $p_v=0$ and $\rho=1000$ kg/m³, $T_b = 0.55$ μ s, we obtain $R_{\max} = 6.56$ μ m by using Eq 6.9. Treating the polystyrene size as the initial nuclei size R_0 of the transient microcavitation, we see that the initial bubble nucleus can expand by

$$\frac{R_{\max}}{R_0} = 6.47, \quad 6.10$$

given the driving frequency of 2.46 MHz, which is within the range of the theoretical predictions computed by Flynn and Church [Flynn and Church, 1988]. The volume of the microbubble will increase by about 271 times; therefore, the density within the microbubble will decrease by the same amount. Such a tremendous decrease in density will cause the transient microcavitation to behave like a void during its collapse.

6.1.4 The Detection of Doppler Frequency

6.1.4.1 Results and Analysis

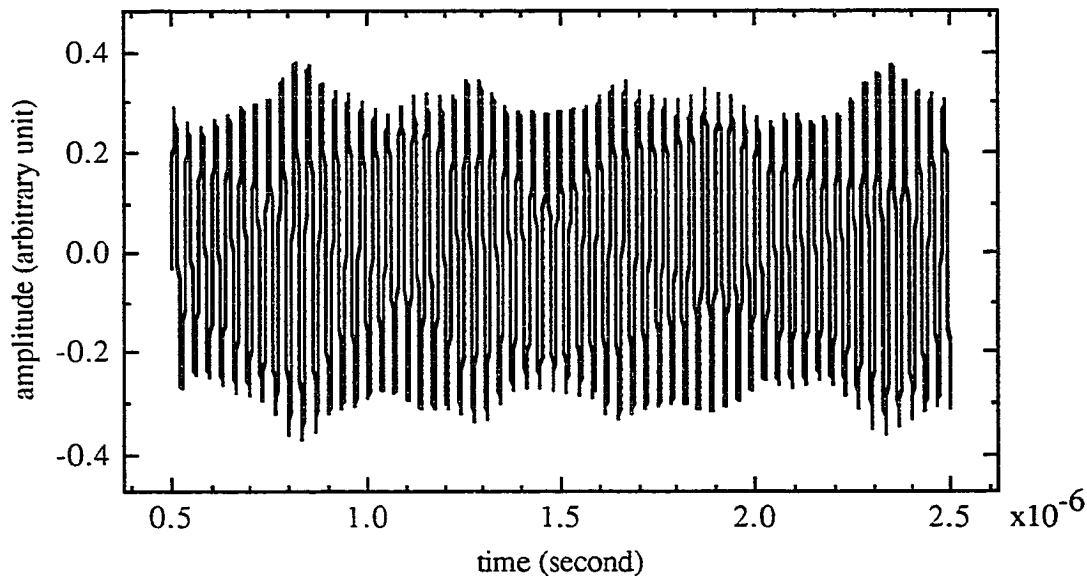


Fig. 6.18a. Backscattered signal from a non-cavitation target

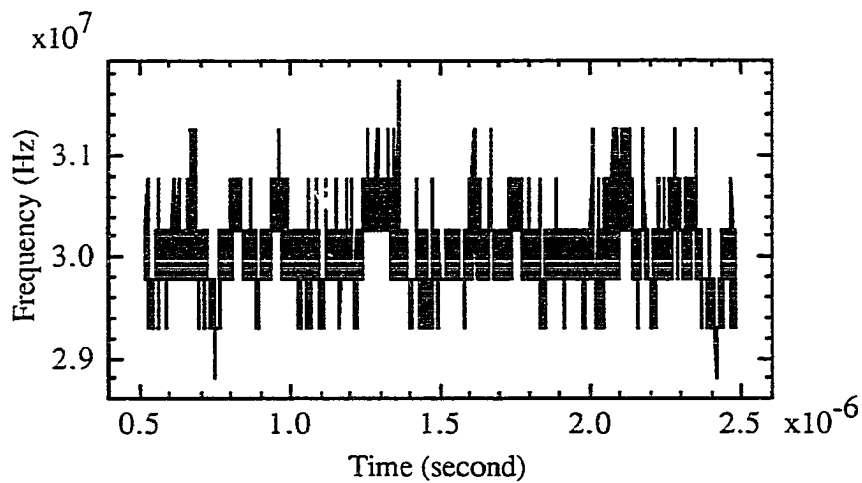


Fig. 6.18b. Frequency estimation of the signal shown in Fig. 6.18a by Prony's method.

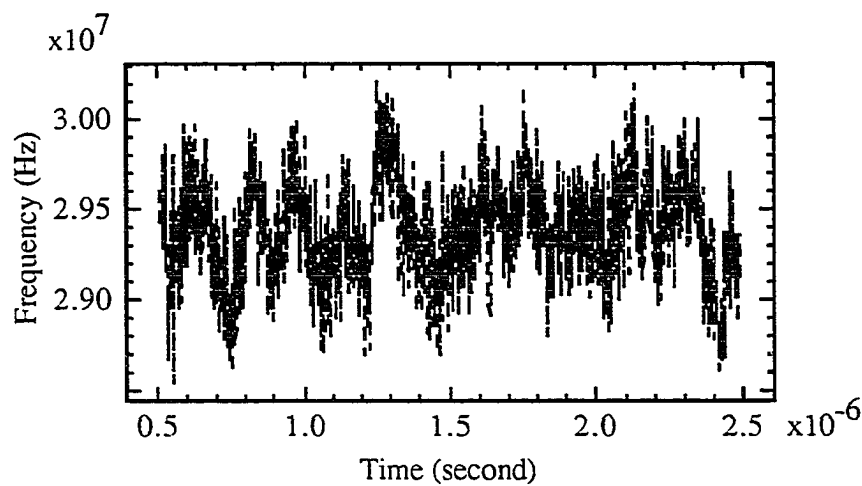


Fig. 6.18c. Frequency estimation of the signal shown in Fig. 6.18a by eigenanalysis method.

Figures 6.18b and 6.18c show the analysis results of the backscattered signal from a non-cavitation target. The estimated frequency with Prony's method shows that the

frequency fluctuation is within -1.0 MHz to 1.0 MHz. Although the actual frequency should be 29.4 MHz, the discrepancy may be attributed to two main factors, 1) the limited duration signal is not a very good stationary signal, which more or less violates the ergodic assumption; 2) the noise presence in the experimental data deteriorates the result because Prony's method is not designed to handle noise. For the result by using eigenanalysis the frequency fluctuation is within -0.4 MHz to 2.1 MHz. This fluctuation may also be caused by two main factors: 1) the signal with limited duration does not meet the ergodic assumption well; 2) rounding errors. The frequency fluctuation is random, and no obvious trend can be seen. Judging from our computation experience, it seems that frequency fluctuation by the eigenanalysis method is more computer-related than Prony's method is. Both methods can give good results. The estimation error for Prony's method is $\pm 3.4\%$, and it is -1.36% to 7.1% for eigenanalysis method.

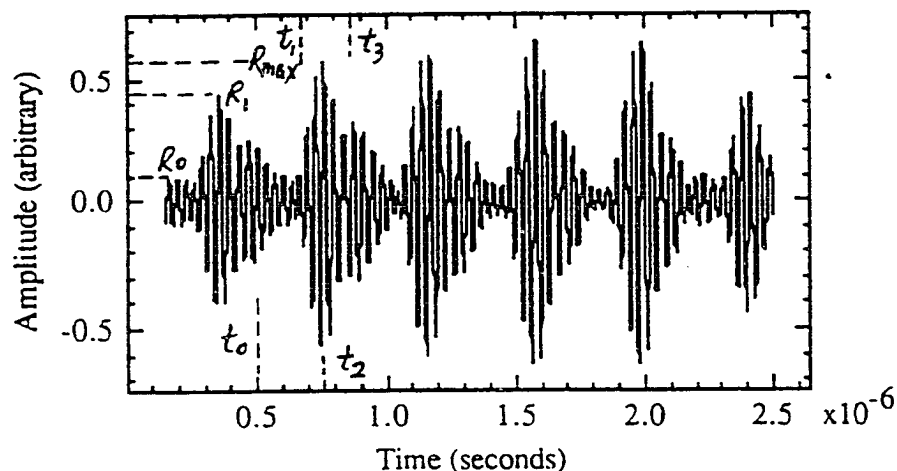


Fig. 6.19, the backscattered signal from the microcavitation of six cycles.

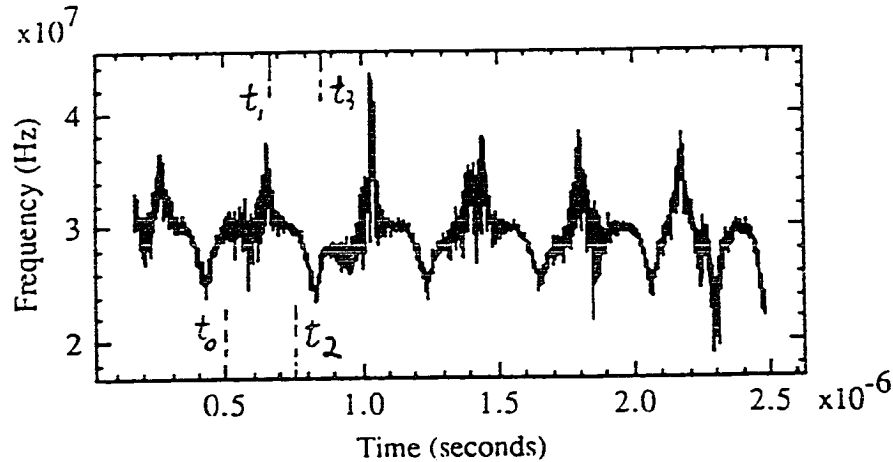


Fig. 6.19b, the time-frequency curve of Fig. 6.18a.

Fig. 6.19a is a backscattered signal from multiple cavitation events. Fig. 6.19b is the corresponding transient frequency estimation. As shown in Fig. 6.19a, the cavitation bubble experiences multiple growth and collapse phases. The transient frequency curve of the backscattered signal also has a pseudo-periodic behavior. This behavior is completely determined by the bubble shrinking and collapsing as shown in Fig. 6.19a. As the bubble starts to grow from time t_0 with an equilibrium radius R_0 to t_1 with radius R_1 , the bubble expands rapidly; and the corresponding velocity of the bubble wall causes the frequency of probing wave to shift up, which is shown from t_0 to t_1 in Fig. 6.19b. From the period of t_1 to t_2 , the bubble still grows but with a decelerated boundary, and the frequency in Fig. 6.19b goes down from its peak to the original frequency of 29.4 MHz. When the bubble reaches its maximum size, the velocity of the bubble wall is zero and the frequency of the probing wave does not change, which is shown in Fig. 6.19b. When the bubble begins to collapse, the boundary velocity is negative, and the frequency of the probing wave is shifted down, corresponding to the period of t_2 to t_3 in Fig. 6.19b. Taking a careful look at Fig. 6.19a and Fig. 6.19b during the period of t_0 to t_2 , we find that the bubble growth

and frequency increase do not match exactly. This is mainly because of the limited bandwidth of the detecting transducer. As the bubble collapse occurs within an extremely short period, this corresponds to a very wide spectrum in the frequency domain. The detecting transducer filters out some of the frequency components, and some distortion is inevitably added to the backscattered signal.

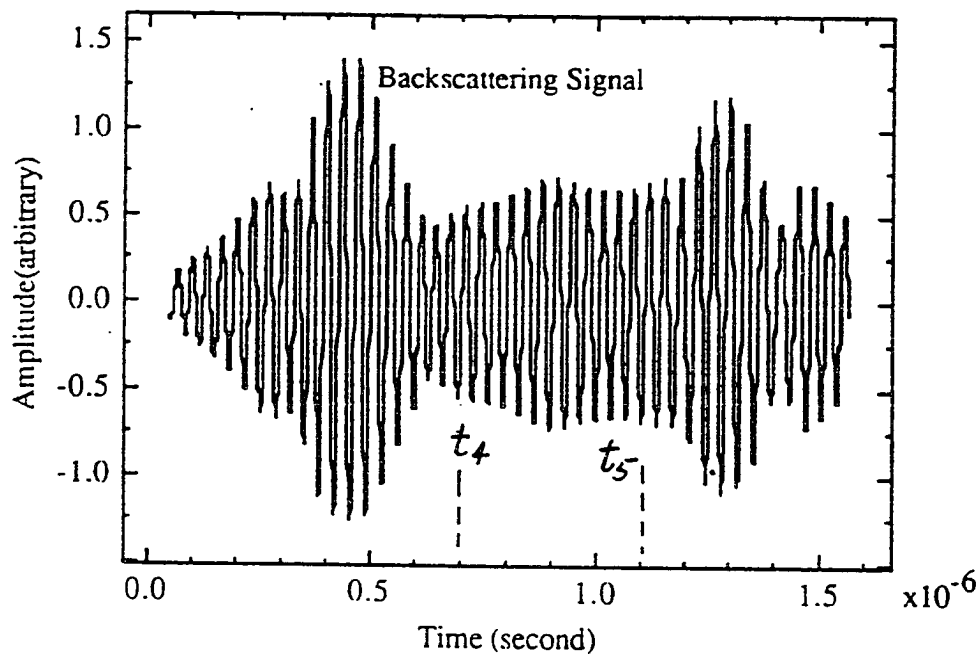


Fig. 6.20a. Backscattered signal from microcavitation

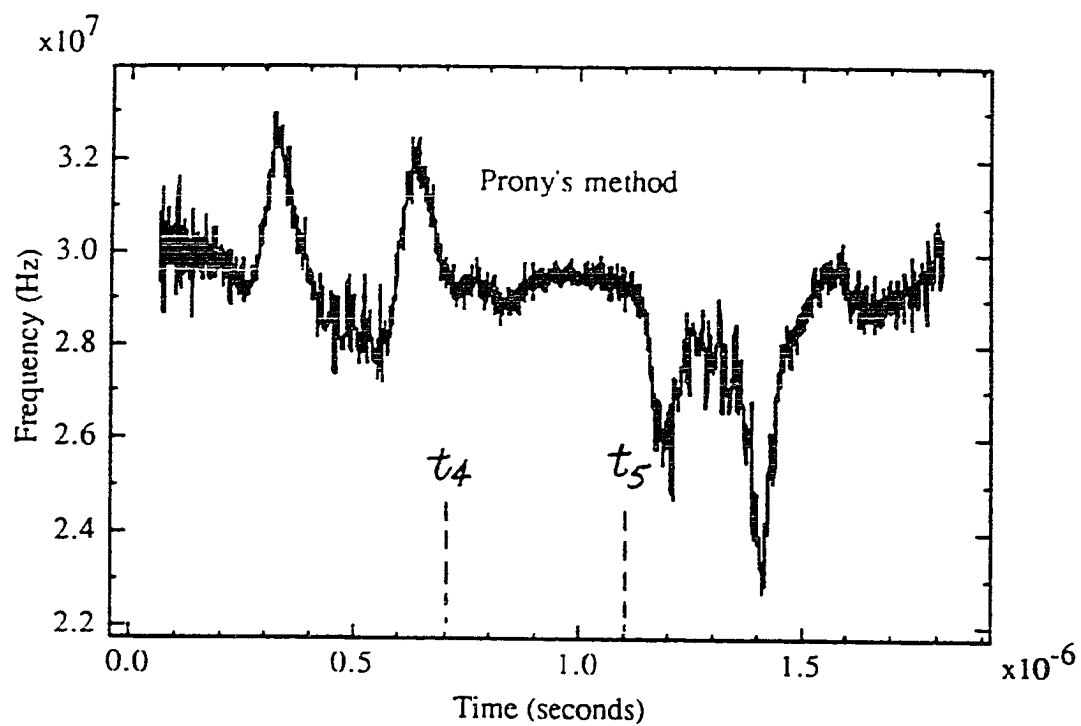


Fig. 6.20b. Estimate of transient frequency vs. time by using Prony's method

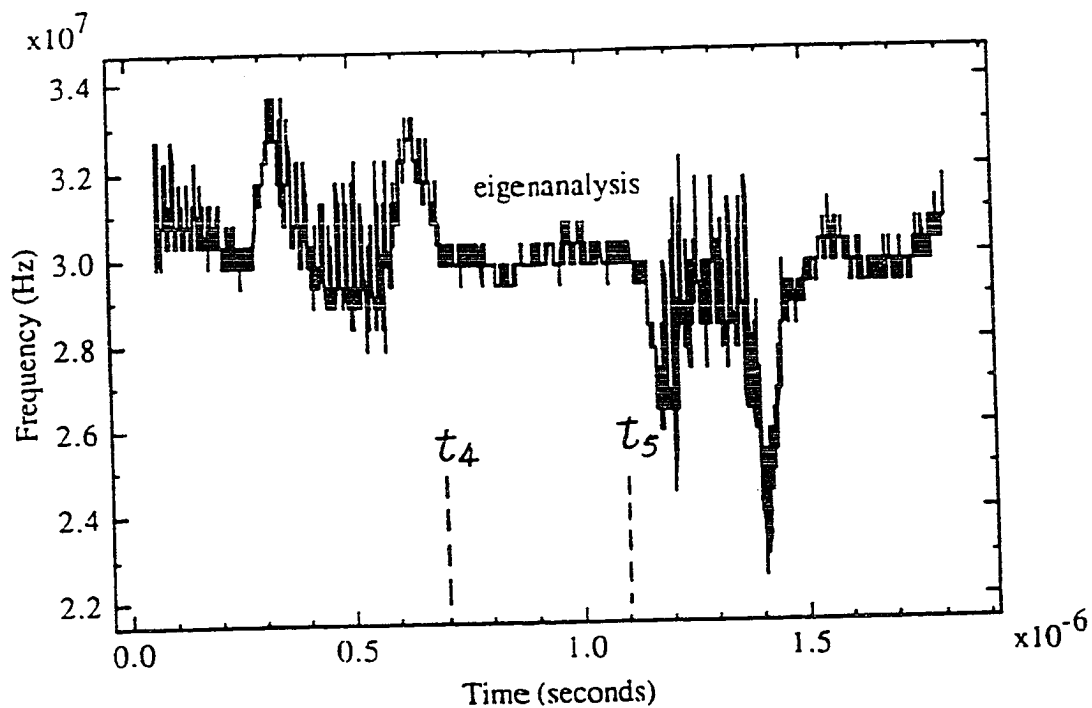


Fig. 6.20c. Estimate of transient frequency vs. time by using eigenanalysis method

Fig. 6.20a is a cavitation event, but apparently with a static scatterer attached during the period of t_4 to t_5 . During t_4 to t_5 , the scatterer did not grow or shrink, and the frequency of the probing wave is a constant as shown in Fig. 6.20b and 6.20c. Before t_4 or after t_5 , cavitation occurred. Figures. 6.20b and 6.20c clearly show that the frequency of the probing wave is shifted up or down for different parts of the cavitation cycle.

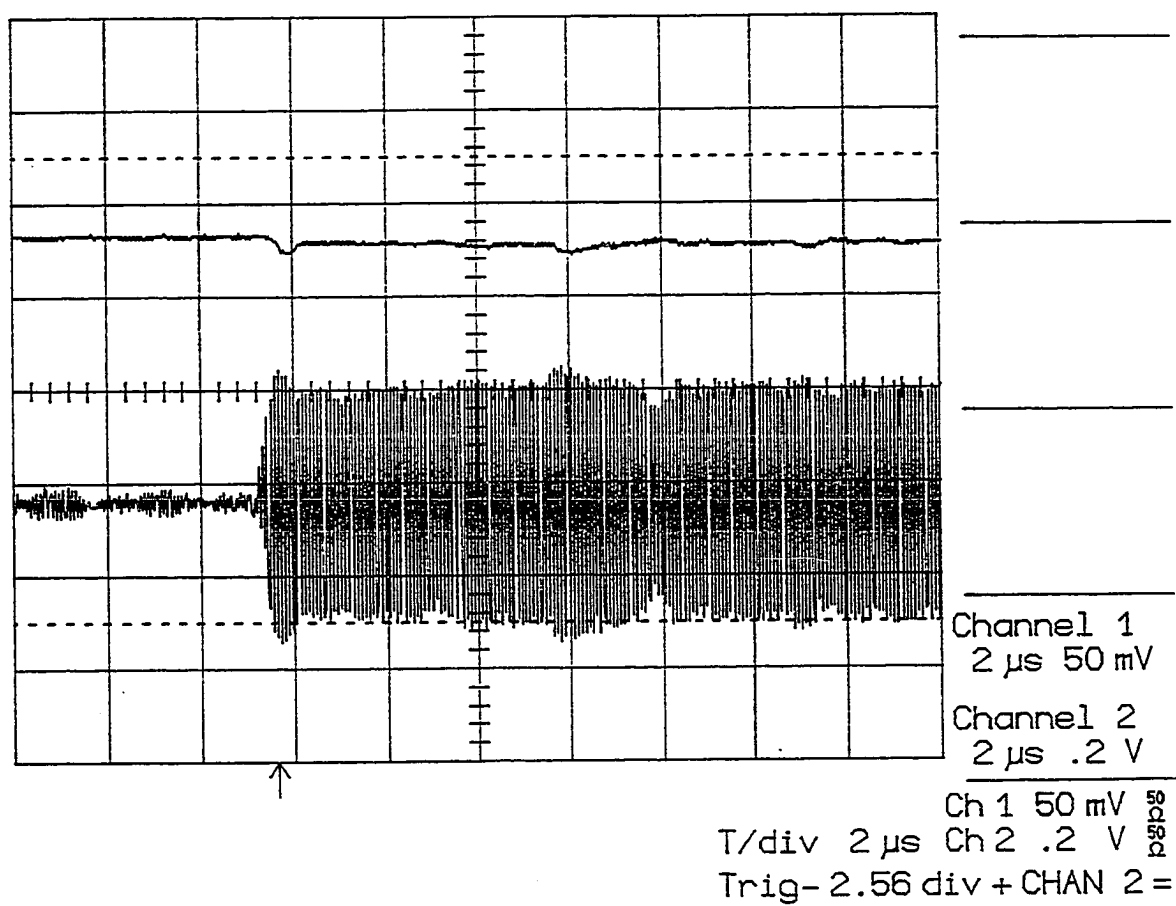


Fig. 6.21. Analog processing results. Lower trace shows the backscattered signal from a polystyrene particle. Upper trace shows the corresponding transient frequency information using analog method.

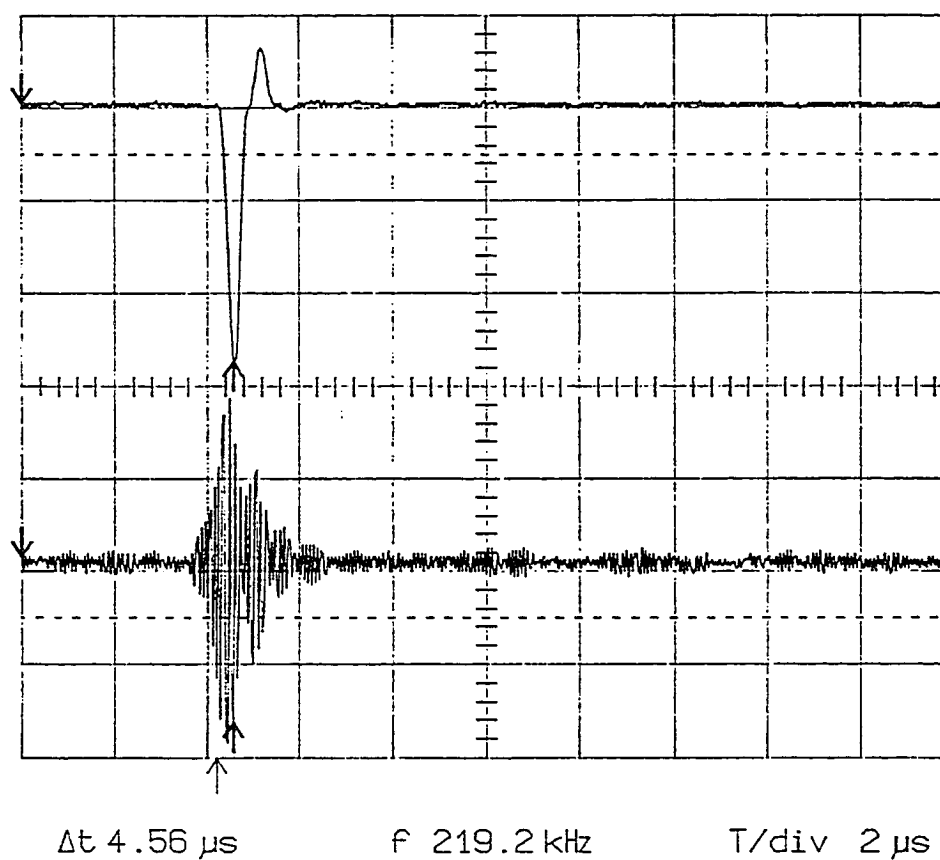


Fig. 6.22a. Analog processing results. Lower trace shows the backscattered signal from cavitation caused by 2.46 MHz acoustic wave. Upper trace shows the corresponding transient frequency information using analog method.

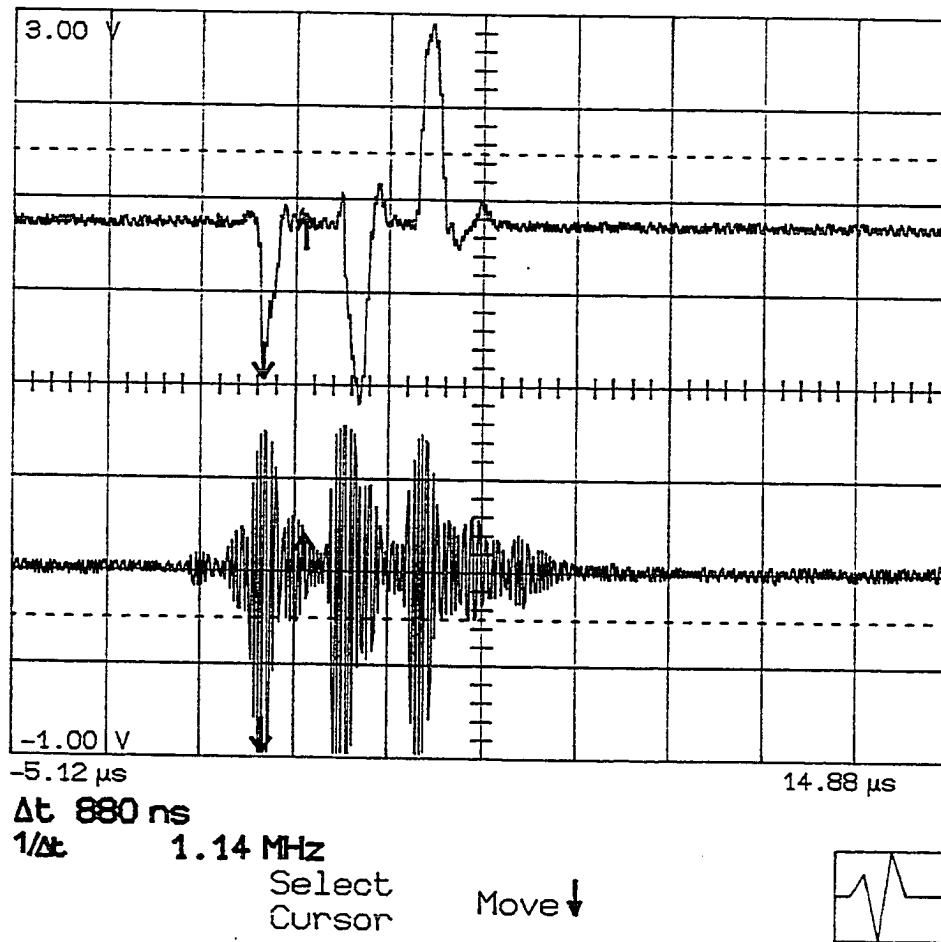


Fig. 6.22b. Analog processing results. Lower trace shows the backscattered signal from cavitation caused by 2.46 MHz acoustic wave. Upper trace shows the corresponding transient frequency information using analog method.

Figures 6.21, 6.22a and 6.22b show the results of analog processing. In these figures, the upper trace shows the backscattered signal from the cavitation generated by a 2.46 MHz acoustical wave. The lower trace shows the result of analog processing. Fig. 6.21 shows the backscattered signal from a polystyrene particle. The corresponding upper trace tells us that the center frequency has not changed. Figs. 6.22a and 6.22b are the results of the backscattered signals from the cavitation of single and multiple events. The upper traces show that the center frequency has changed, and there are Doppler frequency shifts associated with the backscattered with the scatterer.

As given in Eq. 2.94, the initial phase is not known. Therefore, we cannot determine when the frequency is shifted up or shifted down for the backscattered data in Figs. 6.22a and 6.22a. The important thing is that Figs 6.22a and 6.22b tell us when the frequency shifts, which is caused by bubble growth or collapse

6.1.4.2 Conclusion and Discussion on Water Measurements

By using Prony's method or the eigenanalysis method, the frequency shift of the probing wave is clearly shown. The time and frequency correspondence for bubble growth and collapse reasonably match, which is expected. Although Prony's method seems to give more accurate estimations in most cases, Prony's method is not designed to handle noisy data. It is believed that with smaller signal to noise ratio, the eigenanalysis method may outperform Prony's method. As shown in Fig. 6.20, when the backscattered signal is relatively weak or the signal to noise ratio is small, Prony's method gives too much local fluctuation. There is no clear indication of the frequency trend, and the eigenanalysis method gives less fluctuation and a reasonable frequency trend.

6.2 Cavitation in Blood

Although many techniques have been well developed for detecting stable bubble oscillations and transient microcavitation *in vitro*, and considerable experience has been accumulated in detecting such cavitation, still, the detection of transient microcavitation *in vivo* is not a trivial task. First, the size of the transient cavitation bubble is so small as to be below the range of the spatial resolution of diagnostic scanners. Secondly, the life of the cavitation bubble is too short to extract the cavitation signal from its environmental noise. Finally, the most difficult detection problem to be overcome is the scattered acoustic noise or "clutter," which can come from almost any part of the quasi-static materials *in vivo*.

The purpose of doing the experiment in blood is to induce microcavitation by insonifying biological material (human blood) and to measure cavitation thresholds. The experimental data may not directly prove that diagnostic ultrasound will definitely cause cavitation *in vivo*. However, the measured thresholds might indicate that there is a large probability that microcavitation may be induced *in vivo* once the diagnostic ultrasound intensity reaches a certain amplitude. The results obtained and methods employed in this work should pave the way for further understanding of transient microcavitation *in vivo* induced by diagnostic ultrasound.

6.2.1 Experimental Results

The following table and figures show the experimental results in blood,

Blood Sample	HP Output (mV)	CAVITATION THRESHOLD (MPa)		
		NTR	Reflective Target	Absorbing Target
plasma	130	2.805	2.947	3.66
12.5%	95	2.052	2.135	2.68
25%	100	2.16	2.251	2.82
50%	120	2.59	2.715	3.38
100%	180	3.88	4.017	5.061

Table 6.5. Cavitation thresholds for different blood samples. The right most three columns show the threshold values converted with three calibration methods. The column NTR corresponds to the threshold values calibrated with NTR hydrophone probe (NTR system, Seattle, WA). The percentage of plasma and RBC are given in terms of volume concentrations.

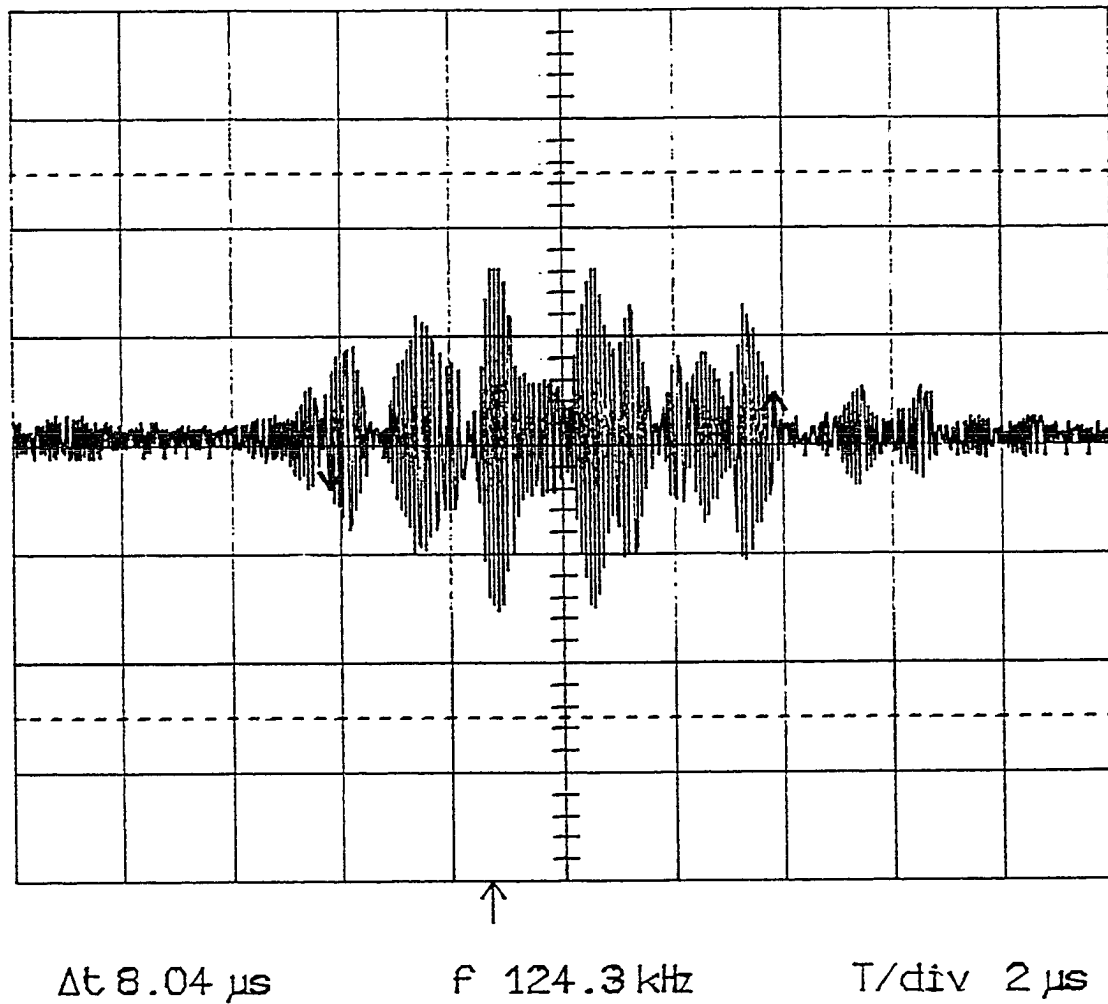


Fig.6.23. Backscattered signal from cavitation caused by a 2.46 MHz acoustic wave in 100% plasma

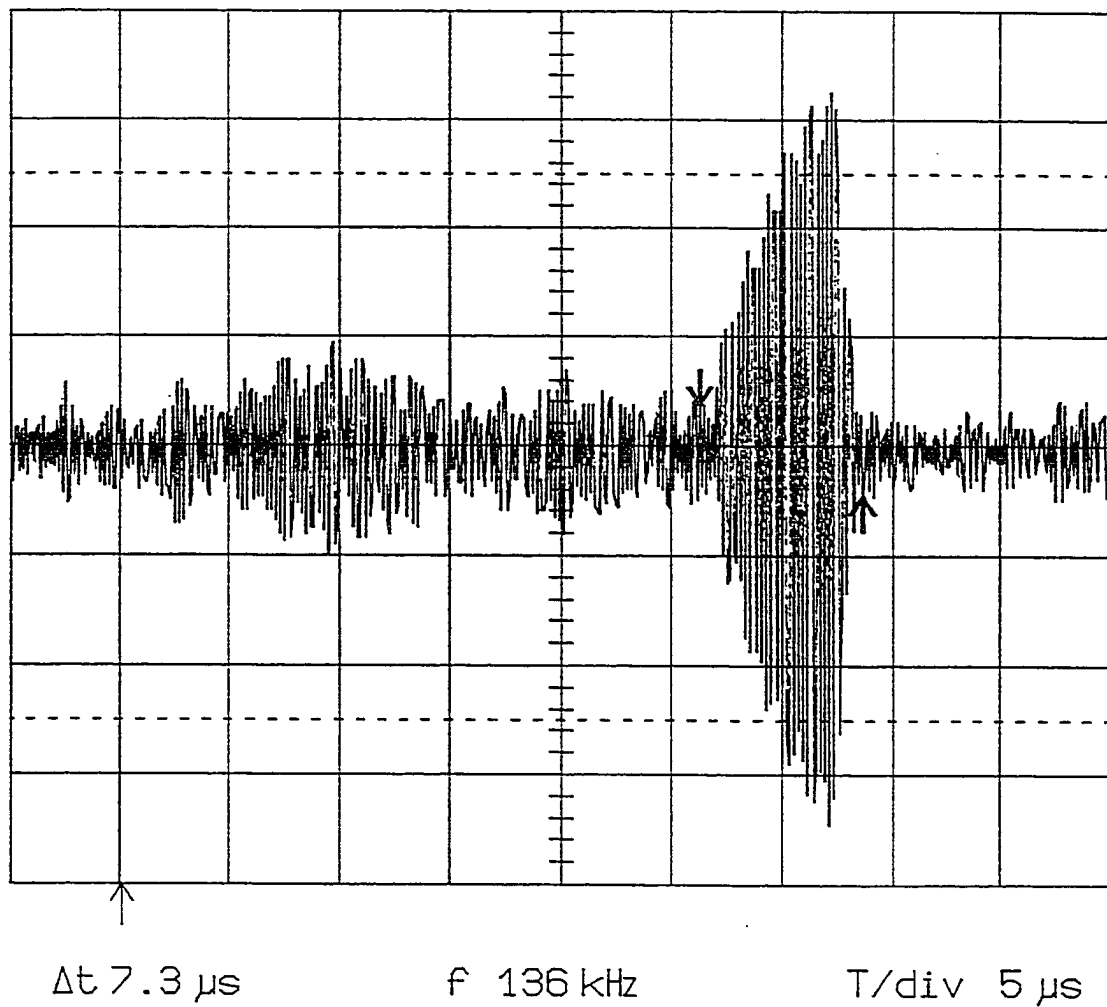


Fig.6.24. Backscattered signal from cavitation caused by a 2.46 MHz acoustic wave in a solution of 12.5% whole blood and 87.5% plasma.

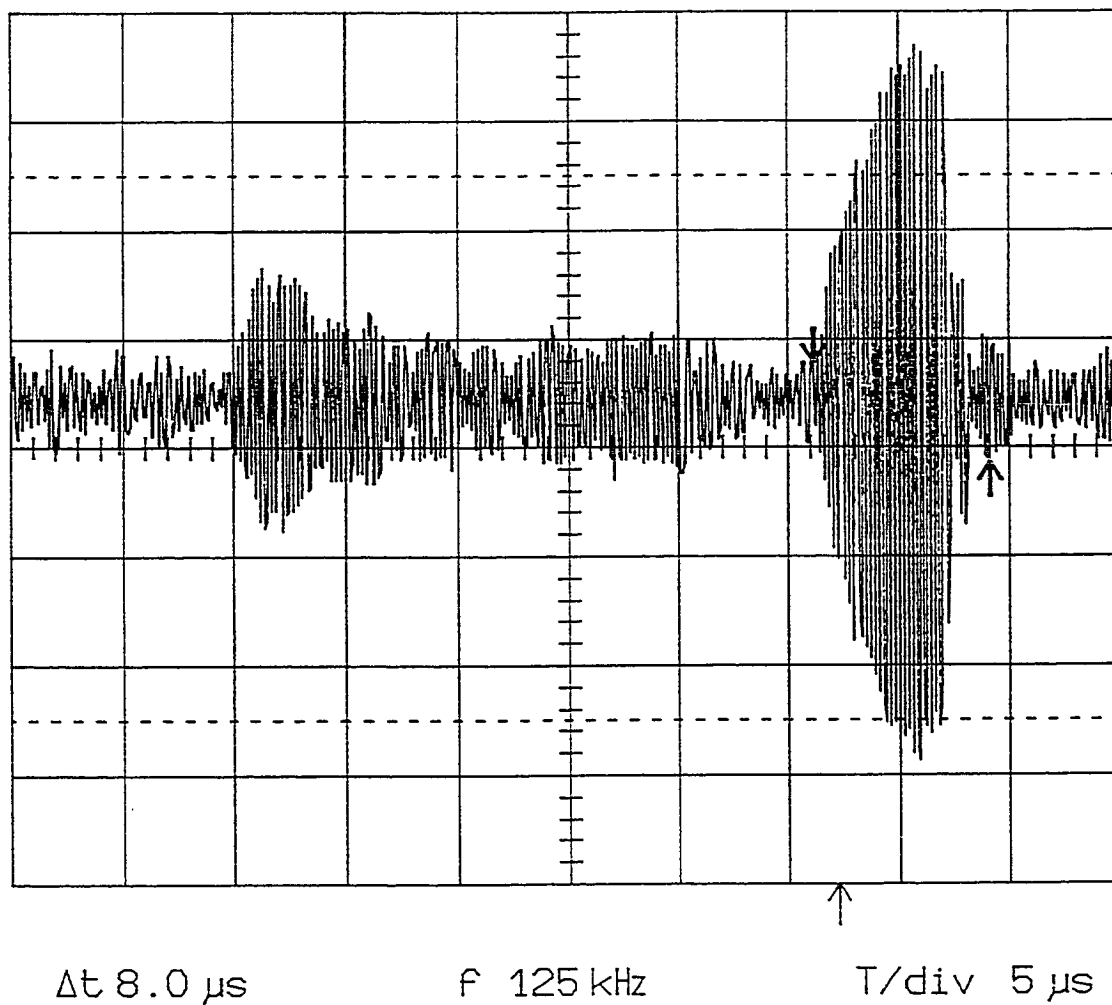


Fig.6.25. Backscattered signal from cavitation caused by a 2.46 MHz acoustic wave in a solution of 25% whole blood and 75% plasma.

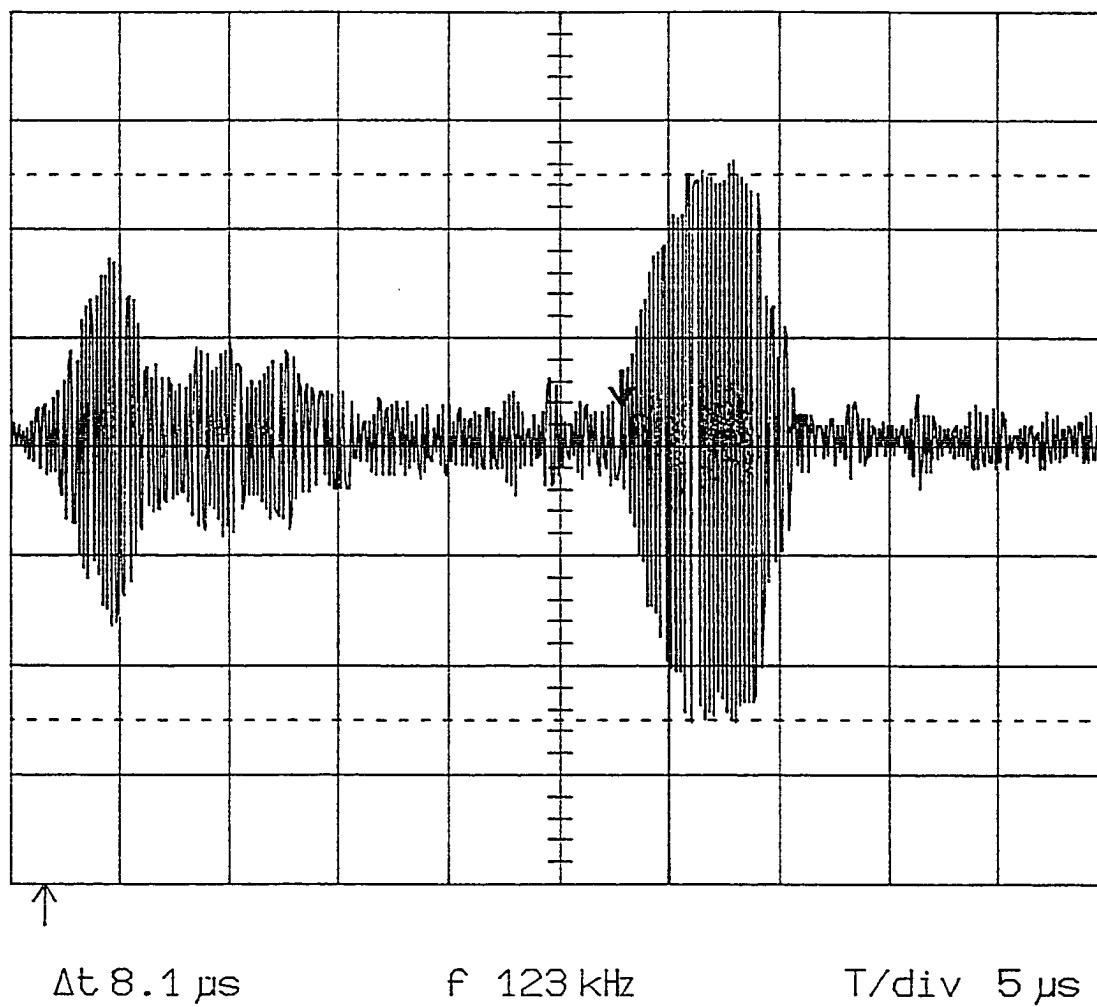


Fig.6.26. Backscattered signal from cavitation caused by a 2.46 MHz acoustic wave in a solution of 50% whole blood and 50% plasma.

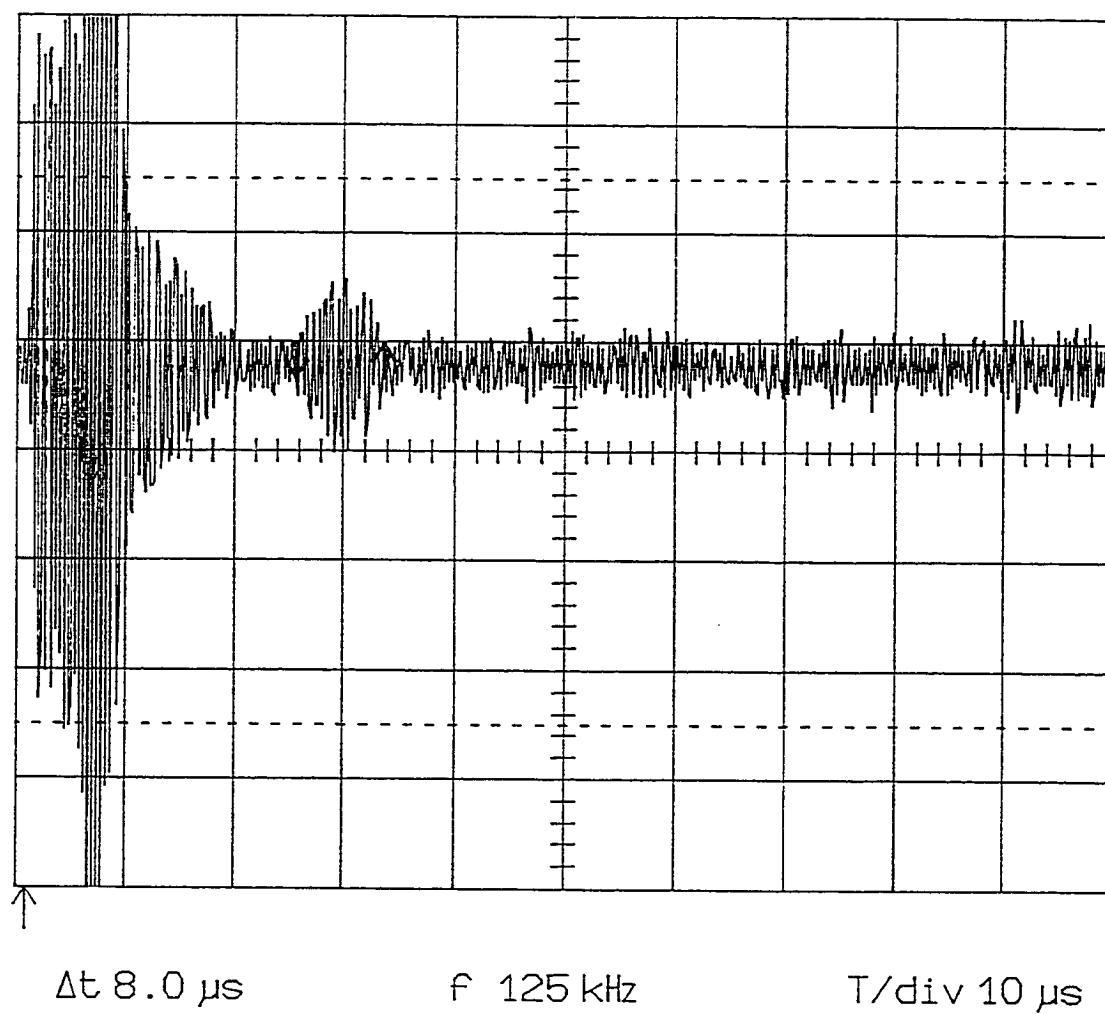


Fig.6.27. Backscattered signal from cavitation caused by a 2.46 MHz acoustic wave in whole blood.

6.2.2 Cavitation Thresholds

In Figs. 6.24, 6.25, 6.26 and 6.27, the first peaks correspond to the leakage through T/R switch from the main "bang", which is the signal exciting the probing transducer. The second peaks are the backscattered signals from microcavitation. Although the volume concentrations of RBC are different, the durations of the microcavitation are all the same as the duration of the driving tone burst, which is $8.1\mu s$.

For plasma, the backscattered signals from the cavitation consistently appeared on the screen of the oscilloscope once the acoustic pressure from the cavitation transducer was high enough. The duration of the backscattered signals was wholly determined by the duration of the driving signal. There is a clear indication for the growth and collapse of a bubble, which behaves very similarly to the cavitation caused in distilled water. According to Rayleigh scattering theory for a small scatterer, the cube root of the scattering strength is proportional to the scatterer size. Therefore, the oscillations of the backscattered signal in Fig. 6.23 indicate the process of bubble growth and collapse. But in Fig. 6.27, this process is not obvious.

Fig. 6.28 shows cavitation thresholds for different volume concentrations. The cavitation thresholds are in the range of the acoustic pressures employed by some medical diagnostic ultrasound instruments [Duck, *et al.*, 1985, 1987]. Therefore, it appears possible, even accounting for attenuation *in vivo*, to generate transient microcavitation with acoustic pressures reported in the literature [Duck, *et al.*, 1985, 1987], given RBCs as nuclei.

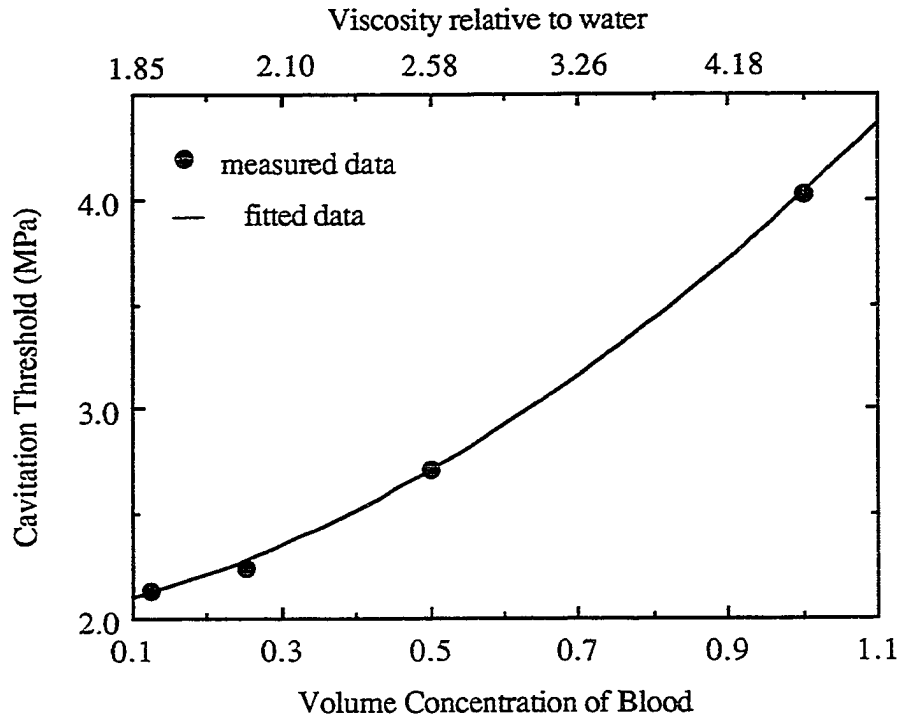


Fig. 6.28. The Cavitation threshold in blood vs the volume concentration of blood.

The cavitation threshold values are converted with NTR calibration.

Viscosity is an very important factor for microcavitation thresholds, generated by megahertz acoustic pressure. Fig. 6.28 shows that the cavitation threshold varies approximately linearly with the volume concentration of RBC. As RBC concentration increases, the viscosity of the blood solution also increases.

6.2.3 Noise

The time history of the noise for the samples of 25% blood and 12.5% blood defined in the table were very similar to the scattered signals caused by placing Albunex® (stabilized microbubbles, MBI, Inc., San Diego, CA) into distilled water. In Fig. 6.24, the noise level is about 45% of the amplitude of the backscattered signal from the cavitation. If the noise level is higher than the signal level of cavitation scattering, we cannot identify the cavitation event from the noise by simply comparing the signal amplitude with the noise level. Larger scatterers cause stronger backscattering. Therefore, the scale of the detected microcavitation bubble must be large enough so that the backscattered signal is well above the noise level. Consequently, the measured cavitation thresholds in our measurements may be a function of noise level.

6.2.4 Conclusion on Blood Measurements

The cavitation thresholds we measured are in the range of the acoustic pressure amplitudes that some medical diagnostic devices can generate *in vitro* and very likely *in vivo* in clinical applications, as envisioned by Apfel [Apfel, 1986]. Considering the effect of noise on the threshold measurements, the real cavitation thresholds may be below the values we measured. The cavitation threshold varies linearly with the volume concentration of RBC.

Chapter 7 Epilogue-Significance, and Implication for Future Work

As diagnostic medical ultrasound market has grown to an unprecedented size (about a billion dollars/year), the image qualities have been improved drastically. Accompanying these the image improvements are higher acoustic pressures from the ultrasound scanners. Some of these scanners can generate acoustic pressures of 4.0 MPa amplitude in water with center frequency at about 5.0 MHz. In this dissertation, we have shown the cavitation thresholds in both water and blood samples using ACD techniques and Doppler frequency principles. We have used a 4.3 MHz transducer to cause cavitation in water, and the corresponding cavitation threshold is about 4.0 MPa. We have also used a 2.46 MHz transducer to cause cavitation *in vitro* blood. These results may not directly indicate that diagnostic medical ultrasound can cause cavitation *in vivo*., but they imply that further work is needed to study transient microcavitation *in vivo*.

To expand the work to a deeper understanding of the potential side effects of diagnostic ultrasound *in vivo*, the following work would be useful:

- 1) Improve the method of analog signal processing discussed in Chapter 6. A more sophisticated analog signal processing circuit is needed to extract the Doppler frequency shift and the direction of the bubble motion, and further to eliminate the DC offset of the input signal. A high quality FM demodulator is required to extract the frequency information from the backscattered signal. To distinguish the direction of bubble motion, we can use the techniques similar to that widely used in Color Doppler flow system which can indicate the direction of blood flow.
- 2) It would be useful to obtain the bubble wall velocity from the backscattered signal. This can be done by calibrating transient Doppler Frequency Shift vs the surface velocity of scattered. A suggested scheme is that by measuring the transient Doppler shift of the backscattered signal from a spherical scatterer which only has normal component of

surface velocity, we can establish a relationship between the Doppler frequency shift and the surface velocity of the scatterer. Once such relationship is established, we can obtain the velocity of the bubble wall from the backscattered signal.

3) A study should be done to investigate how much damage could be caused by *in vivo* cavitation. This work involves the corporation between acousticians and medical professionals.

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